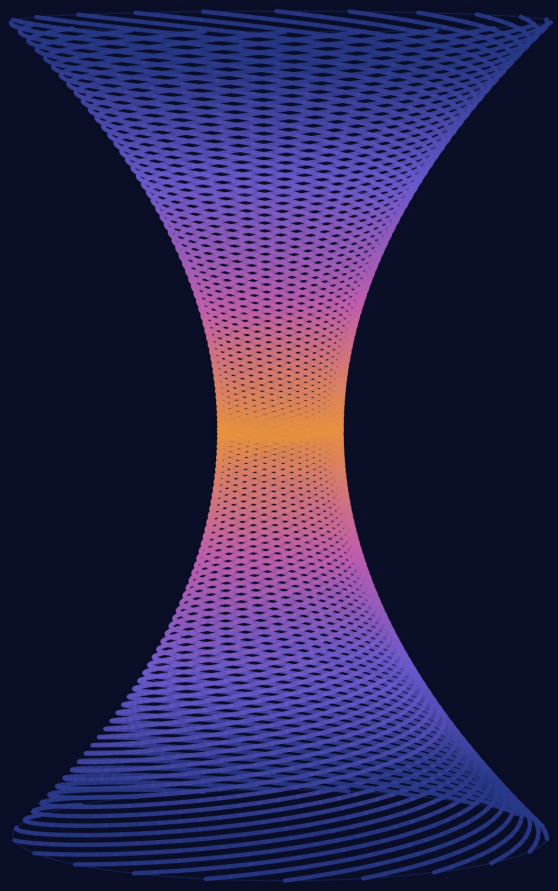


EINSTEIN–CARTAN FRAMEWORK

Architecture of the corpus

Torsion coupled to spin · non-singular bounce · geometric dark sector

Pascal Fichant

Architecture of the Einstein–Cartan Framework

ECF

Scientific synthesis and map of the corpus

Pascal Fichant
Independent researcher

ECF Architecture – version 2 (revision 1.5)
Post-audit v2 scientific baseline
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p.fichant.research@gmail.com

ABOUT THIS VERSION

French and English versions. The Architecture exists in French and in English. The present document is the **English version** of version 2 (internal revision 1.5), translated from the French, which is the source; it supersedes the English version 1 of 15 September 2026 (10.5281/zenodo.22778508), and the differences are recorded in the version register and erratum of the October 2026 deposit. It mainly reorganises the narrative and the scientific navigation. The substantive corrections explicitly introduced during the review are documented in the comment register and in the editorial traceability; they close no open problem by mere rewording. This English version is a translation of the French version 2, checked for structural parity (headings, tables, equations and numerical values); in case of divergence, the French text prevails.

Purpose. Architecture is the map, the synthesis and the navigation tool of the ECF corpus. It makes explicit the genealogy of the results, their dependencies, their epistemic status and the problems that remain open. It is not an additional scientific source: **the papers of the corpus remain the primary scientific sources.**

Resources and public deposit. See the section The ECF corpus at a glance, which gathers the documents, their deposit statuses and the available access points.

Foreword — An invitation to read, critique and comment

The **Architecture** is both a scientific synthesis and an entry point to the whole ECF corpus. It was designed to make visible not only the results and their dependencies, but also their **status**: what is derived, calculated under assumptions, calibrated, phenomenological, conditional, still open or withdrawn.

The corpus is a **research programme in development**. Critical reading is therefore part of it. An objection, an ambiguity, an insufficiently justified equation, a missing reference, a status that seems too strong or simply a passage that is hard to understand are all useful feedback.

You can comment on Architecture, but also on any paper of the corpus. The method is the same: open the relevant PDF with a **PDF reader that allows comments and annotations to be added** — for example Adobe Acrobat Reader, Foxit PDF Reader, PDF-XChange Editor, Microsoft Edge or Preview on macOS. Other readers offering equivalent functions are also suitable. Highlight or annotate the passage directly, then save the PDF with its annotations and send it to:

Pascal Fichant — p.fichant.research@gmail.com

To ease processing, please indicate where possible the **page**, the **section** and, where it exists, the **equation number**. Remarks are attached to the document they concern and examined at that level. A correction of Architecture never replaces a correction of the source paper; conversely, when a remark modifies a primary paper, the Architecture synthesis is updated if necessary. **The papers remain the primary scientific sources; Architecture remains the map of the corpus.**

The detailed procedure, the proposed types of comments and an example annotation are given in **Appendix G — How to review and comment on the ECF corpus**.

> **Processing principle.** An editorial correction may be applied directly after verification. A remark that bears on an equation, a value, a scientific status, a causal dependency or an open problem is checked against the corpus before any modification. A genuinely open problem stays open: it is not closed by rewording.

Confidentiality. Annotated PDFs and comments received are not made public nor communicated to other readers by the author. Only the corrections retained appear in subsequent versions; your name is cited — in the acknowledgements, for example — only with your agreement.

The reader who only wishes to understand the framework can start with the Summary and the Quick start. The reader who wishes to audit it can go directly to the statuses, the scientific locks and the traceability appendices.

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How to read this document. It can be used at **three depths**, and **it need not be read in full nor linearly**.

1 — Overview of the corpus *(Summary · The corpus at a glance · Quick start · In brief)*: understand the idea, the physical chain and the statuses without entering the technical apparatus. **2 — Scientific exposition** *(§1 to §3, §5 to §8)*: follow the construction, the confrontations with data, the limits, the scientific locks and the positioning. **3 — Technical reference** *(§4 and appendices A–F and H)*: find equations, statuses, checks, provenance and revision history. Appendix G separately describes the procedure for returning comments.

The explicitly open points (notably the causality chiral walls / knot populations) remain open and are not resolved by wording.

Summary

The Einstein-Cartan Framework (ECF) explores the cosmological consequences of an extension of general relativity in which the spin of matter sources a torsion of spacetime. At very high density, this contribution modifies the gravitational dynamics and allows a non-singular bounce scenario. The programme then studies how this primordial phase could leave observable effects on the expansion, the dark sector and structure formation.

The ECF does not introduce entirely new physics at each step. It assembles known building blocks — Einstein-Cartan theory, the Weyssenhoff fluid, the Kibble-Zurek mechanism, the Landau-Ginzburg potential, the CPL parametrisation and the Sakharov conditions — around a postulate proper to the framework, the Topological Invariance Principle. The consequences obtained do not all have the same status: some are calculated within the adopted model, others are calibrated on observations, several remain conditional on an effective description that is still incomplete.

In the cosmological sector studied here, the framework produces a bounce at finite density and modifies the expansion history before recombination. The same programme explores a geometric dark sector, a formation of topological defects and a chiral baryogenesis. The **Macro** population (~$10^5\ M_\odot$) is treated there as a distinct sector: its causal scale can be calculated, but its formation and collapse remain a **structural hypothesis**. The confrontations with data bear notably on the sound horizon, H_0, S_8, rotation curves, cosmic voids and the evolution of dark energy. Fitted values and unfitted predictions are explicitly distinguished throughout the document.

Several locks remain open. In particular, the complete dynamical action of the axial sector is not yet derived, the treatment of perturbations does not yet rest on a complete Einstein-Cartan Boltzmann solver, and some structure-formation channels — notably the Macro sector — remain conditional. Architecture's function is therefore to present the current state of the programme without confusing calculated result, calibration, structural hypothesis and open question.

The ECF is a **research programme in development**. Architecture is its organised mirror: it synthesises and links the documents of the corpus, but constitutes neither an independent proof nor an additional scientific source.

The ECF corpus at a glance

*This section is a **condensation of Appendix A**, which gives the full scope of each document. It contains no result that does not already appear there.*

Documentary progression

PIT → Foundation I → Foundation II → Foundation III *Cosmological Physics* → Strong Gravity → Chiral Bounce → Foundation IV

⚠ **The arrows indicate the order in which the programme was built, not a necessary derivation of each branch.** The actual scientific dependencies are those carried by the statuses of each result and the detail of Appendix A: some branches are conditional on links that are still open, others are independent of them.

The documents

Document	What it covers	State
PIT Letter	the Topological Invariance Principle — spatial flatness is *exact* ($\Omega_k \equiv 0$), a falsifiable claim; and the open problem of the link $\Omega_{\text{spin}} \leftrightarrow \Omega_\tau$	v9 (October 2026 deposit; v8 online since 06/09/2026), 6 pp.
Foundation I	the cosmological background: Einstein-Cartan-Kibble-Sciama framework, reduction of the sound horizon ($r_s = 135.65$ Mpc), joint H_0/S_8 treatment via F_{ion}	v8 (October 2026 deposit, internal revision 13; v7 online since 14/09/2026), 87 pp.; PRD Short v8, 8 pp.
Foundation II	the geometric dark sector: chiral baryogenesis, dark matter as micro-knots via Kibble-Zurek, topological zoology of defects, macroscopic structure	v5 (October 2026 deposit, internal revision 20; v4 online since 15/09/2026), 173 pp.
F III — *Cosmological Physics*	BBN constraint, observational comparisons and confrontations of the cosmological sector	v3 (October 2026 deposit, internal revision 1.1 revised; v2 online since 15/09/2026), 41 pp.
F III — *Strong Gravity*	geometry of compact objects and high-density regime — *outside the scope of this document; refer to the document itself*	revision 2.1 + CORE-01, 37 pp., **not deposited**
F III — *Chiral Bounce*	formulation of the bounce as a chiral phase transition — *outside the scope of this document*	revision 0.4, 21 pp., **not deposited**
Foundation IV — *Fertile*	exploratory programme, acknowledged as such by the author — *outside the scope of this document, which covers the cosmology that can be confronted with data*	revision 0.3, 64 pp., **preliminary, not deposited**
Architecture *(this document)*	**map, synthesis, navigation and traceability of the corpus** — organised mirror of the papers, and **not an additional scientific source**	version 2 (revision 1.5); supersedes v1 of 15/09/2026

Zenodo concept DOIs — *they always point to the latest published version, which is not necessarily the current version indicated above; the version-by-version detail is given in Appendix D:*

- **PIT Letter** — 10.5281/zenodo.19798923
- **Foundation I** *(Extended and PRD Short, same deposit)* — 10.5281/zenodo.19577447
- **Foundation II** — 10.5281/zenodo.20629237
- **Foundation III — Cosmological Physics** — 10.5281/zenodo.22359982
- **Architecture** — concept DOI 10.5281/zenodo.22778507, v1 of 15/09/2026: 10.5281/zenodo.22778508; the present version 2 supersedes it.
- **Strong Gravity · Chiral Bounce · Foundation IV** — *not deposited to date*

The states above are those of Appendix A, ***which prevails in case of discrepancy****: this section condenses it as an entry point, it does not replace it. The documents are made available according to their current state of development and dissemination.*

Data and code availability

The ECF corpus (documents, calculation and figure scripts) is made available **according to the state of development and dissemination of each element**; the public part is licensed under **CC-BY 4.0**:

github.com/pfichant/spin-torsion-cosmology

Every quantity cited in this document points to the script that computes it; most are directly executable from the repository for independent verification. *A limitation must be stated explicitly, as in Foundation I*: the repository contains the scripts that **reproduce** the published values, not necessarily those that **produced** them — some upstream parameter searches are not included (no global MCMC fit was run) as such. Each value can therefore be checked for its internal consistency, without the calibration itself always being re-runnable from the repository as it stands.

© 2026 Pascal Fichant. This document synthesises works deposited publicly and time-stamped (see the Zenodo DOIs cited in reference: PIT 10.5281/zenodo.19798923, Foundation I 10.5281/zenodo.19577447, Foundation II 10.5281/zenodo.20629237, Foundation III — Cosmological Physics 10.5281/zenodo.22359982. Distributed for reading and evaluation. Any reuse must cite the author and the original repositories.

Quick start

No time to read 300 pages?

You don't have to. The corpus can be **queried** in natural language: upload the PDFs into **NotebookLM** (Google, free), ask your question, and the answer comes back in seconds **with the source passage**.

Finding where a quantity is defined, derived or calibrated takes tens of minutes of cross-reading. The same question, put to the tool, takes three minutes.

→ **Appendix E** gives the procedure, **three orientation questions** for a first contact, a **worked example**, and how to read the answers without misinterpretation. *Five minutes to set up.*

*This document is designed to be read **on its own**: it carries the methods, inputs and scripts needed to reproduce the calculations without opening the papers. The papers carry the detail of the derivations.*

Three ways in, depending on the time you have:

You have…	Read
5 minutes	"In brief" below, then the **diagram** and its navigation table (§2)
30 minutes	+ the **summary of calibrated quantities** (§2) — how each quantity is obtained — and §1 *The problem*
2 hours	+ §3 *Modified Friedmann*, §4 *Complete derivations*, §5 *Confrontation with data*

Quick access to the deposits. The concept DOIs of PIT, Foundation I, Foundation II, Foundation III — Cosmological Physics and Architecture are gathered in "The ECF corpus at a glance", to avoid having to search for them in the document.

Reading feedback on the whole corpus. You can annotate **Architecture or any ECF paper** directly and send the PDF back to **p.fichant.research@gmail.com**. Indicate where possible page, section and equation number. → **Appendix G** describes the method common to all documents of the corpus.

The full kit, in the suggested reading order:

1. **this document** — the technical entry point;
2. **Foundation I** — cosmological background, calibration, H_0 and S_8;

3. **Foundation II** — the dark sector, and Appendix J *(the derivation of static dark energy)*;
4. **Foundation III — Cosmological Physics** — BBN, DESI, voids, primordial spectrum;
5. *(in reserve)* **PIT Letter** — the postulate and the open question it leaves.

Querying the corpus without reading it end to end. NotebookLM (Google) locates in seconds where a notion is treated, with its source. → **Appendix E**, which gives the method, three orientation questions for a first contact, and how to read the answers without misinterpretation.

Your comments are welcome — on Architecture as on the papers of the corpus. Annotate the relevant PDF directly and send it back to **p.fichant.research@gmail.com**. Each remark is attached to the source document, checked and receives a personal reply. → **Appendix G** describes the common approach.

The two founding equations of Einstein-Cartan

The minimal foundation is set by two field equations. The first is Einstein's metric equation with the effective spin contribution,

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \kappa \Big(T^{\text{matter}}_{\mu\nu} + \Theta^{\text{spin}}_{\mu\nu} \Big) \,. \tag{EC-A}$$

The second is Cartan's equation, which algebraically relates torsion and spin density in minimal ECKS,

$$T^{\lambda}{}_{\mu\nu} + \delta^{\lambda}_{\mu} T_{\nu} - \delta^{\lambda}_{\nu} T_{\mu} = \kappa \, S^{\lambda}{}_{\mu\nu}. \tag{EC-B}$$

In this minimal foundation, torsion is therefore not an independent propagating field. The late sectors of the corpus that use a dynamical axial pseudo-vector constitute an additional effective description; its derivation from the complete PGT action and the identification of its physical pole remain open under **PO-ECF-ACTION / PO-F2-6**.

One reading key, to keep in mind throughout. The results are not placed on a binary "true/false" axis: Architecture distinguishes notably **DERIVED**, **CALCULATED-CONDITIONAL**, **CALIBRATED**, **PHENOMENOLOGICAL**, **CONDITIONAL**, **STRUCTURAL-HYPOTHESIS** and **OPEN**. In the narrative, these statuses are rendered in plain language (derived, calculated under assumptions, calibrated, conditional, structural hypothesis, open). The technical table and the appendices give the precise status, which prevails.

A second convention worth knowing before reading: the "PO-Fx-y" codes. The document names each open problem with a short code rather than repeating a full description each time — for example **PO-F2-8** (the hypothetical 1D string sector) or **PO-F3-XI** (the ξ_{KZ} scale, calibrated rather than derived). The code reads directly:

- **PO** stands for *Open Problem* — an unresolved question, never presented as closed;
- **Fx** indicates the document that carries or introduces the problem (F1, F2, F3, F3-StrongGravity…);
- the **suffix** (a number or a short mnemonic) identifies the specific problem within that document.

A code is therefore never an abbreviation to memorise: at its first occurrence in the text, it suffices to read the sentence around it — the problem is always named there in plain language before or after its code. Codes mainly serve to **link together** distant passages that touch the same problem (for instance, PO-F2-8 recurs in both §4.4 and Appendix A): tracking them down lets a reader gauge a problem's real scope without re-reading the whole document.

In brief (for a quick first reading)

The idea in a few lines

The ECF starts from Einstein-Cartan theory: when matter carries spin, geometry can carry torsion in addition to curvature. This correction is negligible in ordinary regimes, but can become important at extreme densities. The framework

then explores a cosmological history in which the initial singularity is replaced by a bounce, and follows the possible consequences of this primordial phase on the cosmological background, the dark sector and certain observational signatures.

The aim of Architecture is not to present all these consequences as established. It is on the contrary to distinguish, at each step, **what is borrowed**, **what is calculated within the adopted model**, **what is calibrated**, **what is conditional** and **what remains open**.

The physical chain in six steps

1. **Einstein-Cartan:** the spin of matter sources torsion.
2. **Very high densities:** the spin corrections modify the dynamics and allow a bounce in the ECKS sector considered.
3. **Primordial transition:** the corpus applies phase-transition and Kibble-Zurek tools to a proposed chiral freeze-out.
4. **Cosmological background:** the expansion history is calculated with a modified Friedmann equation and parameters whose status is made explicit.
5. **Late sectors:** effective dark energy, geometric dark matter and structure formation are studied at different levels of maturity.
6. **Tests:** the data used for calibration are separated from those that genuinely test the framework, and then from future predictions.

What is established within the framework, and what is not yet

Two torsion regimes must be distinguished from the outset. In minimal ECKS, torsion is **algebraically tied to the spin of matter and does not propagate in vacuum**. Several late sectors of the corpus, by contrast, use a **propagating effective description** of the axial pseudo-vector S_μ (Proca form adopted in some calculations). This is not a "frozen torsion" in the sense of a demonstrated dynamical field: the passage from minimal ECKS to this complete effective action remains the transversal lock **PO-ECF-ACTION**, of which PO-F2-6 is a sub-problem. **This lock and its dependencies are gathered in §7.**

The **bounce** is calculated in the adopted ECKS sector. The H_0/S_8 sectors, the evolution of dark energy and several observational amplitudes involve **explicit calibrations**. The dynamical axial sector remains contingent on the derivation of a complete effective action; results that assume a particular dynamical form must therefore be read with this caveat. The treatment of perturbations is not yet carried by a complete Einstein-Cartan Boltzmann solver. Finally, several structure-formation chains, notably the Macro sector, remain **conditional or structural** rather than derived.

Where to go next

To understand the logic of the framework, read §1 to §3. For the detailed equations, parameters and statuses, go to §4. To find out what is calibrated, tested or genuinely predictive, read §5. The positioning is discussed in §6; the **major scientific locks** are gathered in §7, with cross-references from the relevant sections; the conclusion is in §8. The appendices gather mapping, nomenclature, provenance, versions and revision history.

> **Reading key.** Good agreement with data is not enough to turn an effective description into a fundamental derivation. Architecture maintains this distinction throughout the text.

1. The problem: cosmological tensions

The framework is of interest only if it answers two distinct questions: does it really modify cosmic history, and does this modification leave signatures that distinguish it from a mere readjustment of ΛCDM? The first step is therefore to state precisely the tensions it faces and which of its answers are calibrated.

This is often a cosmologist's first question: what does this framework resolve relative to ΛCDM, and at what price? Here is the balance sheet, with the corpus values.

The standard ΛCDM model faces persistent tensions between probes. The ECF addresses them through its spin-torsion sector with **two free parameters in the global test of Foundation I ($k = 2$ relative to ΛCDM)**: an effective amplitude F_{ion}, calibrated on H_0 (reduction of r_s), and γ_{spin}, calibrated separately for S_8; the joint role of F_{ion} in the growth suppression belongs to the historical chain and is not established under the A-7 closure *(corrected 19/10/2026)*.

Tension	ΛCDM (Planck)	ECF	Local observation
H_0 (Hubble constant)	≈ 67.4 (H_0 tension; Foundation I describes it at 5.0–$7.1\,\sigma$ depending on the combination — not to be confused with the former age tension at $5.7\,\sigma$, **HISTORICAL** ($\Omega_m = 0.315$ branch))	73.04 ± 1.04 (calibrated; conditional: θ_* is preserved only at fixed $\Omega_m = 0.315$, and falls by $\simeq 6.5\%$ under A-7 — `PO-F1-THETA`, open, needs the Einstein–Cartan perturbation solver)	SH0ES 73.04; H_0DN 73.50 ± 0.81
S_8 (structure amplitude)	≈ 0.83	**ECF prediction OPEN**; CAMB A-7 reference $S_8^{\text{ref}} \approx 0.780$	KiDS-1000: $0.766^{+0.020}_{-0.014}$ (observational target; symmetrised to ± 0.019 in the F1 χ^2) *(corrected 22/10/2026)*
Age t_0	13.8 Gyr (Planck reference)	13.33 **Gyr** ($w = -1$); 13.19 Gyr at the calibrated (w_0, w_a) point	above the age of the oldest globular clusters $t_{\text{GC}} = 13.32 \pm 0.10_{\text{stat}} \pm 0.5_{\text{sys}}$ Gyr, a lower bound on t_0; the age of the Universe inferred by the same authors, $t_U = 13.5^{+0.16}_{-0.14} \pm 0.5$ Gyr (Valcin et al. 2020), lies 0.17 Gyr above t_0, within the systematic; at the calibrated point, 13.19 Gyr falls below t_{GC} *(corrected 22/10/2026)*. The active background tension is carried by BAO/DESI; PO-F2-5 remains distinct from the age.
Static dark energy	$w = -1$ (cosmological constant, postulated)	$w = -1$ **derived** from the adopted effective action (static branch of the torsion condensate; F2, Appendix J)	Compatible with $w = -1$; the DESI preference for a dynamical w is addressed in the next row
Dynamical dark energy	Not applicable ($w = -1$)	$(w_0, w_a) = (-0.904, -0.153)$, calibration point **without error bar (CALIBRATED on the DESI preference used by the ECF; not derived and not an independent prediction; `PO-F1-B`)**	DESI prefers dynamical w (DR2: $3.1\,\sigma$ BAO+CMB; 2.8–$4.2\,\sigma$ with SNe, depending on the sample)

Tension	ΛCDM (Planck)	ECF	Local observation
BAO/DESI	Reference fit at $w = -1$	**OPEN** (`PO-F2-5`): lock of the BAO/DESI sector, distinct from the age; the active background tension lies here	BAO/DESI measurements; see §5 and the section on locks (§7)
Early galaxies (JWST, LRD)	Hierarchical formation: galaxies and *Little Red Dots* at high z more massive and more numerous than expected	**Qualitative, conditional**: compact seeds (macro-knots) qualitatively expected; ϵ_{coll} open; no parameter calibrated on these data	JWST surveys (JADES, CEERS, GLASS); see §5
Dark matter	Cold particle, not directly detected	**Structure of topological defects** (micro-knots), not a particle; mass calibrated within the Kibble-Zurek window, aggregation not derived (`PO-F3-C6`); Macro sector: `STRUCTURAL-HYPOTHESIS`	No direct detection; falsification condition set out in §4.4
Baryogenesis	Matter-antimatter asymmetry not explained by the Standard Model	$\eta_{\text{ECF}} \simeq 5.9 \times 10^{-10}$; observed $\simeq 6.1 \times 10^{-10}$: **semi-derived** (normalised on η_{obs}, agreement not independent); full kinetic chain **OPEN** (O8)	$\eta \simeq 6.1 \times 10^{-10}$ (inferred from $\Omega_b h^2$)
Initial singularity	Limit of validity of general relativity (Big Bang)	**Bounce at finite density**, generic in the adopted model; no added exotic field, but the spin–torsion term effectively violates the NEC (§4.1) *(corrected 22/10/2026)*; **shear (BKL) not resolved**; bridge with the calibrated effective branch **open** (`PO-F1-SIGMA0`)	No direct observation of the bounce; consequences for the expansion before recombination (§5)
Horizon, homogeneity and isotropy	Settled by inflation, an added hypothesis	**Open**: the resolution of the horizon problem is no longer established (withdrawal of 30/08/2026); the kinematic route is closed (bound $N_e \leq \frac{1}{3}\ln 4 \simeq 0.46$ e-fold, far below the inflationary requirement); isotropy **open** (`PO-F3-ISO`)	Uniformity of the cosmic microwave background; no direct test (§4.7)

Tension	ΛCDM (Planck)	ECF	Local observation
Primordial spectrum (n_s)	Tilt fitted, stemming from inflation	**Not predicted**: inherited input; the *stiff* era adds a blue component and bounds its duration ($N_{\text{stiff}} \gtrsim 68$, illustrative threshold, not calibrated); **OPEN** (PO-F3-NS)	Measurements of the primordial spectrum; not directly confronted (§4.7)

Tensions in the literature not addressed by the ECF. The table brings together the observational tensions (from H_0 to early galaxies) and problems of the standard model to which the framework provides an answer or which it leaves open (dark matter, baryogenesis, initial singularity, horizon and isotropy, primordial spectrum); these are open questions rather than disagreements between datasets. Other discordances discussed in the literature are not addressed: the sum of neutrino masses, which the corpus does not treat; the lithium-7 anomaly, whose interpretation through late decay of topological defects is a deferred avenue, **not an established result** (§7). This list is not exhaustive.

Former synthesis figure — HISTORICAL. The $\Delta\chi^2 = -39.5$ of the earlier joint chain is kept solely as a historical result. After A-7, the Planck/global block, θ_*, the global $\Delta\chi^2$ and the global AIC/BIC have not been recomputed; no post-audit global preference is therefore claimed.

These results are **calibrated** fits (the framework reproduces the observations), not parameter-free predictions. In the post-Audit v2 baseline, (w_0, w_a) is **CALIBRATED** and $\xi = 1.645$ is calibrated on the DESI preference used by the ECF: this is not an independent prediction prior to DESI. Historical formulations claiming priority over DESI are withdrawn from the active canon. **PO-F1-B** remains a provenance/modelling lock where relevant. The distinction is maintained throughout the document via the statuses.

For each tension, the cause invoked by the ECF, its status and the falsification criterion are gathered in the synthesis table of §2.

Physical signal or systematics?

Each tension taken in isolation *could* stem from an instrumental or modelling effect: a bias in the Cepheid calibration for H_0, or baryonic modelling and the shear pipeline for S_8. The ECF corpus does not claim to settle this debate and explicitly documents this possibility (Foundation I, §motivation).

One element motivated the study of a common mechanism: in the historical chain, F_{ion} **entered both the H_0 and S_8 sectors**, carried by physically different probes (S_8 also requiring γ_{spin}, calibrated separately). Under the A-7 closure, F_{ion} is calibrated on H_0 and this joint role is not established (F1 Short; corrected 19/10/2026). This is a consistency argument to be tested, not a proof that the tensions necessarily have a common physical origin.

> **Genealogy not to be confused — PIT and dark energy.** The **PIT is not the physical source of the dark-energy compensation** and derives neither its density nor its dynamics. Its proper role is the topological constraint (Ω_K≡0). The static dark term and the compensation mechanism are attributed to the **axial/torsion sector** in the effective construction of the Foundations; their microscopic and spectral identification remains conditional, in particular on PO-F2-6. The CPL projection ((w_0,w_a)) is a parametrisation/calibration of the late background, not a direct consequence of the PIT.

2. The proposal: borrowed building blocks, proper hypothesis and consequences

This section gives the physical chain before opening the technical detail: which building blocks are borrowed, which postulate is proper to the ECF, and where the computed, calibrated or still conditional consequences appear.

Where to read what — navigating from the diagram

Each box in the diagram above points to a specific place in this document, and to the paper carrying its full derivation.

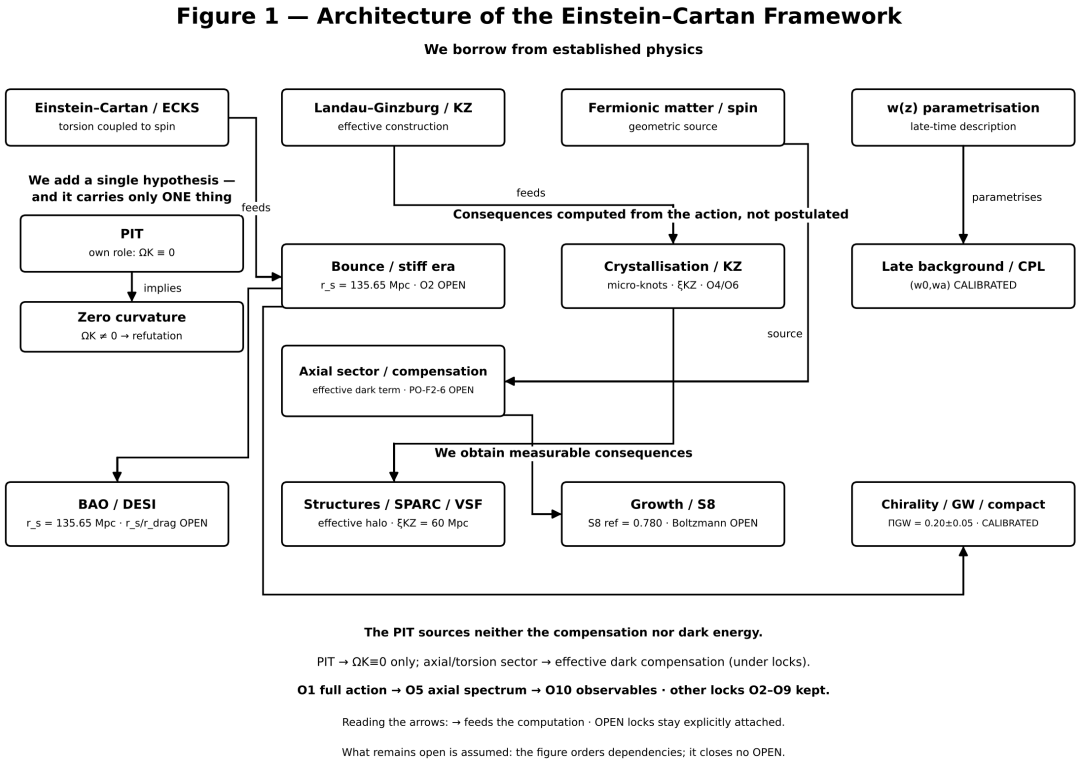


Figure 1: Figure 1 — Architecture of the Einstein–Cartan Framework

Box in the diagram	Status	Developed here	Paper
Einstein-Cartan torsion *(the two equations)*	borrowed	§2 *Borrowed foundation* · §3 *Modified Friedmann*	F1 §2
Spinning matter (Weyssenhoff fluid)	borrowed	§3	F1 § *Weyssenhoff Fluid*
Transitions & defects (Kibble-Zurek)	borrowed	§4.2	F2 §*Geometric DM*
w(z) parametrisation (CPL)	borrowed	§5	F1, F2 §DESI
Topological Invariance Principle	**postulate**	§2 *The only proper addition*	PIT Letter
Zero curvature ($\Omega_K \equiv 0$)	**falsifiable prediction**	§2	PIT Letter
A bounce, not a beginning	**INHERITED / derived in the adopted ECKS sector**	§3 *(Cartan density)*	Foundation I / ECKS; *Chiral Bounce* inherits it and builds the chiral transition conditionally on its existence
Crystallisation of the vacuum	**proposed**	§4.4	F2 §*Geometric DM*
Post-bounce acceleration	**CALCULATED-CONDITIONAL**: $N \simeq 0.153$ e-fold in the F1 Appendix H scheme only; existence in the ECF spin mechanism **OPEN** (`PO-ECF-ACTION-BOUNCE`); horizon **OPEN**	§4.7	F1 Extended Appendix H · F2 Extended Appendix C; F3 *Cosmological Physics* for the external bound
Dark matter = a structure	**proposed; scales calibrated**	§4.4 · §5 *(SPARC, voids)*	F2
Dark energy *(static branch $w = -1$ in the adopted effective action)*	**derived in this reduction**	§4.5 *(reduction of the action)*	**F2 Appendix J**
(its evolution w_0, w_a)	calibrated	§3 · §5	F1 · F2 §DESI
Two tensions, two parameters (H_0 — F_{ion}; S_8 — γ_{spin}; sharing of F_{ion} not established under A-7)	calibrated	§1 *(the problem)* · §5	F1 §5
The spectrum of irregularities (n_s)	**not predicted — treated as an inherited input**	§5 · Appendix B	F3 *Cosmological Physics*
The spin ↔ torsion-debt link	**open**	§2 *Synthesis table*	PIT Letter *(open question)*

Further, depending on what you are after:

You are looking for…	Go to
how a parameter is **obtained** (principle, inputs, script)	the **summary of calibrated quantities**, §2
which **datasets** are used, and for what	§5 *Confrontation with data*
what the framework **does not** do	end of the technical summary · §2 *Synthesis table*
the **definition** of a symbol	**Appendix B** *Nomenclature*
which **document** covers which topic	**Appendix A** *Map of the corpus*
the primary **references**	**Appendix C**

You are looking for…	Go to
the exact **version** of a corpus document	**Appendix D** *Traceability*

*Each building block carries its author and date: all prior (1922-2010). The Topological Invariance Principle carries no borrowing date — it is the ECF's only proper contribution; it is placed **apart**, because it carries a single thing, exact flatness, and does not produce dark energy. The derived mechanisms carry their fathers (Popławski, Kibble-Zurek); what is proper to the ECF is their articulation through the PIT.*

Summary of calibrated quantities

This table is the technical checkpoint for the calibrations. **Calibrated** means fitted or anchored on an explicitly stated datum/constraint; none of these quantities is promoted to an *ab initio* derivation by Architecture.

Quantity	Reference value	Role / anchoring	Status
F_{ion}	1.2765 ± 0.0246	effective background amplitude, H_0/S_8 chain	**CALIBRATED** (F1)
γ_{spin}	0.3116 ($\approx 0.31^{+0.07}_{-0.10}$ by inversion on S_8^{obs}; former ± 0.04 withdrawn)	former growth–torsion efficiency	**HISTORICAL CALIBRATION**; physical value OPEN. Post-A-7 conditional reference: $\gamma \simeq 0.066^{+0.069}_{-0.094}$, CALCULATED-CONDITIONAL / EXPLORATORY
$\Omega_{\text{spin}}(z_{\text{ref}} = 5600)$	0.093 ± 0.004	spin/radiation normalisation at the reference redshift; denoted $\Omega_{\text{spin}}^{\text{peak}}$ in F1	**CALIBRATED** (F1)
τ_{tor}	0.9975 ± 0.0015	stability/correction at the drag epoch in the r_s chain	**CALIBRATED** (F1)
ξ	1.645	factor of the w_a evolution	**CALIBRATED**, not derived (**PO-F1-B**)
ξ_{KZ}	$\simeq 60$ Mpc	comoving scale of the great voids	**CALIBRATED**, microscopic origin open (**PO-F3-XI**)
α_{KZ}	0.8–1.4 (1σ)	amplitude of the VSF excess, constrained by BOSS DR12	**CALIBRATED / PHENOMENOLOGICAL** (F2)

Beware of the count. This summary of seven calibrated quantities in the corpus is not the number of free parameters of any particular fit: Foundation I, for example, uses $k = 2$ additional parameters in its global test, while some of the quantities above are fixed upstream and not varied in that test.

Each row reads: **"the ECF borrows [block] from [author], for [role] — and derives [result]"**, with the ECF status of the result.

Borrowed foundation (established theories)

Borrowed block	Original author(s)	What the ECF makes it do	Status of the result
Einstein-Cartan theory (torsion coupled to spin)	É. Cartan (1922); Kibble (1961), Sciama (1964)	Provides the base geometry: spacetime can *twist*, not only curve. The spin–torsion interaction drives the bounce without any added exotic field; the effective term it produces nevertheless violates the NEC (§4.1)	**Foundation** (established physics)
Weyssenhoff fluid	Weyssenhoff & Raabe (1947)	Describes matter endowed with spin in general relativity — the source of the torsion term. ⚠ *Its cosmological use carries a published reservation: under the Frenkel, Papapetrou-Corinaldesi or Tulczyjew conditions the fluid is incompatible with the cosmological principle (Böhmer & Bronowski 2006). The corpus follows standard averaging without claiming to resolve this tension; tracked as* ***PO-F1-SSC****.*	**Foundation**
Kibble-Zurek mechanism	Kibble (1976), Zurek (1985)	Explains how a rapid phase transition *freezes* topological defects: here, the micro-knots and the ξ_{KZ} scale	**Foundation** (applied), scale **calibrated** (PO-F3-XI)
Topological defect classification (by homotopy)	Kibble (1976), Vilenkin & Shellard (1994)	Provides the *zoology* of defects applied to micro-knots, candidate strings and chiral walls. **Macro-knots are not a block borrowed from this classification: their formation constitutes a distinct sector,** **`STRUCTURAL-HYPOTHESIS`.**	**Foundation** (applied) — *string sector open; Macro sector distinct*
Landau-Ginzburg potential	Ginzburg & Landau (1950)	Models the vacuum phase transition at the bounce: the "splitting landscape" and the choice of a handedness	**Foundation** (applied)
CPL parametrization (w_0, w_a)	Chevallier-Polarski (2001), Linder (2003)	The standard language to describe evolving dark energy — the ECF projects its "frozen debt" onto it	**Foundation** (comparison tool)

Borrowed block	Original author(s)	What the ECF makes it do	Status of the result
Sakharov conditions	Sakharov (1967)	Framework of the matter-antimatter asymmetry, which the ECF's chiral baryogenesis seeks to satisfy	**Foundation**
Raychaudhuri equation	Raychaudhuri (1955)	Derived from the single-density form (3), the acceleration equation (21) gives $\ddot{a} > 0$ at the bounce and changes sign at $\rho_c(1+3w)/(4+6w)$; exact in Einstein-Cartan for $p = 0$, schematic for $w \neq 0$; its link with the ECF spin mechanism is not established (`PO-ECF-ACTION-BOUNCE`)	**Foundation**; ECF application **open**

⚠ **Precision added 29/08 — the vacuum manifold itself is an open problem (`PO-F2-8`).** The homotopy classification is an **applied foundation**, correctly cited; but it applies only to a given manifold, and that manifold is **not derived** from the electroweak PGT extension. · ⛔ The $M \simeq S^2$ hypothesis found in earlier versions has $\pi_0 = \pi_1 = 0$: it admits **neither walls nor strings**, hence two of the three sectors. · ⋆ **Candidate retained on 29/08**: $M \simeq Z_2 \times S^1 \times S^2$ — broken parity of the axial condensate, chiral phase, spin-alignment direction. **Constraint shaping it**: S^2 must enter as a *factor*, never as a quotient, or the winding number of the knots is no longer defined. · ⋆⋆ **The independence of the three orders is not an ECF postulate**: it is observed in condensed matter — 1T-TaS_2 *(arXiv:2601.03379)*, chiral ^{3}He-A under Kibble-Zurek quenching, spinor condensates. · ⚠ **Status**: candidate motivated and supported by analogy, **not derived**. *The problem is narrower than before; it is not closed.*

The only proper addition

Element	Nature	Role
Topological Invariance Principle (PIT)	**Postulate** — the only hypothesis added by the ECF	The PIT **imposes** the global spatial-flatness branch $\Omega_K \equiv 0$. ⚠ *It does not produce dark energy: in the adopted Friedmann equation, the closure relation that follows from it is an identity, used to **extract** the residual from the budget, not to generate it (recentring of 11/08/2026). Static dark energy is derived separately from the action (F2, Appendix J) — independently of this postulate.*

Derived mechanisms (neither borrowed nor postulated: *computed*)

Mechanism	Derived from	ECF status
Non-singular bounce at the Cartan density	Einstein-Cartan alone *(the PIT plays no role)*	**Exact for dust** (form (3)); in the stiff phase the spin term alone does not produce a bounce: **OPEN** (`PO-ECF-ACTION-BOUNCE`, §3)
Vacuum crystallisation (chiral gel)	Chiral gel at the bounce (Landau-Ginzburg); defects frozen at the electroweak freeze-out (Kibble-Zurek)	**CALCULATED-CONDITIONAL** in *Chiral Bounce*: interpolating/Kibble-Zurek solution derived in the effective construction, conditional on the existence of the ECKS bounce; global $O(1)$ prefactor and $k \neq 0$ modes open
Horizon — former scenario withdrawn ($N \approx 60$, **HISTORICAL/WITHDRAWN**; horizon **OPEN**; 0.46 e-fold = **external** upper bound, not an ECF bound; 0.153 e-fold = **conditional** calculation of the F1 scheme)	Modified Raychaudhuri + torsion	⚠ **Withdrawn as established on 30/08** — rested on the assumption that chiral walls are bordered by string loops (Corollary F3-1), forming a connected network; the walls are in fact independent closed bubbles (I69), without this global connection. **Open problem**, dependent on the demonstration of a phase winding for the 1D sector (PO-F2-8, partial progress on 30/08) *and*, separately, on proof that this sector would actually reconnect the bubbles to one another — see the summary register of locks in §7
Parity breaking at the bounce	Chiral crystallisation	**CALCULATED-CONDITIONAL** (inherits the status of chiral crystallisation and of the existence of the bounce)

Observed anomalies, physical causes invoked, status and predictions

Synthesis table (16/10/2026). It replaces the former tables "Results and sectors" and "Point of tension". Each row reflects the state of the papers; instruments and timescales are detailed in §5, locks in §7. Statuses: ***DERIVED*** *(computed from the action),* ***CALIBRATED*** *(value fitted to a datum),* ***CONDITIONAL*** *(computed under an undemonstrated hypothesis),* ***POSTULATE****,* ***OPEN****,* ***WITHDRAWN****.*

Anomaly or question	Physical cause invoked by the ECF	Status	Prediction or falsification criterion	Open problems
Initial singularity	Repulsive spin-spin interaction: non-singular bounce at the Cartan density $\rho_c = c^5/\hbar G^2 \approx 5.15 \times 10^{96}$ kg m^{-3}	**Exact for dust**; in the stiff phase the spin term alone does not produce a bounce: **OPEN**	Blue gravitational-wave background, cutoff near 10^{11}–10^{12} Hz (amplitude not computed); chiral asymmetry $\Delta\chi_{\mathrm{GW}} > 0$ (falsified if < 0.02 at 95 %)	`PO-ECF-ACTION-BOUNCE`

Anomaly or question	Physical cause invoked by the ECF	Status	Prediction or falsification criterion	Open problems
Hubble tension	Stiff phase after the bounce (spin scaling as a^{-6}, 9.3 % of the radiation density at $z_{\rm ref} = 5600$, 0.36 % at recombination): larger $H(z)$ before recombination, r_s reduced from 147.1 to 135.65 Mpc (−7.77 %), hence $H_0 = 73.04$	**CALIBRATED** ($\Omega_{\rm spin}^{\rm peak} = 0.093$, $\tau_{\rm tor}$); **CONDITIONAL**: θ_* is preserved only at fixed $\Omega_m = 0.315$ and falls by ≈ 6.5 % under the closure A-7; **the a^{-6} law must hold up to $z \sim 10^5$ and is excluded by BBN beyond (`PO-ECF-BBN-SPIN`)**; resolving this needs an Einstein–Cartan perturbation solver, not expected in the near term	21 cm acoustic scale $r_s = 135.65$ Mpc (falsified if $r_s > 145$ Mpc, HERA/SKA, after the $r_s \to r_d$ mapping, PO-ECF-BAO-RDRAG); damping-tail deficit $\Delta D_\ell / D_\ell \simeq 2.5$–$6.5\,\%$ at $\ell = 2000$–3000 (illustrative forecast)	`PO-F1-THETA`, `PO-F2-5`, `PO-ECF-BAO-RDRAG`
S_8 **tension**	Closure $\Omega_m = \omega_m / h^2$: a higher H_0 lowers Ω_m (0.267); CAMB ΛCDM reference on that background $S_8 \simeq 0.780$ (0.831 on the Planck background); phenomenological scale-dependent suppression $G_{\rm eff}(k)$	$S_8 = 0.766$ is a **target**, not a prediction; growth mechanism **OPEN**; $\gamma_{\rm spin} = 0.3116$ historical	Suppression of $P(k)$: 7.5 % at $k = 0.1\,h$ Mpc^{-1}, 15 % saturated (parametrisation, Euclid)	perturbative growth (EC solver)
Age of the Universe / BAO	Same background ($w = -1$, $\Omega_m = 0.267$)	$t_0 = 13.33$ Gyr, above the bound $t_{\rm GC} = 13.32$ Gyr and 0.17 Gyr below $t_U = 13.5^{+0.16}_{-0.14} \pm 0.5$ Gyr (Valcin et al. 2020); the tension moves to the BAO ($\Delta\chi^2_{\rm BAO} \simeq +10.3$) and DESI	A stellar age clearly above 13.33 Gyr falsifies the $w = -1$ background	`PO-F2-5` (quadrilemma)

Anomaly or question	Physical cause invoked by the ECF	Status	Prediction or falsification criterion	Open problems
Dark energy; DESI preference for $w \neq -1$	Axial torsion condensate frozen at the minimum of its potential	Static branch $w = -1$ **DERIVED** in the FLRW reduction (F2, Appendix J); evolution **CALIBRATED**: $(w_0, w_a) = (-0.904, -0.153)$ is an adopted point produced by no script (-0.907 ± 0.004 computed), $\xi = 1.645$ calibrated on DESI	Crossing of $w = -1$ at $z \simeq 1.68$ (1.55 with –0.907); point at 2.03σ from the DESI DR1 posterior (Planck prior)	`PO-F1-W0`, `PO-F1-B`
Cosmic birefringence	Photon–axial-torsion coupling (Chern–Simons type)	**Non-zero value predicted** ($\beta \neq 0$); sign not derived; amplitude 0.35° **CALIBRATED** on Minami–Komatsu	Falsified if β is consistent with 0 below 0.1° (LiteBIRD, $\sigma(\beta) \sim 0.02°$)	—
Dark matter	Topological Micro-Knots frozen at the electroweak freeze-out, $M_{\mathrm{MK}} \approx 3.74 \times 10^{24}$ kg; aggregates	**Proposed / CONDITIONAL**; producer of the chiral walls not identified	Excess of microlensing events with $t_E \sim 1$–10 h (Roman); OGLE-IV already limits these aggregates to $f_{\mathrm{DM}} \lesssim 1$–$2\,\%$	`PO-F2-10`, `PO-F2-8`

Anomaly or question	Physical cause invoked by the ECF	Status	Prediction or falsification criterion	Open problems
Early black holes and galaxies (JWST, LRD, UHZ1)	**Macro-Knot** seeds (causal scale $\sim 10^5\,M_\odot$ at the Great Annihilation, $T \sim 1$ MeV); the linear growth boost is **WITHDRAWN** as a mechanism	**CONDITIONAL**: seed formation is a structural hypothesis ($\epsilon_{\rm coll}$ and abundance not derived); growth computed; CMB (Planck 2018, Foundation II): if the seeds accrete at the Eddington rate from recombination, $n \lesssim 2.3 \times 10^{-4}$–$1.3 \times 10^{-3}$ Mpc^{-3}, below the adopted abundance ($n \approx 2 \times 10^{-3}$–$1.6 \times 10^{-2}$ Mpc^{-3}), a working assumption (factor $\approx$ 1.5–70); no constraint if early accretion is less efficient (disk and spherical recipes)	At the Eddington rate from $z = 20$: UHZ1 ($z = 10.1$) and J1342+0928 ($z = 7.54$) reached at $f_{\rm Edd} \simeq 1.1$–1.3 (ECF seed) against 1.9–2.8 (Pop III seed); falsified by the absence of compact objects $\gtrsim 10^5\,M_\odot$ at $z \gtrsim 10$	`PO-F1-BOOST`, `PO-F1-JWST-SRC`, `PO-F1-SIGMA-M`, `PO-F2-SEED-CMB`
Matter–antimatter asymmetry	Chiral baryogenesis (chiral crystallisation + Sakharov conditions)	**Semi-derived**: $\eta_{\rm ECF} \simeq 5.9 \times 10^{-10}$ against 6.1×10^{-10} observed ($\approx$ 3.5 % gap; agreement not independent, normalised on $\eta_{\rm obs}$)	Falsification criterion not stated in the corpus	—
Spatial flatness	Topological Invariance Principle (PIT): $\Omega_K \equiv 0$ exactly	**POSTULATE** (the ECF's only proper addition)	Any robust measurement of $\Omega_K \neq 0$ falsifies it (inflation is unaffected)	$\Omega_{\rm spin}$–Ω_τ link
Black-hole singularities	Cartan core: the bounce applied to collapse	**Proposed** (F3 *Strong Gravity*, out of scope)	see F3	—
Primordial spectrum (n_s)	Not predicted: inherited; bound $N_{\rm stiff} \gtrsim 68$	**OPEN**	—	`PO-F3-NS`

Control without anomaly: recombination on the full ECF background (`script_09_cosmic_ages.py`, branch `with_spin`, spin term included) occurs at 368.4 kyr against 371.3 kyr in ΛCDM (at $z_* = 1089.9$). It is a calculation conditional on the calibrated spin term; $t_{\rm rec}$ is not an observable and no signature is drawn from it.

3. The modified Friedmann equation, complete form

Once the building blocks are identified, the whole background cosmology is played out in a localised modification of the Friedmann equation. The physical role of the spin term is presented first; the parameters and their statuses come next.

Gathered into a single expression, the ECF Friedmann equation reads, in cosmological form (normalised to the present value H_0). Here, each $\Omega_{i,0}$ denotes the reduced contribution evaluated today ($a = 1$), then propagated with its law in a:

$$\boxed{\frac{H^2(a)}{H_0^2} \;=\; \Omega_{r,0}\, a^{-4} \;+\; \Omega_{m,0}\, a^{-3} \;+\; \Omega_{\Lambda} \;+\; \Omega_{\text{spin},0}\, a^{-6}} \tag{1}$$

Note on the sign of the spin term. The **positive** sign of the spin term above is the one **actually implemented** in `Scripts_F1/script_01_solve_sound_horizon.py`, which produced every published value ($r_s = 135.65$ Mpc, $H_0 = 73.04$, S_8, $\Delta\chi^2 = -39.5$; S_8 and $\Delta\chi^2$ belong to the historical chain). It is consistent with the definition $\Omega_{\text{spin},0} = 2\pi G\kappa\sigma_0^2/3H_0^2$ in Foundation I Appendix A (manifestly positive, being built on σ_0^2).

This term is not the same object as the repulsive ECKS correction $-\rho_{\text{spin}}$ of §4 below (fundamental Friedmann equation), which describes the ultradense regime at the bounce. The former is a calibrated late-time effective stiff fluid; the latter is the spin-spin contact correction, repulsive by the field equations. **The microphysical bridge between them — how a repulsive non-polarised correction becomes an effective fluid of opposite sign after coherent torsion domains crystallise — is not established**: open problem PO-F1-SIGMA0, entry I81 in the dead-ends register.

with the closure relation, which is an **identity** obtained by dividing the complete Friedmann equation by H^2, with $\rho_c = 3H^2/(8\pi G)$: $\sum_i \Omega_i + \Omega_K = 1$. The PIT imposes $\Omega_K \equiv 0$; the relation therefore becomes $\Omega_{r,0} + \Omega_{m,0} + \Omega_\Lambda + \Omega_{\text{spin},0} = 1$. This closure is not an additional independent dynamical equation.

Numerically, today ($a = 1$), the active post-A-7 budget closes with $h_{\text{ECF}} = 0.7304$ and $\Omega_{m,0} = 0.2669$; the former Planck convention at $h = 0.6736$ is kept solely as a **HISTORICAL** reference point:

Component	Value	Origin
$\Omega_{r,0}$ (radiation)	7.8417×10^{-5}	**active post-Audit v2 value at h_{ECF}**; former convention $\simeq 9.22 \times 10^{-5}$ at $h = 0.6736$, corresponding to the **same ω_r**: **HISTORICAL / Planck reference**; S-37 closed
$\Omega_{m,0}$ (matter)	0.2669	active budget A-7; 0.315 remains the Planck/historical reference depending on context
Ω_Λ (residual torsion)	0.73302	fixed by closure
$+\,\Omega_{\text{spin},0}$ (late-time effective stiff fluid)	$+2.73 \times 10^{-13}$	**F1 value** (after the S-37 correction of $\Omega_{r,0}$)
Sum	**1.0000**	closure identity under the branch $\Omega_K \equiv 0$ imposed by the PIT

Density convention. $\omega_m = \omega_b + \omega_c = 0.14237$ is the Planck sum of baryons + cold dark matter; the massive-neutrino share ($\sum m_\nu = 0.06$ eV, i.e. $\omega_\nu \approx 6.4 \times 10^{-4}$) is not counted in it, neutrinos being treated as massless radiation ($N_{\text{eff}} = 3.046$). Planck publishes $\Omega_m h^2 = 0.1430$ including them; the $0.4\,\%$ difference would give $\Omega_m = 0.2681$ instead of 0.2669 (t_0 shorter by 0.016 Gyr, diagnostic). This is a convention, not a free parameter; its full treatment requires the Einstein–Cartan perturbation solver, which does not exist yet.

The sum closes by construction, since Ω_Λ is fixed by the closure identity under the branch $\Omega_K \equiv 0$. What is checkable

lies elsewhere: the resulting value in the F1 chain, $\Omega_\Lambda \approx 0.733$, corresponds to about $73.3\,\%$ of the budget in this A7 **CALCULATED-CONDITIONAL** closure; this value is an arithmetic consequence of the adopted budget, not an independent measurement. And the bounce term, $\Omega_{\text{spin},0} \simeq 2.73 \times 10^{-13}$ in F1 (after the S-37 correction), quantitatively confirms that it is without effect at the present epoch, after a^{-6} dilution from the normalisation where the ratio $\rho_{\text{spin}}/\rho_{\text{rad}}$ is about 0.093 at $z_{\text{ref}} = 5600$; this number is not a "peak" in the sense of a dynamical maximum.

Development of the values (reproducible):

- $\Omega_{r,0}$ — **S-37 SETTLED.** The corrected physical density is $\omega_\nu = 1.7106 \times 10^{-5}$. In the active post-A-7 baseline, with $h_{\text{ECF}} = 0.7304$, it leads to $\Omega_{r,0} = 7.8417 \times 10^{-5}$. The value $\simeq 9.22 \times 10^{-5}$ at $h = 0.6736$ is solely the **HISTORICAL / Planck reference** conversion of the same ω_r. A-6 / DEC-07D and A-7 are settled; no upstream follow-up is requested here.
- $\Omega_{\text{spin},0} \simeq 2.73 \times 10^{-13}$ follows from the pair $(\Omega_{\text{spin}}^{\text{peak}}, z_{\text{ref}}) = (0.093,\ 5600)$ **(the reference peak is therefore associated with $z_{\text{ref}} = 5600$ in the current numerical canon)** through $\Omega_{\text{spin},0} = \Omega_{\text{spin}}^{\text{peak}}\,\Omega_{r,0}/(1+z_{\text{ref}})^2$. ⚠ **This pair is not two parameters but one, written as a pair**: only the product enters the solver, so quoting 0.1668 at $z_{\text{ref}} = 7500$ would describe exactly the same background. The choice of z_{ref} is a writing convention, not an extra degree of freedom — the $k = 2$ count of the ECF sector is therefore unaffected. The $\rho_{\text{spin}} \propto a^{-6}$ dilution then suppresses the contribution by dozens of orders of magnitude, leaving $\Omega_{\text{spin},0} \simeq 2.73 \times 10^{-13}$ today (the F1 value, after the S-37 correction). This is what makes the bounce invisible at late times.
- Ω_Λ — **CALCULATED-CONDITIONAL.** In the active post-A-7 budget: $\Omega_\Lambda^{\text{closure}} = 1 - 7.8417 \times 10^{-5} - 0.2669 - 2.73 \times 10^{-13} = 0.73302$. This is an arithmetic closure, not an independent parameter of the baseline. The minus sign comes from isolating Ω_Λ in the closure identity; it does not change the positive sign of the $+\Omega_{\text{spin},0} a^{-6}$ term in the effective Friedmann equation.

$\Omega_{m,0} = 0.2669$ is the active post-A-7 value, obtained by keeping the physical densities consistent with $h = 0.7304$. The former 0.315 is kept as the Planck/historical reference and must no longer close the active ECF budget.

Definition of the terms:

Term	Meaning	Value / status	Source
$H(a)$	expansion rate at scale factor a	—	—
H_0	Hubble constant today	73.04 (km/s)/Mpc — **calibrated ECF value**, the one actually used in $H(a)$ (F1 §*Theoretical Framework*, DOI 10.5281/zenodo.19577447); compatible with the external H_0DN measurement 73.50 ± 0.81 at $< 0.6\sigma$	ECF corpus — F1
$\Omega_{r,0}\, a^{-4}$	radiation	standard	standard cosmology
$\Omega_{m,0}\, a^{-3}$	total matter (baryons + effective dark sector of the model)	$\Omega_{m,0} = 0.2669$	A-7 — active budget; 0.315 historical/Planck reference

Term	Meaning	Value / status	Source
Ω_Λ	dark energy (frozen torsion "debt")	$(w_0, w_a) = (-0.904,\ -0.153)$ (calibration point, without error bar; likelihood profile: -0.907 ± 0.004, -0.153 ± 0.007)	**CALIBRATED** — w_0 empirically linked to Ω_{spin} (not derived), w_a phenomenological; $\xi = 1.645$ is calibrated on the DESI preference used by the ECF and is not an independent prediction prior to DESI (PO-F1-B) (F1 §*Observational context*). *Uncertainty: likelihood profiling on* $\Omega_{\text{spin,peak}}$ *(*$\Delta\chi^2 \leq 1$ *against* $r_s = 135.65$ *Mpc),* ξ *held fixed — floor; borne on the canonical calibrated value, not on the* $-1 + \Omega_{\text{spin}}$ *approximation which differs from it by 0.3,%.*
$+\,\Omega_{\text{spin},0}\,a^{-6}$	**torsion term** — the Einstein-Cartan contribution	$\Omega_{\text{spin}}^{\text{peak}} = 0.093$	**calibrated** on r_s — F1 §*Weyssenhoff Fluid ⚠ not derived*: adjusted to reproduce the sound-horizon reduction

The spin parameter is itself defined by:

$$\Omega_{\text{spin},0} \;=\; \frac{2\pi G\,\kappa\,\sigma_0^2}{3H_0^2} \tag{2}$$

All parameters of this expression, with their meaning, value and its source:

Symbol	Meaning	Value	Source of the value
$\Omega_{\text{spin},0}$	reduced spin density today — 2.73×10^{-13}	**derived** from the pair $(0.093,\ z_{\text{ref}} = 5600)$, which counts as **one** parameter	**calibrated** on r_s — F1 §*Weyssenhoff Fluid and Spin Density ⚠ not derived*
G	Newton's gravitational constant	6.674×10^{-11} m^3 kg^{-1} s^{-2}	CODATA (physical constant)
κ	torsion-spin coupling constant	$\kappa^2 = 2.0766 \times 10^{-43}$ m · J^{-1}	**ECF corpus** — F2 §*Topological Zoology* (published, Zenodo 10.5281/zenodo.20629237
σ_0	spin density of matter	$\sigma^2 = \frac{1}{2}\langle S_{\mu\nu} S^{\mu\nu} \rangle$ — **numerical value not evaluated** (PO-F1-SIGMA0)	**ECF corpus** — F1 §*Theoretical Framework*, DOI 10.5281/zenodo.19577447

Symbol	Meaning	Value	Source of the value
H_0	Hubble constant today	73.04 (km/s)/Mpc — **calibrated ECF value**; compatible with H_0DN 73.50 ± 0.81 at $< 0.6\sigma$	ECF corpus — F1

Note — spin density. The spin density σ_0 is not a free parameter: it derives from the spin tensor $S_{\mu\nu}$ of fermionic matter via $\sigma^2 = \frac{1}{2}\langle S_{\mu\nu}S^{\mu\nu}\rangle$, and does not propagate as an independent field (it is a density algebraically tied to matter, a property of Einstein-Cartan-Kibble-Sciama theory). **Its numerical value is not evaluated from first principles** (F1, explicitly stated): $\Omega_{\text{spin,peak}} = 0.093$ is calibrated first (against r_s), and σ_0 should be deduced from it — that deduction has not been carried out. Microphysical derivation deferred to Foundation II/III (PO-F1-SIGMA0).

Fundamental constants used (standard SI values, source CODATA):

Constant	Value	Source
G (gravitation)	6.674×10^{-11} m^3 kg^{-1} s^{-2}	CODATA
c (speed of light)	2.998×10^8 m s^{-1}	CODATA (exact by definition)
$\hbar$ (reduced Planck)	1.055×10^{-34} J s	CODATA

In the canonical normalisation adopted by the ECF, these three constants fix the Cartan density $\rho_c = c^5/(\hbar G^2)$ with no adjustable parameter in this convention. **The exact coefficient is an ECF canonical normalisation; it is not presented as a universal coefficient of every ECKS realisation.**

The two decisive properties of the torsion term:

1. **Negative sign** — a direct consequence of the Einstein-Cartan-Kibble-Sciama equations: the contact spin-spin interaction between fermions is *repulsive*, and enters the effective stress-energy tensor of the Weyssenhoff fluid with the sign opposite to radiation, matter and Λ.
2. a^{-6} **scaling** — the term decays far faster than all others. At low redshift it is completely negligible (standard Friedmann behaviour is recovered to very high precision); it dominates only at extreme densities, near the bounce.

Strong form (bounce regime). At high densities, this repulsive term is often written in a compact, single-density form:

$$H^2 \;=\; \frac{8\pi G}{3}\,\rho\left(1 - \frac{\rho}{\rho_c}\right) \tag{3}$$

This form is not a mere rewriting of the two-term equation (8). In Einstein-Cartan theory, the Friedmann equation with spin reads $H^2 = (8\pi G/3)(\rho - \alpha n^2)$, where n is the fermion number density (Popławski 2012, eq. 11); it takes **exactly** the form (3) for non-relativistic fermionic matter ($\rho = mn$, $p = 0$). This statement concerns the Friedmann equation itself; the dynamics additionally requires a conservation law. The single-density dynamics used here (usual continuity, hence $n \propto a^{-3}$ for $p = 0$) corresponds to a spin term that behaves as stiff matter, $\tilde{\epsilon} = \tilde{p}$, as in the spin-fluid description of Hehl, von der Heyde and Kerlick cited by Popławski (primary source to be confirmed). In Popławski's Dirac description, the spin term satisfies $\tilde{\epsilon} = -\tilde{p} = -\alpha n^2$ (his eqs. (10), (12)–(13)); na^3 is then not conserved for dust, and the dynamics differs. For the other equations of state, including the stiff phase, the spin term varies as $\rho^{2/(1+w)}$ (linearly in ρ for $w = 1$) and form (3) is a **schema** (`PO-ECF-ACTION-BOUNCE`, OPEN). It is also the form of the effective Friedmann equation of loop quantum cosmology (Agullo & Singh 2017), of different physical origin *(clarification of 04/10/2026)*. The reference density is the **Cartan density**:

$$\rho_c = \frac{c^5}{\hbar G^2} \approx 5.15 \times 10^{96}\ \text{kg} \cdot \text{m}^{-3} \tag{4}$$

(of the order of the Planck density, built on the fundamental constants alone; established in F2 §*Fundamental Scales and Constants of the ECF Bounce*, also present in F1). When $\rho \to \rho_c$, the right-hand side vanishes: $H = 0$, the

expansion stops and reverses — this is the bounce; the bound $\rho \leq \rho_c$ is exact in equation (3). The Planck value of ρ_c is an adopted reference scale; its link to the Einstein-Cartan value m^2/α is not established.

Friedmann equation by regime *(added 05/10/2026)*. A *regime* is set by the ratio $w = p/\rho$ between the pressure and the energy density of matter: $w = 0$ for dust (slow matter, negligible pressure), $w = 1/3$ for radiation (photons, ultra-relativistic particles), $w = 1$ for the stiff fluid (sound travels there at the speed of light). As the Universe grows, the density falls as $\rho \propto a^{-3(1+w)}$: a^{-3}, a^{-4}, a^{-6}. If the fermion number is conserved ($n \propto a^{-3}$, spin-fluid description), the spin term αn^2 always falls as a^{-6}. The bounce is therefore a race: going back in time, the spin term catches up with matter if the latter grows more slowly than it. One obtains, for $w < 1$:

$$H^2 = \frac{8\pi G}{3}\,\rho\left[1 - \left(\frac{\rho}{\rho_c}\right)^{\frac{1-w}{1+w}}\right], \qquad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left[(1+3w)\,\rho - 4\rho\left(\frac{\rho}{\rho_c}\right)^{\frac{1-w}{1+w}}\right]. \tag{3a}$$

Regime	Spin	H^2	Bounce	$\ddot{a}/a$ at bounce	Sign of $\ddot{a}$ changes at
Dust ($w = 0$), form (3) exact	$\propto \rho^2$	$\frac{8\pi G}{3}\rho\,(1 - \rho/\rho_c)$	yes, $\rho = \rho_c$	$4\pi G\rho_c$	$\rho_c/4$
Radiation ($w = 1/3$)	$\propto \rho^{3/2}$	$\frac{8\pi G}{3}\rho\,[1 - (\rho/\rho_c)^{1/2}]$	yes, $\rho = \rho_c$	$\frac{8\pi G}{3}\rho_c$	$\rho_c/4$
Stiff ($w = 1$)	$\propto \rho$	$\frac{8\pi G}{3}(1 - \epsilon_s)\rho$ (ϵ_s constant)	**no**	—	—

In the stiff phase, matter and the spin term fall at the same rate (a^{-6}): their ratio stays frozen and **the spin term alone does not produce a bounce**. The stiff-phase bounce, and the values $0.4\,\rho_c$ and 0.153 e-fold drawn from form (3), are therefore schematic (`PO-ECF-ACTION-BOUNCE`, OPEN). For radiation, the extended form (3) would give $\frac{16\pi G}{3}\rho_c$ and $\rho_c/3$ instead of the values in the table. In radiation, ρ_c depends on the ratio $n/\rho^{3/4}$, hence on the thermal composition (not computed here). These relations hold in the spin-fluid description (primary source to be confirmed), not in Popławski's Dirac description; symbolic check: `check_regimes.py` (to be added to the repository).

A static core requires a twice-as-tight bound. For a *static* compact object (the core of a regular black hole, not the cosmological bounce itself), the same effective density $\rho_{\text{eff}} = \rho(1 - \rho/\rho_c)$ peaks at $\rho = \rho_c/2$ and then decreases: beyond that point $d\rho_{\text{eff}}/d\rho < 0$, adding matter *reduces* the supporting pressure, and no hydrostatic (TOV) equilibrium exists any more. **A bound, static core is therefore constrained to $\rho_{\text{core}} < \rho_c/2$** — a bound **conditional** on the adopted stiff equation of state and on the single-density form $\rho(1 - \rho/\rho_c)$, exact for dust only (see "Friedmann equation by regime" above; for the stiff equation of state, the Einstein-Cartan spin term is linear in ρ; `PO-ECF-ACTION-BOUNCE`), not an order-1 factor left undetermined — F3 *Strong Gravity*, `calculate_core_tov_torsion.py`. The numerical prefactor is only meaningful at the scale where the classical description still holds; only the order of magnitude $R_{\text{core}} \sim l_{\text{Pl}}$ is physical.

Complete derivation: Foundation I (cosmological sector, published) and Foundation III (strong regime, forthcoming).

> **Two possible paths from here.**
>
> **Main path — level 2**: continue directly to §5 *Confrontation with data*, which sets out the calibrations, the tests and confrontations, and the falsifiable channels. **Technical deep-dive — level 3**: continue to §4 *Complete derivations*, which gives the derivations sector by sector.

4. Complete derivations (self-contained)

The preceding sections gave the logic of the framework. This section now opens the black box: each sector is taken up again with its equations, its parameters, its epistemic status and its limits.

4.1 The bounce

This section reproduces the spin-sector derivation as it appears in Foundation I, §Weyssenhoff Fluid and Spin Density. The connecting text is translated; the equations are identical to those of the published corpus. The reader thus has the complete chain without needing to open another document.

(1) Starting point: the spin energy density. In the macroscopic picture, the spin contribution is modelled as the averaged effect of the intrinsic angular momentum of fermions. The energy density associated with spin evolves as the square of the fermionic number density:

$$\rho_{\text{spin}} \propto n^2. \tag{5}$$

This law is that of the macroscopic Weyssenhoff treatment used in Foundation I (§ *Weyssenhoff Fluid and Spin Density*, corresponding equations): the quadratic spin contribution is tied to the square of the fermionic density. For the theoretical foundation, see Weyssenhoff & Raabe (1947) and Hehl *et al.* (1976), referenced in Appendix C.

(2) Evolution with the scale factor. Since the particle number density evolves as $n \propto a^{-3}$, the spin contribution follows the "stiffer" evolution law:

$$\rho_{\text{spin}} \propto a^{-6}. \tag{6}$$

This scaling law is a central ingredient of the ECF background dynamics: the spin sector becomes negligible at late times, but can become dynamically important in the ultradense primordial regime. In effective-fluid language, this behaviour corresponds to a "stiff" component (*stiff*), characterised by an equation-of-state parameter:

$$w_{\text{spin}} = 1. \tag{7}$$

Unlike radiation ($\propto a^{-4}$) or matter ($\propto a^{-3}$), the spin density therefore dilutes extremely fast after the primordial universe. It is precisely this property that allows it to modify the high-density regime, regularise the singular limit, then naturally decouple from standard cosmology at late times. In ratio form, $\rho_{\text{spin}}/\rho_{\text{rad}} \propto (1+z)^2$, normalised to 9.3% at the calibration redshift $z_{\text{ref}} = 5600$; the ratio decays to 0.36% at $z_{\text{rec}} = 1100$, a factor of ≈ 26 below the reference point.

(3) Sign convention and repulsive correction. With the adopted metric and sign conventions, the spin-induced quadratic correction enters with a negative sign. The modified ECF Friedmann equation therefore reads, in dimensional form:

$$H^2 = \frac{8\pi G}{3}\Big(\rho_r + \rho_m + \rho_\Lambda - \rho_{\text{spin}}\Big), \tag{8}$$

where the negative sign of ρ_{spin} is a direct consequence of the ECKS field equations (see also the **note on the sign of the spin term** in §3): the contact spin-spin interaction between fermions is *repulsive*, and enters the effective stress-energy tensor of the Weyssenhoff fluid with the sign opposite to radiation, matter and Λ. At sufficiently high density, this term acts as a repulsive contribution; in the low-density limit, the correction becomes negligible and standard Friedmann behaviour is recovered to high precision.

(4) Bounce condition. As the density approaches ρ_{crit}, the negative quadratic term can compensate the attractive contribution of ordinary matter. Since $H^2 \geq 0$ by definition, the right-hand side cannot become negative. There therefore exists a minimum scale factor a_{min} for which:

$$\rho_r + \rho_m + \rho_\Lambda \;=\; \rho_{\text{spin}}(a_{\text{min}}) \quad \Longrightarrow \quad H = 0. \tag{9}$$

This is the **bounce condition**: $H = 0$ at finite density, with $\dot{H} > 0$ immediately after. The collapse is halted and reversed by the geometric spin pressure.

(5) The bounce is generic, not fine-tuned. This is not a fine cancellation: because $\rho_{\text{spin}} \propto a^{-6}$ always ends up overtaking $\rho_r \propto a^{-4}$ as $a \to 0$, the bounce condition *always* has a solution at a finite a_{min}, independently of the initial values of ρ_r or ρ_m. The bounce is therefore a generic consequence of the ECKS coupling, not a special initial condition. The argument for robustness against spin inhomogeneities rests on the locality of the interaction — ⚠ *Popławski's analyses are carried out in a homogeneous, isotropic FLRW setting, where homogeneity is an assumption rather than a result; the locality argument is the ECF's own, and stability in an inhomogeneous regime remains to be demonstrated (narrowed 12/09/2026)*: the contact spin-spin interaction being purely local (four-fermion term), spatial fluctuations of the spin density average out on super-Hubble scales before the bounce epoch.

(6) Situating the ECKS bounce with respect to the two classical critiques of bounces. Bounce cosmology built on pure general relativity has faced two well-known critiques: (i) the need to **violate the null energy condition** (NEC), with the attendant instabilities, and (ii) the growth of **shear** (anisotropy) during contraction. Here is where the ECKS bounce stands on each, without confusion:

- *On NEC violation: the ECKS bounce does not avoid it; it changes its origin.* It relies on **no added exotic field**: the bounce comes from the contact spin-spin interaction, local and algebraic; torsion does not propagate and vanishes wherever spin vanishes. But, in the dynamics used by the corpus, this term behaves as stiff matter of negative energy density: $H^2 = (8\pi G/3)(\rho - \alpha n^2)$ with $\tilde{\epsilon} = \tilde{p} = -\alpha n^2$ ($w_{\text{spin}} = 1$; §3 above, Foundation I §2 and Appendix H), hence $\tilde{\epsilon} + \tilde{p} = -2\alpha n^2 < 0$ near the bounce. In Popławski's Dirac description ($\tilde{\epsilon} = -\tilde{p} = -\alpha n^2$), one would have $\tilde{\epsilon} + \tilde{p} = 0$: only the weak condition would be violated. Popławski writes that spin avoids the singularity "by violating the energy condition of the singularity theorems", and the review by Battefeld & Peter (2015, arXiv:1406.2790) lists spin–torsion coupling among the "NEC violating" terms. The NEC critique therefore applies to ECKS. The difference from phantom-field bounces lies in the origin of the violation (geometric, with no extra degree of freedom), not in its absence. The stability of this effective term remains to be addressed. *(corrected 22/10/2026: the previous version stated that the ECKS bounce escapes NEC violation, which the cited sources contradict)*
- *On shear (the BKL problem): it is not resolved, and the corpus says so explicitly.* Shear also decays as a^{-6}, on an equal footing with the torsion term; which one dominates depends on initial conditions. This remains a documented **open problem** (isotropisation possibly via post-bounce particle production, but the treatment is still approximate).

In other words, the Cartan bounce effectively violates the NEC without any added exotic field: it falls under the NEC critique as regards the stability of this effective term. It further inherits the shear question, which it makes no claim to close. The useful distinction concerns the origin of the violation, not its absence.

(7) Complementarity with loop quantum gravity (LQC/LQG). The relationship between the ECF and loop quantum cosmology is one of complementarity, not rivalry. Both frameworks predict a bounce near the Planck density — one (LQC) through the quantum discreteness of geometry, the other (ECF) through torsion-spin repulsion, an effective classical mechanism. Put simply: LQC establishes **that** the universe bounces; the ECF proposes **why**; the two agree on **where**. What the ECF adds and LQC does not naturally provide is **chirality**: the bounce order parameter is the axial torsion (a P-odd pseudovector), hence chiral predictions (GW asymmetry, birefringence) absent from a non-chiral geometric bounce. The status of this connection remains **conjectural**: it indicates a convergence and a possible division of roles, not a theorem linking the two theories.

(8) An opening: a possible generative reading. Two bricks already present — the **emergent** metric (in the sense of Sakharov's induced gravity, 1967) and the bounce housed within a **parent black hole** (Popławski) — lend themselves to a *generative* reading of the framework: in this scenario, a universe would be born from gravitational collapse within another, its parameters (parent black hole spin, crystallisation scale) inherited from that event. Extended, this reading meets the evolutionary cosmologies — a cosmos in which universes beget one another — and even, in Smolin's lineage, a form of cosmological selection. This direction is the subject of *Foundation IV* (Fertile Universe), an **exploratory and highly speculative, not-yet-deposited** programme: it is mentioned here to situate the framework's horizon, with no claim of result and no commitment as to its validity. The reader whom this line interests will find it developed there; the one who prefers to stay with testable cosmology may set it aside without losing any of the rest — nothing in the preceding sections depends on it.

(9) Passage to the normalised effective branch. The late-time form used in Foundation I **is not obtained by simple**

division of the ultradense ECKS equation: it corresponds to a calibrated effective branch, with a positive sign. The dynamical bridge with the high-density repulsive correction remains open (**PO-F1-SIGMA0**). Introducing the reduced parameters $\Omega_i = \rho_{i,0}/\rho_{\text{crit},0}$, this branch reads:

$$\frac{H^2(a)}{H_0^2} = \Omega_{r,0}\, a^{-4} + \Omega_{m,0}\, a^{-3} + \Omega_\Lambda + \Omega_{\text{spin},0}\, a^{-6}, \tag{10}$$

the term $\Omega_{\text{spin},0}\, a^{-6}$ being the translation, into a reduced parameter, of $\rho_{\text{spin}} \propto a^{-6}$ established at step (2). The loop is thus closed: from the microphysics of fermionic spin to the measurable background equation.

4.2 The furnace

This section gathers, under its own title, the mechanism already established in §1 and §3 (the definition of $\Omega_{\text{spin},0}$) — without repeating the tables already given there. The confrontation with observed tensions (H_0, S_8) is developed separately in 4.6, once the dark sector has been laid out.

Between the bounce and recombination stretches a hot era in which the spin fluid, still dominant, leaves a measurable imprint visible today: the **furnace**. The mechanism holds in one sentence — $F_{\text{ion}} = 1.2765$, a parameter calibrated on H_0, sets the sound-horizon reduction; in the historical chain it also entered the small-scale power-spectrum suppression (with γ_{spin}, calibrated separately), on the grounds that the same stiff-era physics governs both — this joint role is not established under the A-7 closure *(corrected 19/10/2026)*.

The mechanism, broken down into three physical steps.

1. **A brief stiff phase.** The spin contribution evolves as a^{-6}: it can matter at high density, then dilutes much faster than radiation (a^{-4}) and matter (a^{-3}) (see item 2 for the range that the sound-horizon calibration requires).
2. **Consequence for the sound horizon.** This modified expansion history reduces the integral that defines r_s: the corpus obtains $r_s = 135.65$ Mpc versus 147.07 Mpc in the ΛCDM reference. The reduction comes from the history of $H(z)$ entering the pre-recombination integral; it must not be summarised as an arbitrary variation of the sound speed. This holds only if the a^{-6} law is kept up to $z \sim 10^5$ (spin density about 30 times the radiation there); extended to $T \sim 1$ MeV, the same branch is excluded by BBN, so its domain of validity is open (`PO-ECF-BBN-SPIN`).
3. **Consequence for growth (historical chain).** In earlier versions, the same parameter F_{ion} entered the small-scale suppression, via $G_{\text{eff}}(k)$, with γ_{spin} calibrated separately; under A-7 this role is not established and the ECF growth remains OPEN.

The confrontation with the two probes H_0 and S_8 is developed in 4.6.

4.3 Baryogenesis

Equations extracted from Foundation II, §Chiral Baryogenesis. They describe how the ECF proposes to generate the observed matter-antimatter asymmetry, $\eta = n_B/n_\gamma \simeq 6.1 \times 10^{-10}$, by proposing a mechanism for each of the three Sakharov conditions; their complete realisation is not established (corrected 19/10/2026).

The mechanism rests on the **axial torsion vector** S_μ, the totally antisymmetric part of torsion:

$$S_\mu \equiv \tfrac{1}{6}\, \varepsilon_{\mu\nu\lambda\rho}\, T^{\nu\lambda\rho}. \tag{11}$$

In the minimal propagating extension of the ECF (axial Proca structure, free of *ghost*-type instabilities), the effective Lagrangian reads:

$$\mathcal{L}_{\text{ECF}} = \frac{\sqrt{-g}}{2\kappa}\, \tilde{R}(g,T) - \frac{1}{4}\, F^{(S)}_{\mu\nu} F^{(S)\mu\nu} - \frac{1}{2}\, m_T^2\, S_\mu S^\mu + \mathcal{L}_m, \tag{12}$$

where $F^{(S)}_{\mu\nu} = \partial_\mu S_\nu - \partial_\nu S_\mu$ and m_T is the torsion mass scale. Variation with respect to S^μ gives the massive Proca wave equation:

$$\Box\, S_\mu + m_T^2\, S_\mu = \kappa\, m_T^2\, \tau_\mu^{\text{spin}}, \tag{13}$$

where τ_μ^{spin} is the fermionic axial spin current. Torsion then **propagates** as a massive spin-1 field. The coupling of this field to fermionic chirality is **geometric and universal** (derived from Dirac algebra alone, with no dependence on gauge charges — corrected 04/09/2026): it violates chirality for any Dirac fermion, quarks and leptons alike. **Electroweak sphalerons** (standard physics of the Standard Model, frozen out at $T_* = 131.7$ GeV) then convert this universal chiral bias into the specifically observed baryon asymmetry, by violating $B + L$ while conserving $B - L$ — this second step is not ECF-specific. The temporal orientation of the mean value of S_μ ($\langle S^\mu \rangle = (s_0, \mathbf{0})$) selects a unique time arrow, providing C and CP violation; cosmological cooling provides the departure from thermal equilibrium. The mechanism thus proposes a route for each of the three Sakharov conditions, although their complete realisation is not established (F2); the resulting asymmetry $\eta \simeq 5.9 \times 10^{-10}$ is within $3.5\,\%$ of the observed value (6.1×10^{-10}, **inferred** from $\Omega_b h^2 = 0.0224$ through the standard conversion $\eta \simeq 273.9\,\Omega_b h^2$ — Planck does not publish η directly), an agreement that is not an independent test, since the pre-dilution asymmetry is normalised on the observed value. ⚠ *Semi-derived result (F2, corrected 30/08, then 04/09/2026 for the dilution factor f): in the decomposition of F2, step (i), the universal chiral coupling, is first-principles; step (ii), the pre-dilution asymmetry, is normalised on the observed value ($\eta_{\text{raw}} = \eta_{\text{obs}}/0.50$), neither derived nor taken from the Foundation I background; step (iii), the survival fraction $f \simeq 0.48$, is semi-analytic (clarified 19/10/2026).* The ECKS limit of Foundation I is recovered exactly when $m_T \gg H$, so that all cosmological-sector results remain unchanged. **The derivation of a complete kinetic/Boltzmann chain remains open; this point is summarised in §7.**

4.4 Dark matter

*Equations extracted from Foundation II, §Geometric Dark Matter. Dark matter is not a particle but a structure of topological defects, whose density is **read off** the closure relation, not imposed by it.*

The topological invariance postulate **imposes** `Ω_K ≡ 0`. Dividing the complete Friedmann equation by H^2, with $\rho_c = 3H^2/(8\pi G)$, one obtains the identity $\sum_i \Omega_i + \Omega_K = 1$. Thus, for $\Omega_K = 0$, the sum of all the components contributing to the Friedmann equation equals exactly 1 at every instant. In the effective decomposition adopted here:

$$\Omega_b(a) + \Omega_\gamma(a) + \Omega_{\text{knots}}(a) + \Omega_\tau(a) \equiv 1, \tag{14}$$

where Ω_b is the baryonic contribution, Ω_γ the radiative contribution (photons; the other relativistic species are grouped according to the paper's convention), Ω_{knots} the localised-defect component used as geometric dark matter, and Ω_τ the diffuse torsion component associated with the effective dark sector. **Here γ denotes photon radiation; it does not denote dark energy.** In the adopted Friedmann equation, this closure relation is the expression of the choice $\Omega_K \equiv 0$ imposed by the PIT; it does not constitute a distinct dynamical mechanism that would "prevent" curvature.

The elementary brick of this dark matter is the **micro-knot**, whose characteristic mass is of the order of the horizon mass at the electroweak scale:

$$M_{\text{MK}} = \frac{c^3}{2\,G\,H_{\text{EW}}} \approx 3.74 \times 10^{24}\ \text{kg}, \tag{15}$$

i.e. about 0.63 Earth mass (the value sets an order of magnitude, not an exact Earth mass). The formula sets the order of magnitude ($\sim 10^{25}$ kg); the value 3.74×10^{24} kg is the centre of a Kibble-Zurek log-normal distribution spanning 10^{23} to 10^{25} kg — it is therefore a mass **calibrated within that window**, not a prediction to the exact figure.

Aggregation of micro-knots. Foundation II uses a Poisson parametrisation whose mean λ is not derived from a coagulation dynamics. Compatibility with Subaru/HSC imposes $\lambda \gtrsim 36$; at the threshold $\lambda = 36$, the typical mass is $M_{\text{typ}} \simeq 6.8 \times 10^{-5}\, M_\odot$ (about 23 Earth masses). **This value is a minimal benchmark compatible with HSC, not a coagulation ceiling.** The absolute normalisation linking λ, R_{coal} and n_{KZ} remains suspended on the lock PO-F3-C6.

The Macro sector is distinct. Micro → Macro coagulation is **WITHDRAWN**: no public quantitative chain produces a compact population $\sim 10^5\, M_\odot$. The horizon-mass scale at the Great Annihilation is computable, but the formation and collapse of a Macro object remain `STRUCTURAL-HYPOTHESIS` and depend in particular on a collapse efficiency that is still open. Nor does the corpus, at this stage, allow the causal role of "structuring" the chiral walls to be assigned demonstrably to micro-knots, to their aggregates or to Macro-knots: this chain remains **open** and is not closed here by wording.

The candidate 1D filament sector (torsion strings) was considered but **is not retained as an established component**: it remains an open problem (PO-F2-8, Foundation II), with no established Lagrangian foundation. It must not be confused with the *observed* filaments of the cosmic web, explained by standard anisotropic gravitational collapse (Zel'dovich). **The coherent rotation of these filaments nevertheless remains a candidate signature in F2, via a magnetic coupling; it is not presented as derived. The topology of the defects and the Micro / aggregates / Macro chain are taken up as distinct locks in §7.**

> ⚠ **Falsification condition, stated before the data (note of 08/09/2026).** This claim — dark matter is a structure, not a particle — is falsifiable, and it must be said by what. On 1 September 2026 the LZ collaboration reported **one** event consistent with a nuclear recoil of 248 ± 23 keV (2.84 tonne·year, extended window up to ~270 keV, EFT and inelastic models), in tension with the background-only hypothesis at **2.6σ global** (3.4σ local at most); the collaboration does not claim a detection (arXiv:2609.02823). Two things must be distinguished. **(1)** That event cannot be a micro-knot: a 3.74×10^{24} kg body does not deposit 248 keV in a single nuclear recoil, and at the derived number density it does not traverse the detector on any relevant timescale. **(2)** But if such a signal consolidates as particle dark matter, it **refutes the geometric sector as the sole component**: the ECF would then have to postulate a second, particulate dark matter, which is precisely the economy this sector claims. At 2.6σ on a single event, nothing is settled — the point of recording it here is that the refutation condition should be written **before** the data arrive, not after.

4.5 Dark energy

Equations extracted from Foundation II, §Vacuum Condensation and the Origin of Dark Energy. Dark energy is not an added constant but the energy of a torsion condensate, tied to the topological "debt" of the bounce.

In vacuum (sourceless), the axial torsion field obeys the homogeneous Proca equation:

$$\Box\, S_\mu + m_T^2\, S_\mu = 0. \tag{16}$$

In a spatially flat FLRW background with $H \ll m_T$, the zero-mode solution is a constant field. The effective potential for the cosmological mode $S_0(t)$ takes the form:

$$V_{\text{eff}}(S_0) = V_0 - \frac{m_T^2}{2}\, S_0^2 + \frac{\lambda_T}{4!}\, S_0^4, \tag{17}$$

where $\lambda_T > 0$ is the axial self-coupling. In this effective form, the potential has a non-trivial minimum: the field acquires a **vacuum expectation value** (VEV). **Status caveat: the negative sign of the quadratic term is imposed in Foundation II in order to obtain a non-zero VEV; its geometric origin is not derived there and remains an open problem.**

$$\langle S_\mu \rangle = s_0\, \delta^0_\mu, \qquad s_0^2 = \frac{m_T^2}{\lambda_T/3!} \neq 0. \tag{18}$$

The VEV is oriented along the cosmic time axis δ^0_μ, which **preserves spatial isotropy**: this is a spontaneous breaking of boost invariance (not of isotropy), subject to laboratory bounds on Lorentz violation: if the condensate couples minimally to fermions (SME coefficient $b_\mu = -\frac{3}{4} S_\mu$; Kostelecký, Russell & Tasson, *Phys. Rev. Lett.* **100**, 111102 (2008)), $s_0 \lesssim 3 \times 10^{-27}$ GeV, hence $m_T \gtrsim 4 \times 10^3$ GeV; the quartic potential then gives no perturbative origin

for the VEV (Foundation II, Appendix F). *(Corrected 22/10/2026: this sentence invoked a bound $|s_0|/M_{\rm Pl} \lesssim 10^{-15}$, which has no source and is about thirty orders of magnitude too loose.)*

Normalisation of the condensate on the observed dark-energy density. The energy density of the condensate is the **well depth** of the tachyonic potential:

$$\rho_\Lambda^{\rm eff} = \tfrac{1}{4}\, m_T^2\, s_0^2. \tag{19}$$

Setting $\rho_\Lambda^{\rm eff} = |\Omega_\tau|\, \rho_{\rm crit,0}$ gives the mass-coupling relation:

⚠ **Stated discrepancy.** Identifying $|\Omega_\tau|$ with the **initial** debt $\Omega_{\rm spin,peak} = 0.093$ **does not hold numerically**: $0.093\, \rho_{\rm crit,0} \simeq 3.4 \times 10^{-48}$ GeV^4, a factor ~ 7 below the value used, which in fact equals $\Omega_\Lambda\, \rho_{\rm crit,0}$ to within 5%. The relation therefore **calibrates** $m_T^2 s_0^2$ on the observed dark energy; it does **not derive** it from the torsion debt. The origin of the factor $\Omega_\Lambda/\Omega_{\rm spin,peak} \approx 7$ is an **open problem**.

$$m_T^2\, s_0^2 = 4\, |\Omega_\tau^{(0)}|\, \rho_{\rm crit,0} \simeq 1.2 \times 10^{-46}\ {\rm GeV}^4. \tag{20}$$

Dark energy today is **calibrated** on its observed value, not derived from the torsion debt: ⚠ the identification with $\Omega_{\rm spin,peak} = 0.093$ fails by a factor ~ 7 (see the stated discrepancy above). Projected onto the standard CPL parametrisation, this dynamics generates the trajectory $(w_0, w_a) = (-0.904,\, -0.153)$ — the calibrated trajectory that can be confronted with DESI data. Any revision of the Foundation I calibration would propagate directly into this calibrated trajectory: the dark sector is therefore *tied*, not independently adjusted.

How these two parameters are obtained (traceability). The values $w_0 = -0.904$ and $w_a = -0.153$ are an adopted calibration point — no deposited script produces -0.904 (`PO-F1-W0`, F1, corrected 16/10/2026); the computed value is -0.907 ± 0.004 — and the two parameters do not have the same status, and this must be distinguished precisely:

Parameter	Origin	Status
$\Omega_{\rm spin,peak} = 0.093 \pm 0.004$	spin-density ratio **normalised** at the reference redshift $z_{\rm ref} = 5600$. The spin sector is expected to be sourced at the QCD transition, but 0.093 is not a QCD-epoch value (the ratio scales as $(1+z)^2$); its numerical value is a calibration point at $z_{\rm ref}$. *Uncertainty: likelihood profiling ($\Delta\chi^2 \leq 1$) against $r_s = 135.65$ Mpc.*	CALIBRATED
$w_0 = -0.904$ (calibration point, without error bar; the likelihood profile gives -0.907 ± 0.004)	$w_0 = -1 + \Omega_{\rm spin,peak} = -1 + 0.093 = -0.907$ (i.e. -0.904 to within $0.3\,\%$). The relation $w_0 = -1 + \Omega_{\rm spin}^{\rm peak}$ is an **empirical identification, not a derivation**; since the amplitude 0.093 is calibrated, w_0 inherits that status, uncertainty included.	CALIBRATED (empirical identification)
$w_a = -0.153 \pm 0.007$	$w_a = -\xi \cdot \Omega_{\rm spin,peak}$ with $\xi \approx 1.645$, a phenomenological factor for the dilution of the torsion debt. ξ is not derived from the ECKS action, and its own uncertainty is not included here — the above is therefore a floor.	CALIBRATED
$\xi = 1.645$	its *ab initio* derivation would require the condensate variance $\langle \delta S^2 \rangle(z)$.	OPEN (PO-F1-B)

The distinction is clear: w_0 is tied to the spin density by an **empirical identification**, not a derivation, and w_a goes through $\xi = 1.645$. In the frozen baseline, ξ is **CALIBRATED on the DESI preference used by the ECF**. The pair $(w_0, w_a) = (-0.904, -0.153)$ is therefore **CALIBRATED**, not derived and **not independent of DESI**. The reconstructed confrontation with the 12-observable DESI DR1 block gives $\Delta\chi^2(\text{ECF} - \Lambda\text{CDM}) = +6.21/+38.59$ under Planck/SH0ES priors; the former values $+8.06/+41.94$ are historical.

Degeneracy objection and growth test. Agreement of the pair (w_0, w_a) with background constraints is not enough to identify torsion as the unique physical mechanism. Foundation I explicitly discusses the objection of Petri, Marra & von Marttens (2026), which illustrates a possible degeneracy at the background level, and recalls that the crossing of $w = -1$ is not derived here. The complementary test is then growth: integrating the growth equation on the ECF background gives an ECF/ΛCDM ratio of $f\sigma_8$ that varies by about 9.8% over $0 < z < 0.8$ (0.870 at $z = 0$, 0.959 at $z = 0.8$; F1), reproduced by `compute_fsigma8_ecf.py` *(corrected 16/10/2026: earlier versions gave about 2%)*. **This check is not presented as an observational preference for the ECF: it tests a failure mode of the scenario and does not exclude it in the reported computation.**

Independent compatibility test: DES Y6 (arXiv:2605.27221, May 2026; added to the corpus on 02/09/2026). The Dark Energy Survey published its final-year dataset in the form of a **pivot** parameter w_p (at the redshift z_p where $\text{Cov}(w_p, w_a) = 0$), which avoids the pitfall of the strong w_0–w_a correlation. For DES alone: $w_p = -0.91 \pm 0.04$ at $z_p = 0.19$ — the ECF trajectory evaluated at the same redshift gives $w_p^{\text{ECF}} = -0.928$, a 0.46σ difference. The agreement narrows only mildly with DESI DR2 added (0.28σ) and widens only to 1.45σ for the most constraining combination including the CMB — compatible in every case tested (F1, F2).

Error bar added 02/09/2026, on $\Omega_{\text{spin,peak}}$ alone (`profile_likelihood_calibrated.py`, rigorous profiling) — ξ is held fixed, so the above is a floor, not the complete uncertainty of the pair. Also reproducible, less precisely, via `calibrate_w0wa.py`.

Scripts (public repository). The computation is reproducible via the repository `github.com/pfichant/spin-torsion-cosmology`:

- `plot_ecf_desi_mcmc_v3_0.py` — the confrontation of the point (w_0, w_a) with the DESI contours (full covariance);
- `calibrate_w0wa.py` — the construction of the parameters: w_0 by empirical identification with Ω_{spin}, w_a via the calibrated factor ξ;
- `scan_F_ion_H0_BAO.py` — the scan of F_{ion} on the active background (H_0 and four BAO points), which places the calibrated point relative to the joint compromise (CALCULATED-CONDITIONAL); the former `plot_Optimization_Intersection.py` (function `verify_calibration()`, historical calibration at the "sweet spot") is archived in `Scripts_SUPERSEDED`;
- `plot_cosmic_age_cpl.py` (illustrative CPL proxy $(-1.1; -1.5)$ and calibrated point), `plot_desi_prediction.py` — the CPL trajectory figures.

The background solver itself (**EC-Cosmo**, described in an appendix of Foundation I) chains the sound-horizon solution and then the parameter scan.

Confrontation script. The comparison is carried by `plot_ecf_desi_mcmc_v3_0.py`. In the post-audit baseline, the ECF point is **CALIBRATED on the DESI preference used by the ECF** and does not constitute an independent prediction. The MCMC block has been reconstructed on **12 published DESI DR1 observables**, with the available intra-bin correlations; the former QSO D_M/D_H rows are **WITHDRAWN — MIXED-DATA / PROVENANCE INVALID**.

The effective amplitude F_{ion} and the post-A-7 separation between background and growth. $F_{\text{ion}} = 1.2765 \pm 0.0246$ remains a **CALIBRATED** effective amplitude of the background. The former joint H_0–S_8 intersection is no longer canonical: $S_8 = 0.766$ is the KiDS-1000 observational target, while $S_8^{\text{ref}} = 0.780$ is a CAMB ΛCDM reference on the A-7 background, **CALCULATED-CONDITIONAL**. The ECF physical prediction for growth remains OPEN.

How F_{ion} was determined. In the historical calibration its value was not set by hand: it simultaneously minimised three observational constraints, whose "balance point" was unique. The corresponding script, archived in `Scripts_SUPERSEDED` together with the figure it produced (former Fig. 4 of Foundation I, withdrawn on 09/10/2026), carries the header:

`plot_Optimization_Intersection.py – v2 (referee)` · Foundation I · Fig. 4 (former numbering) · Sources: Sec. 5, Table chi2breakdown, Table priors, App. F

Historical H0–S8 chain. The former optimisation combining SH0ES, KiDS-1000 and a penalty on F_{ion}, with $\gamma_{\text{spin}} = 0.3116$, is **HISTORICAL** and no longer defines the post-A-7 canon. It is kept solely for the traceability of F1. On the active background ($\Omega_m = \omega_m/h^2 = 0.267$), $F_{\text{ion}} = 1.2765$ remains calibrated on H_0; a joint fit of SH0ES and of the four BAO points alone would prefer $F_{\text{ion}} \simeq 1.22$ ($H_0 \simeq 71.8$, $\Omega_m \simeq 0.276$), the calibrated point lying $\Delta\chi^2 \simeq 1.6$ above that compromise (CALCULATED-CONDITIONAL: linear $H_0(F_{\text{ion}})$, no CMB block, r_{drag} identified with r_s; `scan_F_ion_H0_BAO.py`). The H_0–S_8 figure of Foundation I is redrawn on this active background, with no ECF value of S_8 (OPEN).

Consistency note on H_0. Foundation I reports two neighbouring values depending on the calibration used: $H_0 = 73.04 \pm 1.04$ (fit on SH0ES, calibrated ECF value) and $H_0 = 73.50 \pm 0.81$ (H_0 Distance Network value, independent external measurement). The present document uses the first in the equations (the one actually calibrated and used in $H(a)$) and cites the second as external confirmation; the difference between the two is smaller than their error bars ($< 0.6\sigma$). **Perspective.** A future recalibration on this newer consensus value (73.50) is a natural avenue; however, since local H_0 determinations still shift appreciably from one analysis to another, the stable SH0ES anchor (73.04) is retained here as the reference.

4.6 The H_0/S_8 tensions: CMB and large-scale structure

The physical mechanism (the furnace) is laid out in 4.2; this section gives its observational confrontation, once the dark sector (dark matter and dark energy) has been laid out — chronologically, the tensions themselves only manifest through late-time measurements: the cosmic microwave background and the growth of large-scale structure.

The furnace mechanism (4.2) only takes on its full meaning once confronted with the two physically independent probes that put it to the test: the **CMB** (Planck, the acoustic scale and angular distance at recombination) for H_0, and the **growth of large-scale structure** (KiDS-1000, weak gravitational lensing) for S_8. It is this double confrontation that gives the argument its weight: $F_{\text{ion}} = 1.2765$, a calibrated parameter, enters the H_0 tension; its role in S_8 is not established under A-7. For H_0, $H_0 \simeq 73.04$ remains the calibrated background value. For S_8, on the other hand, $S_8 \simeq 0.766$ is no longer an ECF output: it is the KiDS-1000 observational target, on which γ_{spin} is calibrated separately; the reference computed under the A-7 background is $S_8^{\text{ref}} \simeq 0.780$, and the complete ECF growth remains OPEN pending a perturbative/Boltzmann chain. These two targets are measured by two methods, two instruments and at two different epochs, with no necessary logical link between them a priori. A smaller sound horizon, at fixed CMB acoustic angle **and fixed Ω_m**, **mechanically requires** a larger H_0 to compensate (under the closure A-7, θ_* falls by $\simeq 6.5\%$: `PO-F1-THETA`) — exactly the direction needed to relax the SH0ES/Planck tension; in the historical chain, the same F_{ion} also entered, jointly with γ_{spin}, the power-spectrum suppression measured by weak-lensing surveys — a role not established under A-7 (see §1 for the full statistical argument and value table).

The price to pay, stated plainly. With the active A7 baseline, the cosmic age is $t_0 = 13.33$ Gyr for $w = -1$. At the calibrated dark-energy point $(w_0, w_a) = (-0.904; -0.153)$, the same background gives $t_0 \simeq 13.19$ Gyr, 0.13 Gyr below t_{GC} (1.3σ statistical, within the 0.5 Gyr systematic) *(added 19/10/2026; Foundation I)*. The 13.33 Gyr value barely satisfies the lower bound $t_{\text{GC}} = 13.32 \pm 0.10_{\text{stat}} \pm 0.5_{\text{sys}}$ Gyr, which is the age of the clusters without their formation time. The quantity directly comparable to t_0 is the age of the Universe inferred by Valcin et al., $t_U = 13.5^{+0.16}_{-0.14} \pm 0.5$ Gyr, from which t_0 differs by -0.17 Gyr, within the 0.5 Gyr systematic. This is not a high-significance test *(corrected 22/10/2026: the 0.01 Gyr difference compared t_0 with the cluster age, not with the age of the Universe)*. The former $t_0 \simeq 12.74$ Gyr / 5.7σ narrative belongs to the **HISTORICAL** $\Omega_m = 0.315$ branch and is no longer active. The active observational cost lies in the BAO/DESI sector: $\chi^2_{\text{BAO}} = 12.11$ versus 1.79 for ΛCDM, i.e. $\Delta\chi^2 \simeq +10.3$ *(BOSS DR12 values from Alam et al. 2017 checked on 05/10/2026 against Table 7 of the primary source, statistical and systematic errors combined; formerly 16.04 / 2.93 / +13.1)*. PO-F2-5 remains a distinct lock, without being carried by an age deficit.

Status: F_{ion} is **calibrated**, not derived *ab initio* — it is calibrated on H_0 rather than predicted from the action; the S_8 sector relies on γ_{spin}, calibrated separately. See Appendix B for the research programme aiming to derive it from stiff-era thermodynamics.

4.7 The horizon

The detail appears in Foundation I, Appendix H (regularisation of the singularity and bounce); the horizon problem and the primordial spectrum are treated in Foundation III — Cosmological Physics. Here is the essential.

Derived from the single-density form (3) with the usual continuity equation $\dot{\rho} = -3H(\rho+p)$, the acceleration equation reads:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}\left[\rho\left(1-\frac{4\rho}{\rho_c}\right)+3p\left(1-\frac{2\rho}{\rho_c}\right)\right]. \tag{21}$$

At the bounce ($\rho=\rho_c$, $H=0$), $\ddot{a}/a = 4\pi G(\rho_c+p) > 0$; the acceleration changes sign at $\rho^* = \rho_c(1+3w)/(4+6w)$, i.e. $\rho_c/4$ for dust, $\rho_c/3$ for radiation and $0.4\,\rho_c$ for the stiff fluid (the last two values in form (3) alone; the per-regime equations (3a) give $\rho_c/4$ for radiation and no spin-driven bounce for the stiff fluid). This is also the effective Raychaudhuri equation of loop quantum cosmology. For $p=0$, it holds in Einstein-Cartan for non-relativistic fermionic matter whose spin term behaves as stiff matter (spin-fluid description, $n \propto a^{-3}$; primary source to be confirmed), and not in Popławski's Dirac description, where $\tilde{\epsilon} = -\tilde{p}$ and where, for dust, his eqs. (11)–(12) give $\ddot{a} < 0$ at $H=0$; for $w \neq 0$, it inherits the schematic status of (3). For $w=1$, the sign change at $0.4\rho_c$ gives $N = \ln(5/2)/6 \simeq 0.153$ e-fold: a result that is **CALCULATED-CONDITIONAL in this scheme only**. *(Corrected 04/10/2026: earlier versions displayed $\ddot{a}/a = -(4\pi G/3)(\rho+3p)(1-\rho/\rho_c)$, which does not follow from (3) and does not change sign.)* The quadratic term there varies as a^{-12} if $\rho \propto a^{-6}$, whereas the derived microscopic spin term varies as a^{-6} and enters the other Friedmann equation linearly: their correspondence with the ECF action is not established (`PO-ECF-ACTION-BOUNCE`, OPEN).

The former $N_{\text{eff}} \sim 60$ of F2 Extended is **HISTORICAL/WITHDRAWN**; its expression, which has H_{rebond} in the denominator, cannot be evaluated literally at the bounce of the scheme, where $H_{\text{rebond}} = 0$. Foundation III — Cosmological Physics cites $N_e \leq \frac{1}{3}\ln 4 \simeq 0.46$ e-fold for an **external Weyssenhoff spin fluid** with $w \in [0,1]$: this upper bound serves to close this kinematic route and does not bound a demonstrated ECF window. The earlier threshold $\rho > \rho_c/4$ is withdrawn. The existence of a post-bounce acceleration in the ECF spin mechanism, the resolution of the horizon and the treatment of isotropy remain distinct and open questions.

⚠ **Withdrawal of 30/08/2026: the resolution of the horizon problem itself is no longer established.** A distinct result — Corollary F3-1, which asserted that the *a priori* topological coherence of the Kibble-Zurek network functionally replaces inflation for the horizon problem — rested on the assumption that every chiral wall is bounded by loops of 1D strings, weaving the bubbles into a single, causally connected network. This assumption has been shown to be false (Foundation II, I69): chiral walls are independent closed surfaces, without this connection. Without it, the network is not a single object but a multitude of bubbles causally disconnected from one another — precisely what the horizon problem asks to resolve. Corollary F3-1 is therefore withdrawn as established (Foundation III *Cosmological Physics*, corrected 30/08); the horizon problem itself reverts to being **open** *(see §7)*, pending a demonstration, not yet given, that the 1D sector (a candidate, open problem PO-F2-8) actually reconnects the bubbles to one another.

Obtaining the right e-fold number is *necessary* but not *sufficient*: an expansion phase is ultimately judged by its **spectrum** (spectral index, non-gaussianities), not by the expansion factor alone. Yet it is precisely on the spectrum that the framework meets its principal open problem (PO-F3-NS): the large-scale tilt is not predicted: it is treated as an **inherited input** of the scenario, without the inheritance mechanism being derived in the testable cosmology presented here. In other words: the only positive acceleration computation cited here belongs to the conditional F1 scheme; the microscopic ECF dynamics has not yet established this phase. The horizon problem remains open, as does the production of the observed primordial spectrum. The full spectrum computation would require a dedicated Boltzmann solver, not built.

Inflation's full specification. Inflation does not solve only the horizon: it settles several problems at once, and that is its strength. An alternative framework must therefore measure up to *all* its functions, not just one. Here, extracted from Foundation III, is the honest state of each within the ECF:

Inflation function	Treatment in the ECF	Status
Horizon and homogeneity	⚠ **open (30/08)** — *a priori* topological coherence of the Kibble-Zurek network was proposed, but rested on a connection of the walls by 1D strings now invalidated (Corollary F3-1 withdrawn, see above)	**open** (previously declared covered)
Flatness	$\Omega_K \equiv 0$ branch imposed by the Topological Invariance Principle; the corresponding closure relation is an identity of the Friedmann equation	covered at the level of the postulate — dynamical realisation/derivation to be formalised
Isotropy	isotropisation at the bounce (semi-quantitative WKB lemma)	**open (PO-F3-ISO)** — shear and BKL instabilities near the bounce not controlled
Spectrum n_s	not predicted: tilt inherited; the *stiff* era adds a blue component	**open (PO-F3-NS)** — falsifiable form of bound; illustrative threshold $N_{\text{stiff}} \gtrsim 68$, **not yet calibrated**
Primordial fluctuations	defect statistics / KZ network	**OPEN — F2↔F3 connection to be established**: F2 computes the *shot noise* of the granular micro-knot population ($P_{\text{Poisson}} \sim 1.7 \times 10^{-16}$ Mpc^3) and finds it ~20 orders of magnitude below the observed adiabatic power; F3 estimates, from a coarse-grained KZ network of length $\xi_{\text{KZ}} \sim 60$ Mpc, $\delta N/N \sim 2.8 \times 10^{-4}$ before projection/transfer and proposes an **exploratory** order-of-magnitude compatibility. These two computations do not use the same effective normalisation; no demonstrated bridge allows them to be identified as a single mechanism generating the primordial spectrum.
Magnetic monopoles	trans-GUT bounce; the former lead ($\partial \circ \partial = 0$ would forbid the charges) shared the same premise as Corollary F3-1, now withdrawn	**open, with no current fallback lead (30/08)**

Four functions are therefore still open or incomplete in this table — horizon/homogeneity, isotropy, spectrum and monopoles — while flatness remains to be formalised dynamically and the primordial-fluctuation channel has a distinct mechanism to be tested. The ECF does not claim to match inflation across the whole table — it proposes distinct mechanisms, some solid and others still to be established.

4.8 Chiral birefringence

Values and references are detailed in §5; this subsection gathers the mechanism under its own title, rather than leaving it diluted across the data-confrontation tables.

The chiral gel at the bounce — whose defects, frozen later at the electroweak freeze-out ($T \simeq 159$ GeV), make up the dark matter — does not only set the dark sector: because axial torsion is a pseudo-vector (P-odd), the freeze-

out "picks a hand" — spontaneously breaks parity. This single breaking leaves an observable imprint in **two distinct channels**, electromagnetic and gravitational, which are not two independent predictions but two manifestations of the **same** primordial chiral asymmetry.

Electromagnetic channel: CMB birefringence. Residual torsion rotates the polarisation plane of cosmic microwave background light. The ECF predicts a **non-zero** rotation angle, $\beta \neq 0$ — whereas early dark energy (the most directly comparable family of alternative models) predicts $\beta = 0$. This is a test that **separates** the two model families, not a mere additional fit. The expected magnitude, $\beta \approx 0.35°$, is a *consistency check* calibrated on the Minami–Komatsu anchor — not a parameter-free prediction — but the **non-zero-ness** of the effect is the claim made here; the sign is not presented as predicted without further derivation. **Observational status**: Planck PR4 measures $\beta = 0.30° \pm 0.11°$ *(corrected 15/10/2026, formerly 0.32°)*; ACT DR6 measures $\beta = 0.215° \pm 0.074°$ — both compatible with $\beta \neq 0$, without yet settling between the precise values expected from the different candidate chiral mechanisms. Next-generation instruments (LiteBIRD, Simons Observatory) aim for the precision needed for a clear verdict.

Gravitational channel: chiral asymmetry of the gravitational-wave background. The same parity breaking must also show up in the primordial gravitational-wave spectrum, as an asymmetry between left- and right-circular polarisations, $\Pi_{\rm GW} \equiv (\Omega^R_{\rm GW} - \Omega^L_{\rm GW})/(\Omega^R_{\rm GW} + \Omega^L_{\rm GW})$. The calibrated template gives $\Pi_{\rm GW} = 0.20 \pm 0.05$ (**CALIBRATED TEMPLATE**) — an effect well above the expected instrumental noise, unlike most gravitational leptogenesis mechanisms, which predict a negligible asymmetry. **Refutation threshold**: $\Pi_{\rm GW} < 0.02$ at 95% C.L. would refute **the tested chiral tensor sector**, not the ECF as a whole. **Instrument**: LISA, with a caveat: a planar detector is nearly blind to the net circular polarisation of an isotropic background; $\Pi_{\rm GW}$ is accessible only through the kinematic anisotropy (dipole) or a detector network (LISA–Taiji), and the 0.02 threshold is not yet backed by a published forecast *(added 22/10/2026)*; via the primordial stochastic background (`plot_ecf_gw_spectrum.py`) — not yet measurable with current detectors.

Why both matter together. A null signal in one of the two channels would not necessarily invalidate the other taken alone — the two observables depend on different calibrated parameters (the Minami–Komatsu anchor for β, the normalisation of the tensor spectrum for $\Pi_{\rm GW}$) — but a **joint** detection, with consistent signs across both channels, would constitute a cross-confirmation far stronger than either test alone: it would establish that the **same** primordial chiral asymmetry propagates through both the electromagnetic and the gravitational sector, as the mechanism requires.

Back to the main path: §5 *Confrontation with data*.

5. Confrontation with data: calibration, validation, predictions

A coherent construction is not enough. What served to fix the model must be separated from what can genuinely put it to the test. The observational confrontation is therefore organised on three levels: calibration, tests on data not used for the fit, then future predictions.

*To let a reader assess the framework's reproducibility and falsifiability, here is the full set of observational data used across the corpus (Foundations I–III), classified by role. The distinction is essential: what **calibrates** adjusts the parameters; the **tests and confrontations** assess results whose status may be derived, calibrated, conditional or open; the **future instruments** provide channels for testing and refutation.*

Current data — calibration (fix the parameters)

Dataset	What it fixes
SH0ES (Riess et al. 2022)	H_0 sector ($H_0 = 73.04$)
Planck 2018	acoustic scale, ΛCDM baseline ($\Omega_m = 0.315 \pm 0.007$, rounded reference value)
KiDS-1000 (Heymans 2021)	S_8 sector ($S_8^{\rm obs} = 0.766^{+0.020}_{-0.014}$; ± 0.019 symmetrised in F1)

Historical chain of the global H_0/S_8 test. In the earlier analysis, SH0ES/KiDS/Planck fixed $F_{\rm ion} = 1.2765$ and $\gamma_{\rm spin} = 0.3116$. This construction is now **HISTORICAL**: $S_8 = 0.766$ is an observational target, $\gamma = 0.3116$ is no

longer canonical, and ECF growth remains OPEN.

Current data — validation (tests and confrontations)

The principle of each test and the script that produces it (repository `github.com/pfichant/spin-torsion-cosmology`) are given to allow reproduction.

Dataset	Object confronted — principle	Result / status	Script
DESI DR1 then DR2 (BAO)	dark energy (w_0, w_a): the torsion "debt" projected onto CPL fixes the equation of state, confronted with BAO	**CALIBRATED** in the post-Audit v2 baseline; $\xi = 1.645$ is calibrated on the DESI preference used by the ECF and is not an independent prediction prior to DESI. DR2 reports $3.1\,\sigma$ BAO+CMB and 2.8–$4.2\,\sigma$ with SNe for dark-energy evolution; the ECF test gives $2.03\,\sigma$ under a Planck prior and $6.98\,\sigma$ under a SH0ES prior; DR1 comparison on 12 BAO observables with published covariance	`plot_ecf_desi_mcmc_v3_0.py`, `plot_w0wa_DESI_ECF.py`
SPARC (175 galaxies)	geometric dark matter: fit of the rotation curves with the adopted profile $\rho(r) = \rho_0/[1+(r/R_s)^2]$; its integration gives $M(r)$ then $V^2 = GM(r)/r$ (Eq. 70). The link micro-knots / geometric sector $\to (\rho_0, R_s)$ remains **OPEN** (`PO-ECF-HALO`)	$\tilde{\chi}^2_{\rm med} = 0.80$	`plot_F2_sparc_model_comparison.py`, `plot_sparc175_mosaic.py`
VSF / DESIVAST DR1 ($z < 0.24$)	great-void structure: phenomenological VSF template modulated at the scale $\xi_{\rm KZ}$, with calibrated amplitude; the corpus does not derive here a local law "$\Omega_\tau \to 1 \Rightarrow$ widening"	ECF vs ΛCDM comparison	`plot_F3_ecf_vsf_desivast_fits.py`, `plot_F2_vsf_prediction.py`
ACT DR6 + Planck PR4 (CMB polarisation)	chiral birefringence: axial torsion rotates the CMB polarisation plane. **ECF prediction:** $\beta \neq 0$ (early dark energy without a photon coupling predicts $\beta = 0$); the magnitude $\beta \approx 0.35°$ is a *consistency check* calibrated to the Minami–Komatsu anchor, not a parameter-free prediction	Planck PR4 $0.30° \pm 0.11°$, ACT DR6 $0.215° \pm 0.074°$ — compatible with $\beta \neq 0$	`plot_predict_birefringence.py`, `script_05_birefringence_calibration.py`

Dataset	Object confronted — principle	Result / status	Script
eBOSS DR16 (QSO BAO)	conditional comparison of the BAO ruler	published target $D_M/r_d(z{=}1.48) = 30.21 \pm 0.79$ (eBOSS QSO BAO+RSD consensus, Neveux et al. 2020); ECF $D_M/r_s = 31.37$ (+3.8%, 1.5σ)	`chi_carre.py`; **CALCULATED-CONDITIONAL** on `PO-ECF-BAO-RDRAG`
Pantheon+ (SN Ia)	$q(z)$: **late** transition $z_t^{\rm ECF} \approx 0.76$ ($\Omega_m = 0.267$), against 0.63 for the Planck reference ($\Omega_m = 0.315$)	**CALCULATED-CONDITIONAL** on the calibrated background; external comparison, with no ECF fit to Pantheon+	`plot_deceleration_transition.py` (F1 Extended rev. 13, Appendix G; correction of 29/09/2026)
JWST (JADES, CEERS, GLASS; LRD)	Eddington-rate growth of macro-knot seeds from $z = 20$ ($\sim 10^7\ M_\odot$ at $z = 10$)	`CONDITIONAL` — seed formation = structural hypothesis; Planck bound PO-F2-SEED-CMB	`plot_growth_comparison.py`, `plot_ECFLRDJeansMass.py` (F2)
BBN	normalisation of the primordial spin sector at nucleosynthesis	**BBN compatibility not demonstrated**; spin/condensate/cosmology normalisation bridge OPEN	`plot_F3_ecf_bbn_constraint.py`

Scope of the late-time check F1, Appendix G. The $z_t^{\rm ECF} \approx 0.7646$ is computed on the calibrated A-7 branch; the spin contribution is negligible at these redshifts. Arora & Sahoo (2024), for $q_f = -1$, report 0.792 ± 0.073 with Pantheon+ alone and 0.760 ± 0.051 with OHD + Pantheon+ + BAO. The agreement concerns the value of the transition, without fitting their parametrisation to the ECF. Dahiya & Jain (2023) give $0.62 \lesssim z_t \lesssim 0.80$ (from 0.618 to 0.799) as the range of best fits of four distinct models, and not as an ECF confidence interval. This late acceleration is distinct from the open question of post-bounce acceleration.

Two primordial-sector predictions, not yet directly confrontable, also have their script: the n_s spectrum (`plot_F3_ecf_ns_KZ.py`, **to be regenerated** — the figure carries the old result; PO-F3-NS) and the non-gaussianities $f_{\rm NL} > 0$ (`plot_F3_ecf_fNL_KZ.py`), **CONDITIONAL / EXPLORATORY**: the raw 3D Poisson result does not yet constitute the observable CMB $f_{\rm NL}$ without perturbative transfer.

Adopted VSF parameters — post-Audit v2 baseline. $\xi_{\rm KZ} = 60$ Mpc, $\sigma = 15$ Mpc and threshold 40.4 Mpc/h (conversion at $h_{\rm fid} = 0.6732$, reference cosmology of the DESIVAST catalogue). These values are **adopted / calibrated**, not derived; `PO-F3-XI` remains OPEN.

Future instruments — tests and falsifiable channels

Observable	Signal / value	Scientific status	Instrument / channel	Instrument status	Timescale	Adverse result / scope
Direct-detection null	no signal in a weakly-interacting detector	`STRUCTURAL` — a consequence of the geometric nature of the dark sector (F2 §9.2)	LZ / XENONnT / PandaX	operating	ongoing	a particle dark-matter detection confirmed at 5σ refutes the geometric sector as the sole component of dark matter (F2 §9.2)
Void function (VSF)	lensing deficit $\Delta\gamma/\gamma \sim -5\,\%$ at $R \sim 40.4$ Mpc/h	`CALIBRATED PHENOMENO-LOGICAL` — ξ_{KZ} calibrated, VSF semi-analytic (F2, F3)	Euclid DR1-Complete	**mission operating; DR1-Complete product not yet released at the freeze date**	mid-2027 (F3)	no deficit → refutes the **KZ network sector** in this realisation, not the framework

Observable	Signal / value	Scientific status	Instrument / channel	Instrument status	Timescale	Adverse result / scope
Short-timescale microlensing	**benchmark $t_E \simeq 1$ h for one Micro-Knot and $t_E \simeq 6.3$ h at the threshold $\lambda = 36$ (F2)**; for $\lambda > 36$, M_{typ} and $t_E \propto \sqrt{M}$ increase; $\tau_{\text{micro}} = 5 \times 10^{-8}$ *(posited benchmark, not derived; the "stellar-background level" formerly invoked was a Small Magellanic Cloud value — toward the bulge the measured background is ~1.5–1.9 × 10⁻⁶; corrected 15/10/2026)* or 1.5×10^{-7} *(pseudo-isothermal estimate, halo normalised to 0.4 GeV cm⁻³ at the Sun; corrected 15/10/2026, formerly* 1.2×10^{-7}*)*; these values assume a halo made entirely of aggregates ($f_{\text{DM}} = 1$), a case the OGLE-IV bulge survey already disfavours ($f_{\text{DM}} \lesssim 1$–2 % at M_{typ}, order of magnitude; open problem PO-F2-10, added 15/10/2026)	`CALCULATED-CONDITIONAL` — EMF chain under $\lambda \geq 36$ (F2)	Roman, Galactic Bulge survey (GBTDS, defined core survey)	**launched 30/08/2026**; arrival at L2 early December 2026, then 90 days of commissioning	first science data expected early 2027	no excess over the background of **free-floating planets (FFP, *free-floating planets*)** → refutes the **micro-knot sector**
M31 microlensing	same population seen against M31; longer t_E	`PROSPECTIVE` — conditional on an M31 survey being scheduled	Roman, M31 field	**no M31 survey among the three core surveys**; would require a general-astrophysics programme, not scheduled	no date	cannot be dated until the survey is scheduled

Observable	Signal / value	Scientific status	Instrument / channel	Instrument status	Timescale	Adverse result / scope
Astrometric microlensing	source displacement; lifts the mass–distance degeneracy	`PROSPECTIVE CAPABILITY`	Roman, astrometry of the bulge survey (GBTDS)	**astrometry available in the core survey**; a Cycle 1 analysis programme is approved for stellar remnants (no. 19083); **search for the ECF signal not scheduled**	after first data (2027 and beyond)	cannot be dated until the ECF analysis is scheduled
GW chiral asymmetry	$\Pi_{\rm GW} = 0.20 \pm 0.05$ **(CALIBRATED TEMPLATE),** amplitude-independent	`CALIBRATED TEMPLATE` — depends on the chirality channel, open (PO-ECF-ACTION)	LISA (net polarisation via the kinematic dipole or a LISA–Taiji network)	**launch planned ~2035**	verdict after commissioning and data accumulation	$\Pi_{\rm GW} < 0.02$ at 95% → refutes the **chiral tensor sector**
Macro–Macro merger burst	$f \sim 50$ mHz; rate not computed; no ECF-specific feature established (*tail τ_T withdrawn on 18/10/2026*)	`CONDITIONAL` — conditional on the Macro population existing	LISA	launch ~2035	after data	no refutation criterion (conditional channel)
Tensor spectrum	**blue enhancement of the transferred energy spectrum** through the stiff phase — *no canonical value of $n_T^{\rm prim}$ is derived in the corpus; the blue transfer must not be presented as a computed primordial tilt*	`CONDITIONAL` — passes through the stiff transfer	LISA / primordial GW background	launch ~2035	after data	a red spectrum refutes the **stiff-transfer realisation**
C_ℓ suppression	$\sim 18\,\%$ deficit at $\ell \approx 233$ ($\ell_{\rm KZ} = \chi_*/\xi_{\rm KZ}$)	`CALIBRATED` — position set by $\xi_{\rm KZ}$, calibrated (PO-F3-XI)	ACT / SPT / future CMB surveys	**DOE/NSF support for CMB-S4 ended in July 2025**; ACT and SPT operating	timescale to be reassessed	no deficit → refutes the **KZ realisation**, not the framework

Observable	Signal / value	Scientific status	Instrument / channel	Instrument status	Timescale	Adverse result / scope
High-ℓ deficit	2.5 to 6.5 %, peaking near $\ell = 2000$	`CALCULATED-CONDITIONAL` — tied to the stiff era (F1)	ACT / SPT / future high-ℓ CMB surveys	**DOE/NSF support for CMB-S4 ended (2025)**; the associated forecast has lapsed	to be reassessed	absence → refutes the **stiff-era sector**
KZ non-Gaussianity	$f_{\rm NL}^{\rm KZ} > 0$ *(sign conditional on the Poisson→ζ bridge; value open — the former CMB projection* $5/18 \times 1/3 \approx 0.09$ *is withdrawn)*	`CONDITIONAL` — the $\delta N \to B_\zeta \to$ CMB bridge is not closed	Future CMB surveys	**DOE/NSF support for CMB-S4 ended in July 2025**; the $\sigma(f_{\rm NL}) \sim 0.3$ forecast has lapsed	**not fixed**	refutes the **simplified KZ channel realisation**
Equivalence principle violation	no predicted amplitude: no relation to s_0 is established (only $m_T^2 s_0^2$ is fixed); current measurement $\eta(\mathrm{Ti,Pt}) = [-1.5 \pm 2.3_{\rm stat} \pm 1.5_{\rm syst}] \times 10^{-15}$ (MICROSCOPE, Touboul et al. 2022), consistent with zero, i.e. $\lvert\eta\rvert \lesssim$ a few 10^{-15} *(corrected 22/10/2026: "*$\eta < 10^{-15}$*" underestimated the measurement uncertainty,* $\approx 2.7 \times 10^{-15}$*)*	`CONDITIONAL TEST` — the reachable sensitivity depends on the unresolved status of (m_T, s_0)	Future space tests of the EP	**MICROSCOPE 2: concept, no fixed schedule**	**TBD**	**not a falsification criterion** *(corrected 15/10/2026: "*$\eta_{\rm ECF} \sim 10^{-15}$*" was the bound presented as a prediction, and "non-detection at* $\eta < 10^{-16}$ *→ refutes" is withdrawn)*

Observable	Signal / value	Scientific status	Instrument / channel	Instrument status	Timescale	Adverse result / scope
Dynamical dark energy	$w_0 = -0.904$ $w_a = -0.153$	`CALIBRATED` — fitted benchmark (PO-F1-B)	DESI, five-year results	operating	per publication schedule	$w = -1$ confirmed at $5\sigma \rightarrow$ refutes the **calibrated dynamical trajectory** (w_0, w_a); this does not by itself refute the static branch of the condensate, which gives $w = -1$ in the adopted effective model
Compact seeds	$M \gtrsim 10^5\, M_\odot$ at $z \sim 10$	`STRUCTURAL-HYPOTHESIS` — the causal scale is calculable, the formation of a compact object is not ($\epsilon_{\rm coll}$ open)	JWST / ELT, spectroscopy	operating	ongoing	no seeds → refutes the **macro-knot channel**
Cosmic birefringence	$\beta \approx 0.35°$	`CALIBRATED` — **non-nullity** claimed; magnitude calibrated ($g_{\rm CS}$); sign not claimed as derived	LiteBIRD, Simons Observatory	LiteBIRD planned; **DOE/NSF support for CMB-S4 ended (2025)**	per schedules	robust absence of birefringence → disfavours/refutes the **CPT realisation considered**

Observable	Signal / value	Scientific status	Instrument / channel	Instrument status	Timescale	Adverse result / scope
Structure growth	void function, S_8	`OPEN` / `CALCULATED-CONDITIONAL` depending on the observable — the historical growth calibration no longer constitutes a canonical physical prediction; a complete EC perturbation solver is lacking	Rubin / LSST	**operating**	survey ongoing	constrains or refutes the **growth realisation considered**, depending on the observable and its assumptions
21 cm acoustic ruler / BAO	the 21 cm channel constrains the **drag ruler** r_d, distinct from the recombination r_s; $r_s = 135.65$ Mpc remains the F1 baseline, while the ab initio ECF r_d falls under `PO-ECF-BAO-RDRAG`	`OPEN / CALCULATED-DIAGNOSTIC` for the $r_s \to r_d$ mapping	SKA / HERA, 21 cm	under construction	per schedule	a 21 cm measurement constrains r_d; it refutes the F1 r_s branch only after a consistent derivation of the recombination/drag mapping
GRB polarisation	Π_V/I (not derived)	`CONDITIONAL`	POLAR-2	planned	per schedule	no criterion (18/10/2026)
Homochiral	excess of L amino acids	`DETACHABLE` — conditional chain, developed in *Foundation IV*	Mars Sample Return	**programme cancelled (NASA budget, January 2026)**	**no date**	⚠ **not a kill-switch of the ECF core**: detachable channel

How to read this table. *Seven rules, set on 17/09/2026, applying to every row:*

1. **signal and instrument are two independent fields** — each row points to its source Foundation, which remains the scientific reference;
2. **a cancelled or defunded mission does not remove the signal**: it leaves the instrument and the timescale to be reassessed *(CMB-S4, DOE/NSF support ended in July 2025)*;
3. **a mission concept is not a planned mission** *(MICROSCOPE 2)* — no firm date;
4. **launch is not the start of the test, which is not the verdict** *(LISA: launch planned ~2035, verdict after commissioning and data accumulation)*;
5. **a sensitivity forecast is not an ab initio prediction**, and does not transfer from one instrument to a successor without a dedicated forecast;
6. **an adverse result refutes the stated sector or template**, and not ECF as a whole — unless a logical dependency is explicitly demonstrated;
7. **this map summarises, it is not an autonomous scientific source**: the sources are Foundation I, II

and III.

Producing scripts. *The Script column of the previous version is kept here; each row points to its source Foundation for the physics.*

Instrument	Script
LZ / XENONnT / PandaX (direct detection) *(added 08/09/2026)*	*— (structural argument, F2 §9.2; no script: this is a prediction of absence)*
Euclid DR1-Complete *(mid-2027, F3)*	`plot_euclid_pk_S8.py`, `plot_F2_vsf_prediction.py`
Roman (microlensing)	`plot_F2_topological_network.py`
LISA	`plot_ecf_gw_spectrum.py`
LISA / post-merger signal	`plot_ecf_gw_spectrum.py`
LISA / primordial GW background	`plot_F3_ecf_ns_KZ.py`
ACT / SPT / future CMB surveys *(DOE/NSF support for CMB-S4 ended in July 2025)*	`plot_F3_ecf_Cell_suppression.py`
ACT / SPT / future CMB surveys at high ℓ *(DOE/NSF support for CMB-S4 ended in July 2025)*	F1, `plot_planck_tt_residuals_ecf.py`
Future CMB surveys — *DOE/NSF support for CMB-S4 ended in July 2025, no fixed timescale*	`plot_F3_ecf_fNL_KZ.py`
Future space tests of the equivalence principle — *MICROSCOPE 2: concept, no fixed schedule*	F1, F3 physics
DESI, five-year results	`plot_w0wa_DESI_ECF.py`
JWST / ELT (spectroscopy)	`plot_ECFLRDJeansMass.py`
LiteBIRD, Simons Observatory *(DOE/NSF support for CMB-S4 ended in July 2025)*	`plot_predict_birefringence.py`
Rubin/LSST	`plot_F3_ecf_vsf_desivast_fits.py`
SKA / HERA (21 cm)	`plot_21cm_prediction.py`
POLAR-2	— (no script; F2, Appendix H: $\lvert V/I\rvert \simeq \theta_T$)
Mars Sample Return — *programme cancelled (January 2026); conditional and detachable channel, with no identified instrument*	see *Foundation IV*

The LISA test (launch planned ~2035; verdict after data accumulation), in detail: a double discriminant. LISA (networked with Taiji for chirality) can jointly constrain two observables **of distinct physical provenance in the corpus**: the stiff sector for the blue transfer of the tensor spectrum, and chirality for $\Pi_{\rm GW}$. They are not two consequences of a single dynamical origin. *(18/10/2026: the third observable, the post-merger tail $\tau_T \sim 10^6$ s of Macro–Macro mergers, is withdrawn; at the Macro-Knot radius given by Foundation II, $\tau_T \sim R/c \approx 1$–$1.4$ s, indistinguishable from the general-relativity ringdown.)* Three comparable theoretical frameworks can nonetheless be confronted on this set of observables:

Model	n_T	$\Pi_{\rm GW}$
ECF (Foundation II/III)	> 0	0.20 ± 0.05 (**CALIBRATED TEMPLATE**)
Slow-roll inflation	< 0	≈ 0
Hartle–Hawking proposal (Hertog, Janssen & Karlsson, 2023)	≤ 0	≈ 0

The ECF shares its profile with neither of the two other models on these two observables ($n_T > 0$ and $\Pi_{\rm GW} \neq 0$): a sufficiently precise measurement (LISA for n_T, the LISA–Taiji network for $\Pi_{\rm GW}$) does not merely test the ECF in isolation, it **separates** it from both compared scenarios with a single data set. Those two differ from each other only by the sign of n_T (< 0 against ≤ 0), which does not separate them cleanly.

The table distinguishes what is adjusted, what is already confronted with observation, and what remains to be settled by upcoming campaigns. The framework is **falsifiable** — several of these tests carry an explicit refutation threshold.

Prediction and refutation threshold are two distinct numbers. For the chiral gravitational-wave asymmetry, for instance, the **calibrated template gives** $\Pi_{\rm GW} = 0.20 \pm 0.05$ (**CALIBRATED TEMPLATE**); the **refutation threshold** is set much lower, at $\Pi_{\rm GW} < 0.02$ (95% C.L.). This is not a contradiction: the expected value (0.20) and the value below which the chiral tensor sector under test is refuted (0.02) differ by construction — the threshold is deliberately conservative, so as to conclude refutation only on clear failure, safe from measurement uncertainties. The same logic holds for the other tests: the prediction sets the expectation, the kill-switch sets the red line.

On the look-elsewhere effect (test multiplicity). When a programme confronts several signatures with data, the probability of obtaining at least one apparently significant result by chance increases. Here α denotes the **false-positive probability tolerated for one test** (for example $\alpha = 0.05$ for 5%). Not all the tests in the table can, however, be naively added up: some share the same calibrated parameter, and the direct non-detection of particle dark matter follows a logic of experimental coverage of the (σ, m) space rather than a uniform significance threshold. Direct detection therefore remains an important refutation channel, but its absence of signal does not have the same confirming power as tests equipped with a detection statistic. The multiplicity correction must be applied to the families of tests for which a comparable threshold α is meaningful.

Grouping by dependency family (01/09/2026). Two tests that depend on the **same calibrated parameter** are not statistically independent — a bias in that parameter would move them together. Taking each prediction and identifying, in the source text, the calibrated parameter it actually depends on (not its subject or instrument), the tests in the table group, in the partition currently used by the corpus, into **9 dependency families** *(10 until 18/10/2026: the family "τ_T (standalone)" is withdrawn with the post-merger tail)*:

Family	Tests grouped	Shared parameter (source-verified)
Birefringence / MICROSCOPE	CMB birefringence, future space EP tests	$S_{\rm ref} = 4.0456$ (F1 §birefringence; corrected 15/10/2026: 4.3237 was the value at $\Omega_m = 0.315$, replaced on 22/09; the equivalence-principle channel no longer uses $S_{\rm ref}$)
KZ network	VSF (Euclid+Rubin), C_ℓ at $\ell \approx 233$, $f_{\rm NL}^{\rm KZ}$	$\xi_{\rm KZ} \approx 60$ Mpc
Stiff era / $F_{\rm ion}$	SKA/HERA r_s, high-ℓ deficit (F1)	$F_{\rm ion} = 1.2765$
Transferred spectrum (standalone)	Blue enhancement by the stiff phase	sign alone, $w = 1$ — independent of $\Pi_{\rm GW}$ (see note [2])
$\Pi_{\rm GW}$ (standalone)	Chiral GW asymmetry	sign alone, $s_0 \neq 0$ — *explicitly* amplitude-independent (F2)
Roman (standalone)	Microlensing	$M_{\rm MK}$: **micro-knot mass** (and not Macro-knot mass); the aggregate benchmark then depends on λ
DESI (standalone)	(w_0, w_a)	independently calibrated
JWST/LRD (standalone)	Compact seeds	Hubble mass at the Great Annihilation
Mars (standalone)	Homochirality	no shared calibrated parameter

[2] *A choice that deserves justification rather than assumption*: the **blue enhancement of the transferred energy spectrum** comes from the **stiffness** of the post-bounce era ($w = 1$, which any stiff bounce would produce) whereas $\Pi_{\rm GW}$ comes specifically from its **chirality** ($s_0 \neq 0$) — two distinct properties of the mechanism, not two measurements of the same fact.

Result. With this partition into **9 dependency families**, and a usual, uncorrected $\alpha = 0.05$ threshold per family, the probability that **at least one** family appears to "confirm" by chance alone, if the ECF were false, is $1 - (1 - 0.05)^9 \approx 37\,\%$ — non-negligible. The Bonferroni-corrected threshold for the whole programme is therefore $\alpha_{\rm corrected} = 0.05/9 \approx 0.0056$: this level of stringency, not 0.05, is what should apply to any single family for the whole

programme to retain a 5% overall risk.

Limits of this calculation, not to be hidden. Bonferroni is a **conservative, simple** correction, not the most refined method (a Bayesian approach weighting each family's actual degree of independence would be more informative, but requires substantially more work). The split into 9 families reflects the choices made explicit above, not an automatic algorithm — a reader who disagrees with a grouping can recompute with their own partition. This calculation also says nothing about the robustness of each family individually (see each one's status in §5 and Appendix B): it answers only the question of test multiplicity.

What these flagship probes would specifically bring to the ECF, beyond the refutation threshold. The table above is built around adverse thresholds that constrain or refute the indicated sector or template; they do not constitute a global verdict on the ECF — but each instrument also carries a *positive* discovery potential, specific to what it could confirm:

- **Euclid** would directly probe the topological network's structure: a lensing deficit detected at scale ξ_{KZ} would be *compatible with* large voids bounded by chiral torsion walls, and would **strengthen** that reading **provided competing explanations are excluded** (selection bias, baryonic effects, alternative void models), and not mere under-densities — which would **strengthen** the hypothesis of a defect structure at cosmological scale, subject to excluding competing explanations, which would make gravitational lensing a **direct probe** of such a structure at cosmological scale.
- **Roman**, via microlensing, would test the very existence of micro-knots as a point-mass population: a detection in the benchmark $t_E \simeq 6.3$ h at $\lambda = 36$ (and longer durations if λ increases) would **favour** the geometric dark-matter hypothesis, **subject to excluding the free-floating-planet background and other short-timescale lens populations**, in the same way WIMPs should have been established for particle dark matter — with an essential distinction: the micro-knot mass is fixed in the adopted model, whereas the typical aggregate mass depends on λ, free and only bounded by HSC; the Roman scale is therefore not a prediction to the exact figure.
- **LISA** (networked with Taiji for chirality: alone, a planar detector accesses Π_{GW} only through the kinematic dipole) is the richest probe in the programme: two observables of distinct provenance are followed jointly: stiffness for the blue transfer associated with n_T, and chirality for Π_{GW}. A **joint** detection of both, with the expected behaviours in each sub-sector, would not merely be one more confirmation: it would **separate** the ECF from both standard inflation and the Hartle-Hawking-Hertog scenario on a single set of measurements. Macro–Macro mergers (**conditional on that population existing**, rate not computed) would emit in the same band, with no ECF-specific feature established *(the tail τ_T is withdrawn on 18/10/2026)*.
- **Vera Rubin/LSST**, through the depth and volume of its void and weak-lensing surveys, would play a **relay and cross-confirmation** role for the large-scale-structure tests (VSF, S_8): where Euclid opens the decisive test, Rubin brings the statistical depth to consolidate — or refute sooner — a marginal signal.
- **SKA and HERA**, via the 21 cm channel, would probe the acoustic ruler by a route complementary to optical BAO. This channel constrains the **drag ruler** r_d, and not directly the recombination r_s: comparing a 21 cm measurement with the ECF baseline $r_s = 135.65$ Mpc therefore requires a consistent calculation of z_d and of $r_d = r_s(z_d)$. This mapping remains **OPEN** under `PO-ECF-BAO-RDRAG`; no identification $r_s = r_d$ is assumed.

The great-void test (Void Size Function)

This test bears directly on Euclid data. **The complete physical chain between diffuse torsion and the void size function is not derived ab initio in the corpus.** The calculation actually used is a semi-analytic Void Size Function template modulated by a phenomenological amplitude at the Kibble-Zurek scale $\xi_{\mathrm{KZ}} \simeq 60$ Mpc, itself calibrated on void data. Architecture therefore presents this channel as a phenomenological test of the KZ sector, and not as the demonstrated consequence of a local law $\Omega_\tau \to 1$.

The falsification mechanism rests on R_{max}, the radius of the largest observed void, with a **threshold at** $\xi_{\mathrm{KZ}} = 60$ **comoving Mpc** (40.4 **Mpc/**h **at** $h_{\mathrm{fid}} = 0.6732$): if, in a survey of sufficient volume, no void were detected beyond this threshold, the Kibble-Zurek realisation of this sector would be constrained on this point. A nuance of status is called for here, in a spirit of transparency: since ξ_{KZ} is **calibrated on void data** (DESIVAST), the Euclid test is not a purely *a priori* prediction but rather a **robustness test** — it checks that the value calibrated on a first dataset (DESIVAST, small volume) holds against a second, independent and far larger one (Euclid). It is a test of extrapolative consistency, not a blind prediction; the threshold nonetheless remains a sharp constraint criterion. The state of the tests is as follows:

- **DESIVAST DR1** (June 2026) measures $R_{\max} = 31.7$ Mpc/h, but the 40.4 Mpc/h threshold is **out of reach** of the BGS sample ($z < 0.24$): too small to decide, whatever the answer. The method is validated, the verdict deferred.
- **Euclid DR1-Complete** *(mid-2027, F3)*[1], with a volume $\times 20$, is the **decisive test**: it expects to be able to probe $R_{\max} > 60$ Mpc/h (to be confirmed), confronting the prediction with the threshold at 3–5σ. If Euclid, despite this large volume, found no voids beyond the threshold (40.4 Mpc/h), their absence would constrain the KZ sector in this realisation.

  [1] *ESA split the Euclid DR1 milestone on 15/06/2026: a "Foundation" part (raw data, images, catalogues, ~1900 deg²) in November 2026, and the complete part — including the level-3 products (void catalogues, weak lensing) these tests depend on — pushed to DR1-Complete, mid-2027 (F3). The originally announced October 2026 deadline no longer applies to the tests above.*

The microscopic origin of $\xi_{\rm KZ} = 60$ Mpc is not derived from first principles — this is open problem **PO-F3-XI**. Specifically, the Kibble-Zurek dilution applied from the electroweak transition ($T_{\rm EW}$) does *not* reproduce 60 Mpc: the value (within the 50–80 Mpc range) is an **observational calibration** on the Void Size Function, not an *ab initio* prediction. The corpus identifies three resolution paths, rather than hiding the discrepancy: (i) the relevant phase transition would not be electroweak but lower (QCD ~ 200 MeV, or a torsion-specific transition) — a path **quantitatively excluded** by F3: moving down to $T_{\rm QCD}$, or even ~ 1 MeV, regains only 2 to 3 orders of magnitude out of the ~ 19 required *(corrected 19/10/2026)*; (ii) the topological network would remain locked to the causal horizon (*scaling* behaviour, standard for defect networks) until a late freeze-out — recomputed on 30/08 on the ECF background itself (rather than on a ΛCDM background, an error found and corrected that day), the comoving particle horizon reaches $\xi_{\rm KZ} = 60$ Mpc at $z \approx 5145$ — within $8.8\,\%$ of the spin-sector normalisation $z_{\rm ref} = 5600$ (the one that gives $r_s = 135.65$ Mpc; $6.3\,\%$ for the Hubble radius, reached at $z \approx 5976$; F3), and $z \approx 5145$ falls inside the HeIII→HeII recombination range ($z \sim 5000$–8000), near its lower edge, a standard thermodynamic transition independent of the ECF. *(Corrected 15/10/2026: this passage gave $z \approx 5153$ and "a factor 1.46 from the window $z \approx 7500$"; F3 corrected the anchor to $z_{\rm ref} = 5600$ on 05/09.)* The proximity remains a suggestive coincidence between a calibrated quantity ($\xi_{\rm KZ}$) and another, not evidence, and F3 states that the path is not strengthened by the horizon argument (PO-F3-XI unchanged; corrected 19/10/2026); *why* the network would freeze precisely at that epoch remains unmodelled (chiral-annihilation dynamics, PO-F2-3); (iii) $\xi_{\rm KZ}$ would remain a phenomenological parameter whose microscopic value would require a dedicated perturbation solver. This does not weaken the test's falsifiability — the 40.4 Mpc/h threshold remains a clear criterion — but it would be dishonest to present $\xi_{\rm KZ}$ as a derived result. **The origin of this scale is therefore kept among the open fronts of §7.** Finally, the corpus distinguishes two scales not to be confused: this late-time $\xi_{\rm KZ}$ (~ 60 Mpc, void regime) and a GUT-scale $\xi_{\rm KZ}$ (at the bounce, relevant for monopoles) — two distinct calculations, unrelated to each other.

Early galaxies (JWST)

The James Webb Space Telescope observes, at high redshift, galaxies more massive and more numerous than standard hierarchical structure formation predicts — an excess that grows as surveys (JADES, CEERS, GLASS) extend to higher z. In the ECF, under the structural hypothesis, structure seeds (macro-knots) would be compact objects with a scale tied to $M_H(T_{\rm GA})$; their formation is not derived ($\epsilon_{\rm coll}$ open), and not the late outcome of hierarchical gravitational collapse: if a macro-knot population forms (structural hypothesis), massive seeds at high z would be a **conditional** consequence of it, rather than an anomaly to be explained after the fact *(corrected 19/10/2026)*.

The "Little Red Dots" (LRD) case. A distinct, compact, red population of high-z sources (observed mostly at $z \sim 4$–9, spectroscopically confirmed up to $z = 9.29$) shows obscured active-galactic-nucleus (AGN)-like signatures — one of the most puzzling discoveries of the early JWST surveys, with no established consensus on its origin. The standard astrophysical explanation invokes rapid hyper-accretion onto stellar remnants. The ECF proposes a non-exclusive alternative reading: primordial macro-knots ($M \sim 10^5$–$10^6\,M_\odot$), already present as structure seeds before any gravitational collapse, could serve as gravitational anchor points for these compact engines, the obscuring cocoon arising from rapid baryonic infall onto the topological seed — a **conditional** seeding channel for the LRD mass range — *conditional because the formation of a compact macro-knot is not derived, and the fraction of the horizon mass converted into an object remains an open problem* — see §7.

Seed growth, comparison with UHZ1 and J1342+0928 and falsification criterion: synthesis table of §2.

Status: the qualitative comparison is compatible under the structural hypothesis. The Planck bound on early accretion (PO-F2-SEED-CMB) constrains the abundance: under Eddington accretion from recombination, the adopted abundance ($\mathcal{O}(1)$ macro-knot per galactic volume, not derived) exceeds the bound by a factor ≈ 1.5 to ≈ 70, while the LRD abundance remains allowed *(added 19/10/2026)*. Moreover, a quantified test now exists (compact seeds $M \gtrsim 10^5\, M_\odot$ at $z \sim 10$, JWST/ELT — see the future-instruments table above), but no parameter is calibrated specifically on this data, and the full quantitative confrontation (the high-z galaxy mass function, and its precise overlap with the LRD mass function) remains to be carried out. *Two recent results narrow, without settling, this comparison*: decomposing the LRD Hα profile into a narrow Doppler core and electron-scattering wings revises the naive virial mass of a typical object downward by roughly two orders of magnitude ($\sim 10^8\, M_\odot \rightarrow\sim 10^6\, M_\odot$), a value landing at the upper edge of the macro-knot window rather than far above it; and a stacking analysis of 217 LRDs (COSMOS-Web) detects faint extended emission consistent with a compact host galaxy (~ 200 pc, stellar mass $\sim 10^9\, M_\odot$), ruling out a fully isolated supermassive black hole. Neither constitutes a confirmation of the ECF channel — the reference explanation (standard plasma cocoon) remains operative in both cases — but both point toward a converging narrowing of the gap to be closed. *(Added 23/10/2026)* **What the ECF does not predict**: no redshift, no redshift window and no number-density evolution for LRDs. Observed LRDs lie mostly at $z \sim 4$–9, their number density falls steeply below $z \approx 4$, and local analogues are extremely rare; why LRD-like phases would concentrate at these epochs is not derived (no accretion timescale, no duty cycle). The comparison concerns mass scale and abundance, not redshift behaviour.

6. Positioning: the ECF alongside other models

Comparing the ECF with other scenarios only makes sense if its theoretical costs and its dependencies remain visible. This section therefore places the mechanisms side by side without turning a calibration or a conditional signature into a demonstrated advantage.

To place the framework in the theoretical landscape, here is how it compares to the main serious alternatives. These tables are taken from the comparisons established in Foundation I (tension sector) and Foundation II (dark sector).

Dark sector: positioning without a scorecard. Foundation II stresses that WDM, SIDM, FDM and the ECF geometric sector answer different physical questions and must not be turned into competitors on a single grid of "problems solved". Architecture therefore keeps here only the necessary distinctions: WDM/SIDM/FDM introduce additional matter degrees of freedom, whereas the ECF explores geometric torsion structures. In the ECF, the micro-knot mass is tied to the adopted electroweak scale, but the aggregate population depends in particular on λ, **free and not derived**, and the scale ξ_{KZ} is calibrated. $H_0 = 73.04$ is the **output of a calibrated chain**, not a parameter-free prediction; $S_8 = 0.766$ is the KiDS-1000 target, the ECF growth remaining OPEN *(corrected 19/10/2026)*. Roman microlensing constitutes a conditional test of the micro/aggregate population; it is not sufficient to establish a general hierarchy between dark-matter models.

Tension sector: ECF and early dark energy (EDE). It was the early-dark-energy programme (Poulin, Smith, Karwal and Kamionkowski, 2019) that established the founding idea of this family of solutions: modifying physics *before* recombination to reduce the sound horizon r_s and raise the inferred H_0. The ECF stands in this "*early-time*" lineage and owes it its conceptual framework; it differs in the mechanism invoked. EDE remains the most studied solution and the natural reference point: like the ECF, it reduces r_s and raises H_0 through a pre-recombination modification. The two approaches thus share the same **phenomenological lever** — modifying pre-recombination physics in order to reduce r_s — but **not the same physical mechanism**; their contents and their treatment of S_8 are summarised below.

Criterion	EDE (axion)	ECF (spin fluid)
H_0	can be raised depending on the calibration of the EDE sector	$H_0 = 73.04$ in the **calibrated** background chain; not an ab initio prediction; conditional on θ_* (`PO-F1-THETA`)
S_8	in tension with LSS in some EDE realisations	ECF: **physical prediction OPEN**; KiDS-1000 target 0.766, CAMB A-7 reference 0.780

Criterion	EDE (axion)	ECF (spin fluid)
r_s reduction	yes	yes ($r_s = 135.65$ Mpc)
Equation of state	transient scalar field: $w \simeq -1$ during the relevant EDE phase, but not identically -1	$w_0 = -0.904$, $w_a = -0.153$
Added degree of freedom	scalar field	torsion (geometric)
N_{libre} (tension sector)	3 (m_{EDE}, f_{EDE}, θ_i)	**historical count from the F1 test: 2** (F_{ion}, γ_{spin}); in the active baseline, the physical γ_{spin} is **OPEN**, so this "2" is not a canonical count of predictive parameters

The two frameworks answer the same question by distinct routes: EDE through a transient scalar field, the ECF through a geometric torsion contribution. Tensions with large-scale-structure data are documented for some EDE realisations and depend on the model and on the datasets considered; the ECF approaches this sector differently, but its own physical growth prediction remains **OPEN** as long as a complete EC perturbative solver is missing. This table establishes no hierarchy: it gives the reader the elements to place the two approaches side by side.

7. The major scientific locks of the ECF

This section gathers the **questions that condition the reach of the programme**. They are deliberately brought together here so that a reader can distinguish, in a single view, what belongs to an already calculated or calibrated result and what still requires a physical derivation. The preceding sections refer to it when a result depends directly on one of these locks. **None of these points is closed by the writing of Architecture.**

The ten cross-cutting locks

Lock	Question still open	Reach in the corpus
O1 — `PO-ECF-ACTION`	Derive, from $S_{\text{EC+Dirac}}$ or from an explicitly specified extension, the effective action Γ_{eff} containing the kinetic, gradient and potential terms needed by the dynamical torsion sectors.	Cross-cutting lock of the propagating axial sector, of the chiral dynamics, of spatial Kibble-Zurek and of several transfers to the observables. `PO-F2-6` is a sub-problem of it, not a substitute.
O2 — microphysics of the bounce	Connect the effective description of the bounce to a complete microphysics beyond the ECKS / spin-fluid reduction used in the corpus.	Conditions the fundamental interpretation of the bounce, without cancelling the effective calculation presented in §4.1.
O3 — bridge between the two spin branches	Establish the link between the repulsive high-density ECKS correction and the positive branch used in Foundation I, without identifying the two by mere notation.	Prevents turning a proximity of variables into a dynamical derivation.
O4 — topology and defect network	Derive the vacuum manifold and the topological structure that would establish which 0D/1D/2D defects are actually produced, as well as their spatial network.	Affects `PO-F2-8`, the 1D sector, the connectivity of the chiral walls, horizon/isotropy and the Kibble-Zurek interpretation.

Lock	Question still open	Reach in the corpus
O5 — perturbations and axial pole (`PO-F2-6`)	Identify the physical axial degree of freedom: unitary pole, kinetic term, relation between coefficient mass, pole mass, static range and cosmological mode.	The static background $w = -1$ remains distinct from this question; the perturbations and some dynamical observables depend on the effective choice, notably on the Proca form adopted in some calculations.
O6 — dark matter / structures chain	Establish quantitatively the chain between micro-knots, aggregates and macroscopic structures, and derive the formation/collapse of a Macro population.	The Micro $\rightarrow$ Macro coagulation remains **WITHDRAWN**; $\lambda \geq 36$ is an HSC bound/threshold value and not a dynamical ceiling; ϵ_{coll} remains open. The question "what structures the chiral walls?" also remains unresolved.
O7 — compact objects and child universes (`PO-F3-1`)	Demonstrate, beyond the adopted effective model, the existence, stability and complete evolution of the compact cores and of the child-universe channel.	Sector carried mainly by *Strong Gravity* / F4; **STRUCTURAL-HYPOTHESIS / OPEN**, outside the observational core of Architecture but dependent on the same high-density chain.
O8 — complete baryogenesis	Go from the semi-derived result to a complete kinetic chain, including the Boltzmann treatment and the ab initio provenance of the still-calibrated normalisation.	The current calculation gives $\eta \simeq 5.9 \times 10^{-10}$ from a pre-dilution asymmetry normalised on the observed value: the agreement is not an independent test and does not close the complete microphysics.
O9 — chirality and global memory (`PO-F3-3`, Strong Gravity / F4, not deposited)	Derive the global chirality bias and determine whether a permanent chiral memory survives the transitions and the cosmological evolution.	Conditions several parity signatures, notably CMB/GW, without turning their conditional calculation into a proof of the mechanism.
O10 — transfer to the observables	Connect the effective sectors to the observables through a complete perturbation theory and controlled transfer functions.	The dedicated Einstein-Cartan Boltzmann solver remains to be built; several downstream signatures are therefore conditional tests rather than independent validations.

Explicitly tracked open fronts

Several problems moreover have their own code in the Foundations and must remain visible even when they attach to the cross-cutting locks above:

- `PO-F3-ISO` — **isotropy:** the complete resolution of primordial isotropy is not established; the window calculated in the F1 scheme is not sufficient to establish it.
- `PO-F3-NS` — **primordial spectrum:** n_s is not predicted by the ECF; it is treated as an inherited input. The *stiff* era produces a blue component and a bound on its duration, but not the observed value of n_s.
- `PO-F3-XI` — **Kibble-Zurek scale:** $\xi_{\text{KZ}} \simeq 60$ Mpc is calibrated on the void size function; its microscopic origin remains underived.
- **Macro formation / ϵ_{coll}:** $M_H(T_{\text{GA}})$ provides a **CALCULATED-CONDITIONAL** causal scale, not the demonstrated mass of an object; the collapse fraction and the formation of a compact Macro-knot remain open.
- **R-13 / I-2 — chiral walls:** the corpus has not yet identified in a demonstrated way whether the walls are structured by the micro-knots, their aggregates, the Macro population or the defect network itself. This point remains **blocked** as long as its physical producer is not found.

Named open problems of Foundation II to keep explicitly

- **PO-F2-2** — **helical pitch:** the pitch length of the 1D sector requires solving the cylindrical field equations; it is not derived.
- **PO-F2-4** — **fluid / fluctuation partition:** the fluid fraction and the normalisation of the axial fluctuations are not derived *ab initio*; this lock overlaps the perturbative treatment and **PO-F2-6**.
- **PO-F2-5** — **lock of the BAO/DESI sector:** the active observational cost lies in the BAO/DESI sector; this problem remains **OPEN** and distinct from the age, without being carried by an age deficit.
- **PO-F2-7** — **formation of compact objects / hedgehog–Kerr-ring channel:** the formation dynamics is not established.
- **PO-F2-9** — **detection by deflection:** channel kept as an open problem, not as an established prediction; the geometric assumptions and the experimental reach remain to be consolidated.
- **Lithium-7:** the possible interpretation of the anomaly through late decay of topological defects is a deferred lead, **not an established result** of the current corpus.

Post-audit v2 completeness check

This section serves as a documentary safeguard: it creates no new result, but checks that the conclusions of audit v2 that condition the reading of the corpus remain visible in Architecture.

- **O1 — effective action / complete PGT: OPEN.** The Proca, Strong Gravity and Chiral Bounce reductions remain effective models as long as their derivation from the complete PGT action is not established (PO-ECF-ACTION).
- **O2 — microphysics of the bounce: OPEN.** The bounce is calculated in the adopted ECKS reduction; isotropisation and extension beyond the effective spin fluid remain to be closed. PO-ECF-ACTION-BOUNCE separately tracks the underived correspondence between the quadratic scheme of F1 Appendix H and the linear microscopic spin term.
- **O3 — matching and normalisation of spin: OPEN.** PO-F1-SIGMA0, PO-F1-SSC, PO-ECF-SPIN-NORM and PO-ECF-BBN-SPIN prevent identifying the F1/F2/F3 normalisations without proof. BBN compatibility is therefore not demonstrated. Quantitatively (23/10/2026): extended as a^{-6} to $T \sim 1$ MeV, the calibrated Foundation I branch gives a spin-to-radiation ratio of 10^9–10^{11}, which BBN excludes, while the sound-horizon calibration needs that law up to $z \sim 10^5$; the Foundation I and II normalisations differ by about 22 orders of magnitude (Foundation I, *Domain of validity of the calibrated branch*).
- **O4 — vacuum and topology: OPEN.** The chain action → effective potential → vacuum manifold → homotopy → defects is not derived in full (PO-F2-8). The old global scenario associated with (N≃60) is HISTORICAL/WITHDRAWN; the external upper bound of (0.46) e-fold is not an ECF bound. The (N≃0.153) e-fold window of the F1 scheme remains CALCULATED-CONDITIONAL, with no demonstrated bridge to the microscopic spin (PO-ECF-ACTION-BOUNCE).
- **O5 — axial spectrum: OPEN.** PO-F2-6 requires establishing the possible identification between coefficient mass, pole mass, static mass and cosmological mass, with Hessian, constraints, poles, residues, ghosts/tachyons and mixings.
- **O6 — dark matter / structures: OPEN.** The chain (ξ/KZ}→n{KZ}→λ→R_{coal}→) population is not closed (PO-F3-C6). For SPARC, the adopted effective density profile (ρ(r)=ρ_0/[1+(r/R_s)^2]) is integrated into (M(r)), then the Newtonian relation (V^2(r)=GM(r)/r) leads to the velocity law used in the fits (Eq. 70): this derivation is therefore not OPEN. PO-ECF-HALO bears only on (i) the ECF microphysical justification of this density profile and (ii) the quantitative derivation connecting the properties of the geometric sector / of the micro-knots to the fitted parameters ((*ρ0,R_s)). The (?*{KZ}) procedure remains PO-F3-XI.
- **O7 — compact objects / child universes: OPEN.** The Strong Gravity results are conditional on the effective model; stability/TOV/collapse/junctions remain to be established (PO-F3-1).
- **O8 — complete baryogenesis: OPEN.** The result is SEMI-DERIVED; the normalisation and the complete kinetic network remain open, notably PO-F2-NU.
- **O9 — chirality / memory: OPEN.** PO-F3-3 must distinguish origin of the sign, initial amplitude, dilution/thermalisation, conservation of memory and transfer to the observables.
- **O10 — perturbations / observables: OPEN.** A controlled Boltzmann/CLASS-EC chain remains necessary for (C_ℓ), (P(k)), growth and (S_8). The KiDS value (S_8=0.766) is an observational target; (S_8^{ref}≃0.780) is a

conditional CAMB reference, not an ECF prediction.

Complementary observational checks of audit v2. Roman microlensing must use the corrected durations (t_E≃1,{h}) for a Micro-Knot and (≃6.3,{h}) for (λ=36) (F2). The GW polarisation (Π_{GW}=0.20±0.05) is a **CALIBRATED TEMPLATE**, not an ab initio prediction. The proposed links with ($\Delta\alpha$) remain conjectural. The BAO comparisons remain **CALCULATED-CONDITIONAL** as long as `PO-ECF-BAO-RDRAG` is OPEN.

How to read the results that depend on them

These locks do not mean that all the results of the corpus have the same degree of uncertainty. They impose a reading rule: **a result calculated in an effective sector remains calculated in that sector, but it must not be promoted to a fundamental derivation as long as the corresponding upstream lock remains open**. Likewise, a downstream signature does not constitute an independent validation of the mechanism on which it depends.

This separation is the raison d'être of the statuses used in Architecture: **DERIVED**, **CALCULATED-CONDITIONAL**, **CALIBRATED**, **PHENOMENOLOGICAL**, **CONDITIONAL**, **STRUCTURAL-HYPOTHESIS**, **OPEN** and **WITHDRAWN**.

Mirror check of the primary papers

This version was re-swept against **PIT v8 revised (preprint v9), Foundation I rev. 13 (preprint v8), Foundation II rev. 20 (preprint v5), Foundation III — Cosmological Physics rev. 1.1 (preprint v3), and the undeposited documents Strong Gravity rev. 2.1, Chiral Bounce rev. 0.4 and Foundation IV rev. 0.3** — internal revisions of October 2026, public Zenodo numbers in parentheses. Architecture does not silently settle divergences between papers: it flags them. In particular, (i) F2 and F3 use two Poisson descriptions that do not produce the same amplitude of primordial fluctuations; the matching remains OPEN (S-28); (ii) the provenance of ξ is contradictory in F2 relative to F1 (S-29); (iii) the arithmetic check of ω_ν (S-37) was settled by A-6 / DEC-07D and integrated into the frozen baseline; no rerun of the r_s solver is active; (iv) the uses of $\Omega_m = 0.315$ (**HISTORICAL / Planck reference; active:** 0.2669) and of the consistent budget $\Omega_m \simeq 0.268$ must remain distinguished (S-38).

For *Chiral Bounce*, Architecture explicitly follows the paper's status table: the order parameter, the potential coefficients and the interpolating solution are **derived within the construction**, but the whole constitutes a **conditional theorem** on the existence of the ECKS bounce; the global $O(1)$ prefactor, the inhomogeneous $k \neq 0$ modes and the complete TOV-torsion solution remain open. For *Strong Gravity*, the bound $\rho < \rho_c/2$ of a bound static core is derived in the adopted effective model, while the Hayward metric is modelled and the complete derivation of the core from the ECKS action remains OPEN.

Points subject to author arbitration before the FR freeze

The post-audit v2 pass replaces the former S-29/S-37/S-38 arbitrations with the baseline decisions: $\xi = 1.645$ is CALIBRATED on the DESI preference used by the ECF; $\omega_\nu = 1.71061 \times 10^{-5}$ and $\Omega_r = 7.8417 \times 10^{-5}$ are propagated; $\Omega_m = 0.2669$ is the active A-7 budget, while 0.315 remains the Planck/historical reference. **S-28**, the matching between the two F2/F3 Poisson fluctuation calculations, remains OPEN.

8. Conclusion

The ECF is a cosmological programme built around a geometric idea: in Einstein-Cartan theory, the spin of matter can modify gravitational dynamics at extreme densities. The corpus explores the consequences of this modification from the primordial bounce to the late-time observables, combining this geometry with known mechanisms of phase transition, defect formation and observational cosmology.

The framework today contains several levels of results. Some follow directly from the adopted model, such as the bounce mechanism in the ECKS sector considered. Others rest on parameters calibrated on data, notably in the H_0/S_8 sectors and in the late-time evolution of dark energy. Finally, several signatures — in particular those that depend on the Macro population, on the axial dynamics or on certain topological defects — remain conditional.

The decisive question is therefore not whether all the elements of the programme are already derived: they are not. It is whether the calculated and calibrated sectors lead to tests that can fail. Architecture lists these tests and, with the same importance, the locks that still prevent turning the programme into a complete dynamical construction. **Their synthesis is given in §7.**

For the reader who wishes to check each number, equation or dependency, §4 and the appendices give the provenance paths. For the reader who wishes to judge the reach of the programme, the distinctions between **calculated**, **calibrated**, **conditional**, **structural hypothesis** and **open** are the ones that are authoritative.

Critical remarks are welcome on the whole corpus. Architecture or any ECF paper can be annotated directly; the feedback procedure is described in **Appendix G**.

Appendices

*The appendices gather what documents the corpus rather than the physics: its map, its nomenclature, its references and its traceability, as well as the way to comment on **Architecture or any other paper of the corpus**. They are placed after the scientific argument so as not to interrupt it.*

Appendix A. Map of the corpus: scope of each document

> **Purpose.** This map states, for each document of the ECF corpus, its exact scope, its audience, its status, and the topics it covers. It lets a reader find immediately *where* a question is developed, without going through the whole. It rests on governance rules applied across the corpus (below) and on the four-audience split established for Foundation III.
>
> **State: 22 October 2026** (updated), after finalising the structuring of Foundation III, restructuring the split documents into paper form, cleaning the bibliographies (uncited references removed from the entire split corpus) and checking references and external values at source (22/10/2026).

The governance rules

1. **Zero duplication.** A result is developed in a single document. The others refer to it rather than copying it.
2. **Targeted cross-reference.** Every cross-reference points to the document *and* the section where the topic is actually treated — never to a document that lacks it.
3. **Scope fixed in advance.** Each document carries a charter ("covers / excludes") at its head that delimits its content; anything out of scope goes to the appropriate document.

Overview: the full chain

```
PIT Letter (postulate: exact flatness, 6 pp.)
    |
    ├──► Foundation I  (cosmological background: H0, S8, sound horizon)  v8, 87 pp.
    |
    ├──► Foundation II (geometric dark sector)                           v5, 173 pp.
    |
    ├──► Foundation III ──► 3 documents:
    |         ├── Cosmological Physics  v3,   41 pp. (testable physics, referees)
    |         ├── Strong Gravity        r2.1, 37 pp. (Popławski; undeposited)
    |         └── Chiral Bounce         r0.4, 21 pp. (Barrau / LQG; undeposited)
    |
    └──► Foundation IV (Fertile)  r0.3, 64 pp. (exploratory; undeposited)
```

PIT Letter — the postulate, and the question it leaves open

Status: deposited (v8, 6 September 2026); v9 in the October 2026 deposit (corrected Mahalanobis distance). 6 pages. **Audience**: all. **Covers**: the Topological Invariance Principle — spatial flatness is **exact** ($\Omega_k \equiv 0$), a **falsifiable** claim; and the **open problem** of the $\Omega_{\text{spin}} \leftrightarrow \Omega_\tau$ link. *(**Static** dark energy is derived from the action — F2, Appendix J — independently of this postulate.)* **Excludes**: any detailed cosmological development (→ F1), any dark sector (→ F2).

Foundation I — the cosmological background

Status: deposited (Zenodo v7, 14 September 2026); v8 in the October 2026 deposit (internal revision 13). 87 pages. **Audience**: cosmologists, referees. **Covers**: Einstein-Cartan-Kibble-Sciama framework; sound-horizon reduction ($r_s = 135.65$ Mpc); joint H_0/S_8 treatment via F_{ion}; numerical results and optimisation (former $\Delta\chi^2 = -39.5$, **HISTORICAL / NOT RECALCULATED after A-7**); DESI/SH0ES/KiDS confrontation; dark-energy sector (w_0, w_a) with status PO-F1-B; **cosmic birefringence mechanism** β_{ECF} (rotation of the linear CMB polarisation plane). **Excludes**: baryogenesis and geometric dark matter (→ F2); the high-density regime / exploratory (→ F3 / F4).

Foundation II — the geometric dark sector

Status: deposited (Zenodo v4, 15 September 2026); v5 in the October 2026 deposit (internal revision 20). 173 pages. **Audience**: cosmologists, referees. **Covers**: chiral baryogenesis (Sakharov, $\eta_{\text{ECF}} \approx 5.9 \times 10^{-10}$, to be compared with $\eta_{\text{obs}} \approx 6.1 \times 10^{-10}$); geometric dark matter (micro-knots via Kibble-Zurek); topological zoology of defects; macroscopic structure (JWST, cosmic web); predictions for future missions; **circular polarisation of ORCs** Π_{circ} — **conditional** synchrotron signature: the Foundation II calculation assumes the existence of a Macro-Knot and does not establish this population independently. **Excludes**: the base background (→ F1), the high-density regime (→ F3). *Note: **Appendix C of Foundation II** ("Cosmic Inflation in the ECF") is deprecated: its kinematic estimate is kept for traceability but is no longer a current result of the corpus; the corresponding kinematic route is closed in the current state of Foundation III.*

Foundation III — Cosmological Physics

Status: deposited (Zenodo v2, 15 September 2026, 10.5281/zenodo.22359982); v3 in the October 2026 deposit (internal revision 1.1 revised). 41 pages, 40 references. **In English**, bilingual keywords. **Audience**: referees, Zenodo/arXiv deposit (intended for cosmologists and specialist readers of the corpus). **Covers**: BBN constraint; DESI DR1/DR2 comparison (full covariance); spectral index n_s: status of the primordial spectrum (tilt inherited, not predicted; bound $N_{\text{stiff}} \gtrsim 68$); VSF threshold vs DESIVAST (robustness test); PO-F3-ISO (isotropy); f_{NL} non-Gaussianities of the KZ network; primordial tensor spectrum; structured table of kill-switches; positioning vs Hartle-Hawking-Hertog. **Paper structure** (6 level-1 sections). **Excludes**: detailed Landau-Ginzburg (→ Chiral); the geometry of compact objects (→ Strong Gravity); the exploratory programme (→ F4).

Foundation III — Strong Gravity

Status: revision 2.1 + CORE-01, 37 pages, undeposited. In French. **Audience**: Popławski, strong gravity. **Subject**: geometry of compact objects and the high-density regime. **Outside the scope of this document** — refer to the document itself.

Foundation III — Chiral Bounce

Status: revision 0.4, 21 pages, undeposited. In English. **Audience**: loop quantum gravity community. **Subject**: irreducible decomposition of torsion and the Landau-Ginzburg potential of the bounce. **Outside the scope of this document** — refer to the document itself.

Foundation IV — Fertile

Status: revision 0.3, 64 pages, undeposited. In French. **Exploratory programme**, acknowledged as such by the author. **Outside the scope of this document**, which covers the cosmology confrontable with data. *Listed so that the map of the corpus is complete.*

Two observables with similar names — never to be confused

Two polarisation signatures, real and distinct, carry close names. Any index or summary must specify the symbol and the document:

- β_{ECF} (Foundation I) — cosmic birefringence: rotation of the **linear** CMB polarisation plane by torsion-photon Chern-Simons coupling. Targets: LiteBIRD, ACT DR6, Planck PR4. Status: *consistency check* calibrated on Minami-Komatsu. *(The mechanism is established in Foundation I; Strong Gravity uses it to constrain the spin a_0^*, referring to Foundation I.)*
- Π_{circ} (Foundation II, Appendix F) — **circular** polarisation fraction of ORCs (Odd Radio Circles), **CONDITIONAL** synchrotron signature: the calculation assumes the existence of the Macro-Knot population. For ORCs, the synchrotron Π_{circ} is of order 10^{-10} and is not a realistic target for the planned instruments; the discriminant retained is rather the **coherent alignment** (ASKAP / ngVLA). The circular polarisation of gamma-ray bursts, $|V/I| \simeq \theta_T$ with θ_T not fixed by the framework, is a distinct and **conditional** observable, accessible to **POLAR-2**; no threshold is attached to it *(corrected 19/10/2026)*.

Inverse thematic index ("who covers what")

Topic	Document
H_0, S_8, F_{ion}	Foundation I
(w_0, w_a), PO-F1-B	Foundation I
Linear birefringence β_{ECF}	**Foundation I** (mechanism and main calibration); Foundation II (reuse in the dark sector); Strong Gravity (use)
Baryogenesis (η)	Foundation II
Dark matter (micro-knots)	Foundation II
Circular ORC polarisation Π_{circ}	Foundation II
n_s, primordial spectrum (inherited), N_{stiff} bound, PO-F3-NS	**Cosmological Physics**
DESI DR1/DR2, BBN	**Cosmological Physics**
VSF / Euclid kill-switch	**Cosmological Physics**
f_{NL}, blue tensor spectrum	**Cosmological Physics**
Landau-Ginzburg, parity breaking	**Chiral Bounce**
Experimental spin-torsion detection	**POC (separate, never cited in cosmology)**

Appendix B. Complete nomenclature (symbols, values, sources)

To allow a reader to reconstruct everything, here is the full set of the document's symbols, with their definition, numerical value and the origin of that value. Symbols without a numerical value are those that remain algebraic (defined by a relation, not by a number).

Fundamental constants (CODATA)

Symbol	Definition	Value
G	gravitational constant	6.674×10^{-11} m^{3}kg^{-1}s^{-2}
c	speed of light	2.998×10^8 m s^{-1} (exact)

Symbol	Definition	Value
$\hbar$	reduced Planck constant	1.055×10^{-34} J s

ECF-proper parameters (derived / from corpus)

Symbol	Definition	Value	Source
κ	torsion-spin coupling	$\kappa^2 = 2.0766 \times 10^{-43}$ m J^{-1}	**corpus** F2 §Topological Zoology
$\Omega_{\text{spin}}(z_{\text{ref}} = 5600)$	spin/radiation normalisation at the reference redshift *(F1 notation: $\Omega_{\text{spin}}^{\text{peak}}$)*	0.093 ± 0.004 *(profile interval, other calibrated constants held fixed; corrected 19/10/2026)*	**CALIBRATED** by profiling on the acoustic target r_s; provenance traced, not derived from the ECKS microphysics. The physical relation $F_{\text{ion}} \leftrightarrow \Omega_{\text{spin}}$ remains **OPEN / UNTRACED**
w_0	dark energy, present value	-0.904	**calibrated**: empirical identification with $-1 + \Omega_{\text{spin}}$, not derived; amplitude inherited from Ω_{spin} (F1)
w_a	dark energy, evolution	-0.153	**calibrated** via ξ (F1, PO-F1-B)
ρ_c	Cartan density (bounce threshold)	5.15×10^{96} kg m^{-3}	**derived** $c^5/\hbar G^2$, F2 §Fundamental Scales
M_{MK}	micro-knot mass (⚠ *indexed `MK` since 12/09/2026; earlier versions used `\mu K`, which reads as "microkelvin" and sat next to CMB quantities quoted in μK*)	$\approx 3.74 \times 10^{24}$ kg	**calibrated** (centre of KZ window; order set by $c^3/2GH_{\text{EW}}$), F2 §Geometric DM
m_T	axial torsion mass	algebraic: $m_T^2 s_0^2 \simeq 1.2 \times 10^{-46}$ GeV^4	**corpus** F2 §Vacuum Condensation
s_0	torsion field VEV	algebraic: $s_0^2 = m_T^2/(\lambda_T/3!)$	**corpus** F2 §Vacuum Condensation
λ_T	axial self-coupling	> 0 (potential parameter)	**corpus** F2 §Vacuum Condensation
σ_0	spin density	algebraic: $\sigma^2 = \frac{1}{2}\langle S_{\mu\nu} S^{\mu\nu} \rangle$ — **not numerically evaluated** (PO-F1-SIGMA0)	**corpus** F1 §Theoretical Framework
S_μ	axial torsion vector	$S_\mu = \frac{1}{6}\varepsilon_{\mu\nu\lambda\rho} T^{\nu\lambda\rho}$	**corpus** F2 §Chiral Baryogenesis
z_{ref}	normalisation redshift of the spin sector	5600 (the value in `script_01_solve_sound_horizon.py`; it is what reproduces $r_s = 135.65$ Mpc)	**corpus** F1
H_{EW}	Hubble rate at the electroweak scale	fixes M_{MK} (horizon mass)	**corpus** F2 §Geometric DM

Symbol	Definition	Value	Source
η	baryon asymmetry	6.1×10^{-10} (observational value inferred from the Planck 2018 parameters); 5.9×10^{-10} (ECF, **semi-derived**, 3.5 deviation)	**external** (observation) / **semi-derived** by F2: universal chiral coupling from first principles (Dirac algebra, no gauge dependence); pre-dilution asymmetry normalised on the observed value ($\eta_{\text{raw}} = \eta_{\text{obs}}/0.50$; F_{ion} not used); survival fraction $f \simeq 0.48$ semi-analytic; agreement not independent (19/10/2026); calculation reproduced by `Scripts_F2/derive_f_dilution_baryogenesis.py`
F_{ion}	effective amplitude of the background/acoustic sector; it enters the effective parametrisation of r_s and G_{eff}, but **does not by itself constitute a prediction of** S_8	1.2765 ± 0.0246	**CALIBRATED** — F1. The former construction by H_0–S_8 intersection (`plot_Optimization_Intersection.py`) is **HISTORICAL**; physical growth and S_8 remain OPEN in the active baseline
γ_{spin}	growth-torsion coupling efficiency (S_8 sector)	0.3116 ($\approx 0.31^{+0.07}_{-0.10}$ by inversion on S_8^{obs}; former ± 0.04 withdrawn by F1)	**HISTORICAL CALIBRATION**; conditional reference $\gamma \simeq 0.066^{+0.069}_{-0.094}$; physical value OPEN
n_s	scalar spectral index	**not predicted** — inherited input; bound $N_{\text{stiff}} \gtrsim 68$	PO-F3-NS, restated

External parameters (non-ECF)

Symbol	Definition	Value	Source
H_0	Hubble constant	73.50 ± 0.81 (km/s)/Mpc	H_0 Distance Network (also 73.04 SH0ES fit)
$\Omega_{m,0}$	active ECF matter density	0.2669	A-7; 0.315 = Planck/historical reference

Decision A-7 — closure of S-38. The active ECF budget adopts $\Omega_m = 0.2669$ with $h = 0.7304$ and physical densities held fixed. $\Omega_m = 0.315$ remains a Planck/historical reference, not the active ECF closure. Earlier global results that depended on 0.315 are HISTORICAL until they are recomputed. | $\Omega_{r,0}$ | radiation density | 7.8417×10^{-5} **active** at $h_{\text{ECF}} = 0.7304$; 9.22×10^{-5} at $h = 0.6736$ = **HISTORICAL / Planck reference** conversion of the same ω_r; S-37 SETTLED | F1 + A-6 / DEC-07D + A-7 |

S-37 — SETTLED. The correction of ω_ν is integrated into the frozen baseline. With $h_{\text{ECF}} = 0.7304$, the active value is $\Omega_{r,0} = 7.8417 \times 10^{-5}$. The former conversion at $h = 0.6736$ remains **HISTORICAL**.

No arbitration request or solver rerun is active in Architecture.

What remains algebraic or open

Three quantities receive **no** closed numerical value in the published corpus, and this must be said:

- m_T**,** s_0**,** λ_T individually: only the product $m_T^2 s_0^2$ is fixed, **by normalisation on the observed dark-energy density** (as Λ is in ΛCDM). Lifting this degeneracy belongs to the microphysics of torsion, not yet constrained. ⚠ m_T **sector: laboratory bound and open spectral identification.** The halo bound $m_T \gtrsim 6.4\times10^{-28}$ eV and the degeneracy $m_T^2 s_0^2 = 1.2\times10^{-46}$ GeV^4 give $s_0 \lesssim 1.4\times10^{-6}\, M_{\rm Pl}$ (**non-reduced** Planck mass, 1.22×10^{19} GeV; $7.0\times10^{-6}\, M_{\rm Pl}$ with the **reduced** mass), The laboratory bound on Lorentz violation is far stronger: under minimal coupling to fermions, $s_0 \lesssim 3\times10^{-27}$ GeV, i.e. $m_T \gtrsim 4\times10^3$ GeV ($\gtrsim 6\times10^4$ GeV including the motion of the Sun relative to the CMB); it is compatible with the halo bound and the degeneracy if m_T is at least at the multi-TeV scale, but the quartic potential then gives no perturbative origin for the VEV ($\lambda_T \gtrsim 10^{61}$): the origin of the VEV under minimal coupling is an open point (PO-ECF-ACTION). The former illustrative value ($s_0 \simeq 4.8\times10^{-3}$ eV) and the electroweak scale $m_T \sim \eta_{\rm EW}$ are then excluded (Foundation II, Appendix F). *(Corrected 22/10/2026: this line compared with an unsourced bound $|s_0|/M_{\rm Pl} \lesssim 10^{-15}$, which required only $m_T \gtrsim 10^{-18}$–10^{-17} eV.) (Corrected 15/10/2026: this line declared the three constraints in tension (I71, 29/08; gap "7 to 713 times") by confronting them with a "MICROSCOPE window" $s_0 \sim 10^{-5}$–$10^{-3}\, M_{\rm Pl}$ that has no source in the public corpus.)* What remains open: the corpus does not establish that the m_T of the degeneracy is the same physical pole as the one that sets the halo-scale range. ⚠ **Consequence to be stated clearly: as long as this spectral identification is not established, the massive torsion sector must be regarded as an effective parametrisation and not as a fully identified physical sector.** The normalisation $m_T^2 s_0^2$ reproduces the adopted dark-energy density; it does not thereby constitute an independent derivation of that density. Separate constraints or estimates exist on m_T, s_0 and the associated range, but the corpus does not yet establish that they bear on a single canonical mode or on a single physical pole. This identification belongs to PO-ECF-ACTION / the spectral lock of the axial sector.
- $\xi_{\rm KZ}$ (Kibble-Zurek scale): calibrated, not derived microscopically — this is open problem **PO-F3-XI**.
- n_s: the large-scale primordial spectrum **is not predicted** by the framework (it is treated as an inherited input; the interpretation in terms of a parent universe remains exploratory); what the *stiff* era generates is a blue component bounding its duration, $N_{\rm stiff} \gtrsim 68$ (PO-F3-NS, restated 10/08).

This last list is not a hidden gap: these are the ECF's open research fronts, documented as such.

The research programme: from calibrated to derived

Beyond these individual fronts, several quantities of the framework are today **calibrated** on data rather than derived *ab initio* — $F_{\rm ion}$ (H_0 sector), the (w_0, w_a) evolution of dark energy, and others listed in Appendix B. Their status is stated unambiguously, document by document. Dark matter is partly an exception: it is partly calibrated (mass at the centre of the Kibble-Zurek window; $\lambda \geq 36$, HSC benchmark), and the micro-knot → halo link remains open (conjectural link **PO-F2-1**).

These statuses are **acknowledged limits**, not concealed gaps: the framework continually distinguishes what it derives from what it fits — the condition for any future confrontation with data to carry the weight of a test.

Appendix C. Founding references (verified)

The primary sources of each block, to facilitate investigation. **These references have been verified; they describe the borrowed blocks, not the ECF articles themselves** (the latter are on Zenodo, see the end of the document).

Block	Verified primary reference
Einstein-Cartan (foundation)	É. Cartan, *C. R. Acad. Sci. (Paris)* **174**, 593 (1922); and *Ann. Éc. Norm. Sup.* **40**, 325 (1923)

Block	Verified primary reference
Einstein-Cartan (modern formulation, gauge)	T. W. B. Kibble, "Lorentz invariance and the gravitational field", *J. Math. Phys.* **2**, 212 (1961); D. W. Sciama, *Recent Developments in General Relativity* (Pergamon, 1962), p. 415
Einstein-Cartan (reference review)	F. W. Hehl, P. von der Heyde, G. D. Kerlick, J. M. Nester, "General relativity with spin and torsion: Foundations and prospects", *Rev. Mod. Phys.* **48**, 393 (1976)
Weyssenhoff fluid	J. Weyssenhoff & A. Raabe, "Relativistic dynamics of spin-fluids and spin-particles", *Acta Phys. Pol.* **9**, 7–18 (1947)
Kibble mechanism (topological defects)	T. W. B. Kibble, "Topology of cosmic domains and strings", *J. Phys. A: Math. Gen.* **9**, 1387 (1976)
Zurek mechanism (quench rate)	W. H. Zurek, "Cosmological experiments in superfluid helium?", *Nature* **317**, 505 (1985); review: *Phys. Rep.* **276**, 177 (1996)
Defect classification (homotopy)	T. W. B. Kibble, *J. Phys. A* **9**, 1387 (1976); A. Vilenkin & E. P. S. Shellard, *Cosmic Strings and Other Topological Defects* (Cambridge Univ. Press, 1994); review: A. Vilenkin, *Phys. Rep.* **121**, 263 (1985)
Landau-Ginzburg potential	V. L. Ginzburg & L. D. Landau, *Zh. Eksp. Teor. Fiz.* **20**, 1064–1082 (1950)
CPL parametrization	M. Chevallier & D. Polarski, *Int. J. Mod. Phys. D* **10**, 213 (2001); E. V. Linder, *Phys. Rev. Lett.* **90**, 091301 (2003)
Sakharov conditions	A. D. Sakharov, *JETP Lett.* **5**, 24 (1967)
Black holes → bounce (ECF lineage)	N. Popławski, "Cosmology with torsion: An alternative to cosmic inflation", *Phys. Lett. B* **694**, 181 (2010)
Spin-torsion bounce, two-term Friedmann equation	N. Popławski, "Nonsingular, big-bounce cosmology from spinor-torsion coupling", *Phys. Rev. D* **85**, 107502 (2012), arXiv:1111.4595
Same effective form in loop quantum cosmology (comparison)	I. Agullo & P. Singh, "Loop quantum cosmology: A brief review", in *Loop Quantum Gravity: The First 30 Years*, ed. A. Ashtekar & J. Pullin (World Scientific, 2017), arXiv:1612.01236
Zel'dovich collapse (cosmic web, generic non-ECF mechanism)	Ya. B. Zel'dovich, "Gravitational instability: An approximate theory for large density perturbations", *Astron. Astrophys.* **5**, 84 (1970)
Critique of classical bounces (null energy condition, shear)	D. Battefeld & P. Peter, "A critical review of classical bouncing cosmologies", *Phys. Rep.* (2015), doi:10.1016/j.physrep.2014.12.004, arXiv:1406.2790 *(volume and pagination not verified)*
Chirality of the gravitational-wave background (LISA–Taiji network)	G. Orlando, M. Pieroni & A. Ricciardone, "Measuring parity violation in the stochastic gravitational wave background with the LISA–Taiji network", *J. Cosmol. Astropart. Phys.* **03** (2021) 069, doi:10.1088/1475-7516/2021/03/069, arXiv:2011.07059

Note on verification. Authors, titles, journals and paginations have been checked. This bibliographic verification is distinct from the verification of **scientific scope**: an accurate reference does not guarantee that the source article establishes the whole interpretation the corpus associates with it. This second check is recorded in `Qualite/ECF_REGISTRE_valeurs_empruntees.md` and led to narrowing the attributions to Fabbri, Carroll & Field, Popławski and Kibble-Zurek.

Useful paginations. Weyssenhoff-Raabe (1947): *Acta Phys. Pol.* **9**, 7–18. Ginzburg-Landau (1950): *Zh. Eksp. Teor. Fiz.* **20**, 1064–1082; English translation in *Collected Papers of L. D. Landau* (Pergamon, 1965, p. 546).

Entering Einstein-Cartan theory itself

A reader wishing to master the spin-torsion formalism (beyond its use within the ECF) may follow this reading order, from overview to full treatise:

1. **A. Trautman**, "Einstein-Cartan theory", *Encyclopedia of Mathematical Physics* (Elsevier, 2006), arXiv:gr-qc/0606062 — a concise encyclopaedia article, ideal as a first overview.
2. **N. Popławski**, "Cosmology with torsion: An Alternative to Cosmic Inflation", *Phys. Lett. B* **694**, 181 (2010), arXiv:1007.0587 — the cosmological core: torsion generates a repulsion that prevents the singularity and resolves flatness and horizon. Direct building block of the ECF bounce.
3. **N. Popławski**, "Spacetime torsion as a possible remedy to major problems in gravity and cosmology", arXiv:1106.4859 (2011) — an accessible panorama of what torsion addresses (singularities, inflation, dark energy, baryogenesis, dark matter).
4. **F. W. Hehl, P. von der Heyde, G. D. Kerlick, J. M. Nester**, "General Relativity with Spin and Torsion: Foundations and Prospects", *Rev. Mod. Phys.* **48**, 393 (1976) — the founding review, full formalism of the spin-torsion coupling and derivation of the U_4 theory as a Poincaré gauge theory.

For a textbook treatment: V. de Sabbata & C. Sivaram, *Spin and Torsion in Gravitation* (World Scientific, 1994); and M. Blagojević & F. W. Hehl (eds.), *Gauge Theories of Gravitation: A Reader with Commentaries* (Imperial College Press, 2013), a commented collection of the founding texts.

Observational references (datasets confronted)

The framework is calibrated and tested against published data. To let the reader trace the sources of the confrontations reported in §5 and in the positioning of §6, here are the primary references for the main datasets used. **These papers describe the observational measurements, not the ECF.**

Dataset	Role in the ECF	Primary reference
SH0ES (local H_0)	H_0 sector calibration (73.04 ± 1.04)	A. G. Riess *et al.*, *Astrophys. J. Lett.* **934**, L7 (2022), arXiv:2112.04510
Planck 2018 (CMB)	ΛCDM base, $\Omega_m = 0.315$, acoustic scale	Planck Collaboration (Aghanim *et al.*), *Astron. Astrophys.* **641**, A6 (2020), arXiv:1807.06209
KiDS-1000 (weak lensing, S_8)	S_8 sector calibration ($0.766^{+0.020}_{-0.014}$)	C. Heymans *et al.*, *Astron. Astrophys.* **646**, A140 (2021), arXiv:2007.15632
DESI DR1 (BAO)	(w_0, w_a) confrontation	DESI Collaboration (Adame *et al.*), "DESI 2024 VI: Cosmological Constraints from the Measurements of Baryon Acoustic Oscillations", *J. Cosmol. Astropart. Phys.* **02** (2025) 021, doi:10.1088/1475-7516/2025/02/021, arXiv:2404.03002
DESI DR2 (BAO)	dynamical dark energy ($3.1\,\sigma$ BAO+CMB; 2.8–$4.2\,\sigma$ with SNe)	DESI Collaboration (Abdul Karim *et al.*), *Phys. Rev. D* **112**, 083515 (2025), arXiv:2503.14738
SPARC (175 galaxies)	geometric dark-matter test ($\tilde{\chi}^2_{\rm med} = 0.80$)	F. Lelli, S. S. McGaugh & J. M. Schombert, *Astron. J.* **152**, 157 (2016), arXiv:1606.09251

Dataset	Role in the ECF	Primary reference
DESIVAST DR1 (voids)	ξ_{KZ} calibration, void-function test	H. Rincon *et al.* (DESI), *ApJ* **982**, 38 (2025), arXiv:2411.00148
ACT DR6 (birefringence)	chiral birefringence test ($\beta \neq 0$)	P. Diego-Palazuelos & E. Komatsu, "Cosmic birefringence from the Atacama Cosmology Telescope Data Release 6", *Phys. Rev. D* **113**, L101302 (2026), doi:10.1103/pbc3-t52s, arXiv:2509.13654; $\beta = 0.215° \pm 0.074°$ *(corrected 22/10/2026: the analysis, previously cited through the MPA note of 1 February 2026, is published)*
Pantheon+ (SN Ia)	Hubble diagram, $q(z)$; external comparison of the late F1 z_t, with no ECF fit	D. Brout *et al.*, *Astrophys. J.* **938**, 110 (2022), arXiv:2202.04077; S. Arora & P. K. Sahoo, *Phys. Dark Univ.* **45**, 101510 (2024), arXiv:2312.06464; D. Dahiya & D. Jain, *Res. Astron. Astrophys.* **23**, 095001 (2023), doi:10.1088/1674-4527/ace17a

Note. The DESIVAST and ACT DR6 rows follow the references of Foundation III and Foundation II, where these confrontations are carried out in detail *(completed 19/10/2026)*. The present document refers to the Foundation papers (F1–F3) for the full bibliographic apparatus; this table lists only the foremost observational sources.

Appendix D. ECF corpus used — exact versions (traceability)

This synthesis document rests on the following versions of the ECF corpus. Stating them precisely makes it possible, at any later date, to establish the **delta** between the state described here and a more recent version. The version is authoritative via the **file name**; the Zenodo **concept DOI** (not the frozen DOI of a single version) always points to the latest published version.

Paper	Version used here	Zenodo concept DOI	Freeze of this version
PIT Letter (the postulate)	v8 revised (preprint v9)	10.5281/zenodo.19798923	v8 deposited 6 September 2026; v9: October 2026 deposit
Foundation I — Extended (tensions)	rev. 13 (preprint v8)	10.5281/zenodo.19577447	v7 deposited 14 September 2026; v8: October 2026 deposit
Foundation I — PRD Short	rev. 7.1 revised (preprint v8)	10.5281/zenodo.19577447 *(same concept deposit as F1 Extended)*	v7 deposited 14 September 2026; v8: October 2026 deposit
Foundation II — Extended (dark matter, structure)	rev. 20 (preprint v5)	10.5281/zenodo.20629237	v4 deposited 15 September 2026; v5: October 2026 deposit
Foundation III — Cosmological Physics (testable physics)	rev. 1.1 revised (preprint v3)	10.5281/zenodo.22359982	v2 deposited 15 September 2026; v3: October 2026 deposit

Paper	Version used here	Zenodo concept DOI	Freeze of this version
Foundation III — Strong Gravity (companion, strong-field sector)	rev. 2.1 + CORE-01	—	*(not deposited)*
Foundation III — Chiral Bounce (companion, parity breaking)	rev. 0.4	—	*(not deposited)*
Foundation IV — Fertile (exploratory)	rev. 0.3	—	*(preliminary, not deposited)*

Method note. **This document cites concept DOIs throughout**, never version DOIs. A concept DOI always resolves to the current version of a deposit: it does not go stale, so Architecture need not be reissued whenever a paper is revised. ⚠ *The trade-off must be clear to the reader: the numerical values quoted here correspond to the state of the corpus at the date given below, and a paper consulted through its concept DOI may have revised them since.* To cite exactly the version used here, refer to the version number (column 2) and to the producing script named in the parameter tables. The fields marked "—" will be filled at the corresponding deposit; while empty, the version (file name) remains the unambiguous reference.

Date of this corpus state: 22 October 2026.

Appendix E. Querying the corpus with NotebookLM

A method guide, to move fast without losing reliability.

The public corpus runs to roughly 315 pages across five documents (count of 23/10/2026 on the deposit PDFs: PIT 6 p. + Foundation I Extended 87 p. + Foundation I PRD Short 8 p. + Foundation II 173 p. + Cosmological Physics 41 p.). Finding where a quantity is defined, derived or calibrated can take tens of minutes of cross-reading. **NotebookLM** (Google, free) lets you upload PDFs as sources and query them in natural language; every answer cites the exact passage it comes from.

> **The tool removes the navigation time; it does not replace critical reading.** It renders faithfully what the sources say; it does not judge whether they are physically correct. That remains the reader's role.
>
> *Caveat: this is unpublished work; uploading documents to a third-party service is a personal choice. The nomenclature in Appendix B serves the same indexing purpose with no external dependency.*

E.1 Sources to upload

Upload the whole public corpus, **starting with this document**: it carries the overall map, the technical table (principle, inputs, script and status for each quantity) and refers to the Foundations for detail.

E.2 First contact — three questions to orient yourself

For a reader meeting this framework for the first time, these three give its shape in about ten minutes.

1. **"What problem is this framework trying to solve, and what does it add to general relativity?"**
2. **"Which quantities are derived from the action, which are calibrated, which are predictions?"** — *this is the question that matters most, and this corpus is built to answer it.*
3. **"What does this framework not explain?"**

If the answers to questions 2 and 3 seem evasive, that is a finding about the corpus. They are not meant to be.

E.3 A worked example

Ask: *"How is dark energy obtained? Distinguish what is derived from the action from what is calibrated, and say whether (w_0, w_a) were fixed before or after the DESI data."*

What the answer should contain, sourced: the **static** part ($w = -1$) is **derived** — FLRW reduction of the axial sector, Foundation II Appendix J; the **fluctuations** of the same field behave as radiation ($w = +1/3$ **in the adopted Proca form; conditional on PO-F2-6**); the **evolution** (w_0, w_a) is **CALIBRATED** and, in the active baseline, $\xi = 1.645$ is calibrated on the DESI preference used by the ECF: the pair $(w_0, w_a) = (-0.904, -0.153)$ is therefore **not a prediction independent of DESI**; and the relation $w_0 = -1 + \Omega_{\text{spin,peak}}$ is an **empirical identification**, not a derivation.

E.4 Questions that work well

On the mechanism — the modified Friedmann equation and the origin of the spin term's sign · the derivation of the bounce, step by step · the sound-horizon reduction.

On the status of a result *(the most useful class of questions)* — "is ξ_{KZ} derived or calibrated, and on which data?" · "which script produces (w_0, w_a), and what are its inputs?" · "is the $0.35°$ birefringence a prediction or a consistency check?"

On falsifiability — "what observation would refute this framework?" · "what does it predict that ΛCDM does not?"

On comparisons with ΛCDM — for each announced difference, ask at the same time for its status: `DERIVED`, `CALIBRATED`, `CALCULATED-CONDITIONAL`, `STRUCTURAL-HYPOTHESIS` or `OPEN`.

On the open problems — "list the open problems and say which is the most severe" · "what has been tried to derive the dark-energy evolution, and why did it fail?"

*To locate a quantity, ask for its definition **and** its status: "where is σ_0 defined, and is it a free parameter?"*

E.5 Reading the answers without overinterpretation

The corpus applies a strict rule: **every statement carries an explicit status, and nothing is presented beyond it.** An automatic summary can flatten these distinctions — keep the scale in mind: **borrowed** (established physics, not claimed) · **derived** (follows from the action with no adjustment — *two quantities only*) · **calibrated** (fitted to named data) · **prediction** (an unadjusted consequence: this is where the framework is at stake) · **open** (an unresolved problem, documented as such).

Three points of vigilance.

(a) *A conjecture is not a result.* Where the corpus proposes rather than demonstrates, it says so; check the status in the source.

(b) *The open problems are real and stated.* The principal one is the **mapping between the F1 spin sector and the F2 propagating axial sector**: their identification as two regimes of a single canonical field **is not established**. The connection action → irreducible variables → physical poles → cosmological variable remains to be derived. *(Updated 12/09/2026: this problem is now framed as a spectral question — is the axial degree of freedom employed a unitary pole of the theory, and do the constraints bearing on it probe the same mode? It is tracked under `PO-F2-6` and has been submitted to specialists in Poincaré gauge gravity; its resolution is not claimed here.)* The **evolution** of dark energy is calibrated; three routes were tested and reported as failing. The ξ_{KZ} scale is calibrated (PO-F3-XI).

(c) *On the primordial spectrum — a point where earlier versions were wrong.* The large-scale spectral index n_s is **not predicted**: it is **inherited**, an input of the model on the same footing as Ω_b. Earlier versions claimed a prediction of $n_s \simeq 1$ and a tension of 8.4σ with Planck; **both rested on an incorrect identification of the Wands duality and are withdrawn.** *If a query still returns that figure, the source is a superseded version: check the document date.*

Appendix F. Methods behind each result, calibration or test

Added 01/09/2026, at the explicit request to anchor every result, calibration or test in §5 to its precise computational technique — a methodologically demanding reader is not satisfied by a script reference without knowing what technique it implements. Every row was verified directly in the corresponding script or source section, not inferred from the file name.

Result / test	Method	Nature
Euclid — lensing deficit / S_8	Analytic transfer function (BBKS 1986) + regularised Lorentzian torsion suppression; compared against a simulated Euclid forecast (15 log-spaced points, 3% noise)	**MODELLED / CALCULATED-CONDITIONAL**; full physical ECF growth **OPEN**
VSF — large voids (Euclid, Rubin/LSST)	Sheth–van de Weygaert void statistics (ΛCDM) modulated by a Gaussian amplification factor $A_{\rm ECF}(R, z)$; amplitude fitted on BOSS DR12 (Rubin/LSST, Euclid) and directly confronted with the public DESIVAST DR1 catalogue (FITS)	Void statistics + amplitude fit
Roman — microlensing	Standard analytic point-mass microlensing optical-depth formalism	Direct analytic calculation
LISA — $\Pi_{\rm GW}$, n_T	WKB analysis of the bounce action (Vilenkin/Hartle-Hawking tunnelling formalism) for amplitude; Giovannini-type (1999) stochastic-background spectrum for the stiff era	WKB analysis + analytic stochastic spectrum
C_ℓ suppression at $\ell \approx 233$	Analytic window function (spherical Bessel form factor), Limber approximation	Analytic model
High-ℓ deficit ($\ell \gtrsim 2000$, F1)	PCHIP interpolation over four calibrated points (F1 Appendix K), derived from Silk-damping physics and the recombination phase shift	Interpolation over calibrated points
$f_{\rm NL}^{\rm KZ}$	Bispectrum statistics of a 3D Poisson point process, projected to 2D (Sachs-Wolfe)	Poisson statistics
Space EP tests / $\eta_{\rm WEP}$	No estimate: s_0 is not determined, only $m_T^2 s_0^2$ is fixed; the SME coefficient $b_\mu = -\frac{3}{4} S_\mu$ (minimal coupling) acts on spin, whereas EP tests with unpolarised bodies probe other coefficients; the relevant bound is the laboratory one, $s_0 \lesssim 3 \times 10^{-27}$ GeV (Foundation II, Appendix F) *(corrected 22/10/2026: "only the SME bound $\lvert s_0\rvert/M_{\rm Pl} \lesssim 10^{-15}$ applies" — unsourced bound; corrected 15/10/2026: this line announced an algebraic estimate from $\kappa_{\rm spin}$, an undefined coupling, and $S_{\rm ref}$)*	Conditional test with no predicted amplitude

Result / test	Method	Nature
DESI, five-year results — (w_0, w_a)	Point **CALIBRATED on the DESI preference used by the ECF**, plotted against DESI's published MCMC posterior contours (DR1/DR2); not an independent *a priori* prediction	Comparison to an external MCMC posterior
JWST/ELT — compact seeds (LRD)	Causal scale $c^3/(2GH)$ computed analytically; conversion into a compact object (ϵ_{coll}) not derived at the freeze-out epoch ($M_K = c^3/(2GH(z_t))$), under assumptions	formation = **STRUCTURAL-HYPOTHESIS**
Birefringence (CMB: LiteBIRD, Simons Observatory)	Chern-Simons-type coupling integral over the expansion history, calibrated on the Minami–Komatsu anchor	Calibrated analytic integral
SKA/HERA — 21 cm acoustic ruler r_d	Computation of the drag ruler $r_d = r_s(z_d)$: acoustic integral with modified $H(z)$ **and** consistent determination of the drag epoch z_d; the baseline $r_s = 135.65$ Mpc must not be identified with r_d by assumption	`OPEN / CALCULATED-DIAGNOSTIC` under `PO-ECF-BAO-RDRAG`
Mars Sample Return — homochirality	Qualitative observational criterion (presence/majority of one enantiomer); no associated numerical calculation, probabilistic threshold not yet derived (PO-F3-CHIRAL)	Qualitative, unquantified criterion

What this table does not say. An identified method is not a guarantee of robustness — several of the analytic models above rest on calibrated rather than derived parameters (see each quantity's status in §5 and Appendix B). This table answers one precise, limited question: *by what technique was this number obtained?*, not *is this number solid?*.

A methodological finding, while verifying this table (01/09/2026): two scripts (`plot_F3_ecf_Cell_suppression.py`, `plot_F3_ecf_ns_KZ.py`) carried header documentation describing an earlier, since-corrected version of the calculation (the old error on the C_ℓ suppression scale, the old withdrawn n_s tension at 8.4σ). The calculation **executed** in the body of each script was already correct and consistent with the text above; only the header documentation was not — **corrected the same day**, with no bearing on the results already cited.

Appendix G. How to review and comment on the ECF corpus

*The procedure described here applies to **Architecture as well as to each of the papers of the ECF corpus**. It requires neither a particular account nor specialised software: a **PDF reader that allows comments and annotations to be added** is enough. Adobe Acrobat Reader, Foxit PDF Reader, PDF-XChange Editor, Microsoft Edge or Preview on macOS are suitable, as is any other reader offering equivalent functions. Adobe Acrobat Pro is not needed: the free annotation functions of a PDF reader are enough.*

G.1 General principle

Choose the **relevant PDF** — Architecture, Foundation I, Foundation II, Foundation III — Cosmological Physics, Strong Gravity, Chiral Bounce, Foundation IV, or any other document transmitted in the corpus — then annotate directly the

passage you wish to discuss.

If possible, begin each note with its type, followed by a dash or a colon:

Start of the note	For…
`ERROR —`	a value, a sign, a reference or a claim that seems inaccurate to you
`QUESTION —`	a point you do not understand or that seems insufficiently justified to you
`SUGGESTION —`	an addition, a clarification or a cross-reference that would improve the document
`FORM —`	a typo, a turn of phrase or a layout problem
`… Proposal: …`	*(optional)* the text you would propose instead

If possible, indicate the **page**, the **section** and, where there is one, the **equation number**. In Architecture, displayed equations are numbered to make this kind of exchange easier. In a paper where the equation is not numbered, the page, the section and a few words of the passage are enough.

G.2 Commenting on Architecture

Architecture is a **synthesis and a map**. A remark may therefore concern the clarity of the narrative, the consistency of a status, a missing cross-reference or a gap between the synthesis and a primary paper.

When a remark reveals a problem in the source paper, the correction is **not** made in Architecture alone: it is attached to the primary document, then propagated to Architecture if the synthesis must change.

[Appendix G illustration: graphic resource not included in the current source package.]

*Fictitious example: five comments placed on a single paragraph — examples of a question, a suggestion, a form remark and a free note; the image will be replaced by a version that also explicitly shows the **ERROR + Proposal** convention placed on the page.*

G.3 Commenting on a paper of the corpus

The procedure is identical:

1. open the **PDF of the relevant paper with a PDF reader that allows comments and annotations to be added**;
2. highlight or annotate the passage;
3. indicate as far as possible page, section, equation or figure;
4. save the PDF with its annotations;
5. send it to **p.fichant.research@gmail.com**.

The remark is then examined **at the level of the source paper**. If it changes a value, a status, a dependency or a conclusion taken up elsewhere, the derived documents — notably Architecture — are checked in turn. This rule prevents Architecture from becoming a circular source of authority: **the papers remain the primary scientific sources**.

G.4 What happens after sending

Each remark is checked in the corpus and then receives one of the following states:

- **corrected** — the correction is applied and the version carrying it is indicated;
- **set aside** — the remark does not lead to a correction, with the reason;
- **evolution** — the proposal is retained for a later version;
- **open** — the question touches an unresolved scientific problem;
- **blocked** — an editorial change would be premature as long as a producer, a derivation or a scientific arbitration is missing.

A personal reply may be sent by e-mail, remark by remark, with the decision and its justification.

Confidentiality. The annotated PDFs and comments received are not made public or communicated to other readers by the author. Only the corrections retained appear in subsequent versions; your name is cited — in the acknowledgements, for example — only with your agreement.

Good to know.

- The `ERROR / QUESTION / SUGGESTION / FORM` convention is not mandatory: a free comment is read too.
- The name displayed on your notes is the one set in your PDF reader; you can change it in its preferences.
- *Zotero, Mendeley*: annotations stay in their database; use "Export PDF with annotations" before sending.

Appendix H. Editorial traceability and post-audit v2 baseline

H.1 — Editorial traceability

This appendix indicates **where to find the history**, without reproducing the old scientific formulations in Architecture. The aim is to preserve traceability while preventing a reader from confusing an outdated state with the current state of the framework.

- Earlier versions of Architecture remain archived in the corpus repository and in the Zenodo/GitHub history.
- Scientific and editorial corrections are recorded in the audit logs and the quality register of the corpus.
- Withdrawn claims remain marked `WITHDRAWN` in the governance documents; they are not reproduced as active results in the present text.
- For any divergence between a historical formulation and the present document, **the current state of the body of Architecture and the corresponding Foundation sources are authoritative**.

This choice is deliberate: Architecture is an **organised mirror of the current state of the corpus**, not an exhaustive log of its revisions.

H.2 — Canonical post-audit v2 baseline (22/09/2026)

Background. $F_{\text{ion}} = 1.2765 \pm 0.0246$; $r_s^{\text{ECF}} = 135.65$ Mpc; $H_0 = 73.04$ km s^{-1} Mpc^{-1}; $\Omega_m = 0.2669$; $\Omega_r = 7.8417 \times 10^{-5}$; $\omega_\nu = 1.71061 \times 10^{-5}$.

Chronometry. $t_0 = 13.33$ Gyr ($w = -1$; 13.19 Gyr at the calibrated (w_0, w_a) point); $t_{\text{rec}} = 368.4$ kyr (with the spin term; 371.3 kyr in ΛCDM); $t_{\text{GC}} = 13.32 \pm 0.10_{\text{stat}} \pm 0.5_{\text{sys}}$ Gyr (Valcin et al. 2020). t_{GC} is a lower bound on t_0; the corresponding age of the Universe, $t_U = 13.5^{+0.16}_{-0.14} \pm 0.5$ Gyr (same source), lies 0.17 Gyr above t_0, within the systematic.

BAO — CALCULATED-CONDITIONAL. eBOSS DR16 QSO target $D_M/r_d(z = 1.48) = 30.21 \pm 0.79$ (eBOSS QSO BAO+RSD consensus, Neveux et al. 2020); former 30.85 ± 0.80 = WITHDRAWN / UNTRACED HISTORICAL TARGET; $\chi^2_{\text{BAO}}(\text{ECF}) = 12.11$, $\chi^2_{\text{BAO}}(\Lambda\text{CDM}) = 1.79$, $\Delta\chi^2 = +10.3$ (diagonal, indicative; corrected on 04/10/2026: the script's $z = 0.61$ value, 15.45, replaced by 15.61; on 05/10/2026, errors 0.166 / 0.206 / 0.251 (statistical and systematic combined, table 7 of Alam et al. 2017; the one at $z = 0.61$ was the statistical error alone)). `PO-ECF-BAO-RDRAG` remains OPEN: no identification $r_s = r_d$ is made.

DESI DR1. $\xi = 1.645$ is **CALIBRATED on the DESI preference used by the ECF**, not an independent prediction. $\chi^2 = 55.28$ refers to a subset of **10 DR1 observables in the diagonal approximation**, without BGS, QSO or covariance; it is not the global DESI DR1 χ^2. Reconstructed MCMC block: **12 observables**; $\Delta\chi^2(\text{ECF} - \Lambda\text{CDM}) = +6.21/+38.59$ (Planck/SH0ES).

Growth. $S_8 = 0.766$ is a **KiDS-1000 observational target**, not an ECF prediction. $S_8^{\text{ref}} = 0.780$ is a CAMB ΛCDM reference on the A-7 background, CALCULATED-CONDITIONAL. $\gamma_{\text{spin}} = 0.3116$ is HISTORICAL; $\gamma_{\text{spin}} \simeq 0.066^{+0.069}_{-0.094}$ is CALCULATED-CONDITIONAL / EXPLORATORY, physical value OPEN. $A_{\text{sup}} = 0.15$ is adopted; its spin mechanism remains OPEN.

Structuring OPEN items. Complete PGT action; PO-F2-6; spin/BBN normalisation; PO-ECF-BAO-RDRAG; PO-F2-5; PO-F1-B; Boltzmann/CLASS-EC; γ_{spin}; $T^2(k)$; PO-F3-XI; PO-ECF-HALO; I-2 (BLOCKED).

NOT RECALCULATED. Global Planck, θ_*, global $\Delta\chi^2$, global AIC/BIC after A-7. The old values remain HISTORICAL.

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Independence, funding and open science

The author is an **independent researcher** and declares no conflict of interest. This work was carried out **without institutional, commercial or external funding**. The costs required for the research, for the use of scientific and documentary tools, for the preparation of the documents, and for their dissemination and archiving are **borne entirely by the author**.

The ECF corpus is part of an **Open Science** approach: the works are made publicly accessible, their evolution and epistemic status are documented, and critical remarks, independent checks and reproduction attempts are encouraged.

The full reproducibility material is available at github.com/pfichant/spin-torsion-cosmology (CC-BY-4.0).
